

READING THE FIVE FIELDS

200 Graded Passages That Build What the SAT Assumes

Two hundred passages · five fields · fifty strands · four levels

A companion to Words in Context

What this book is for

The SAT does not test knowledge of history or science. It does test reading, and it does so with short passages drawn from history, social studies, the humanities, literature and science. A student who has met the subject before reads such a passage faster, guesses unknown words better, and is not stopped by a name.

That is not a claim about intelligence. It is the most reliable finding in the research on reading comprehension. Recht and Leslie, working with sixty-four junior-high students in 1988, found that weak readers who knew baseball recalled more of a passage about a baseball game than strong readers who did not. The study has been over-read and criticized, and the direction of the effect has held up: a systematic review of twenty-three studies in 2021 found more domain knowledge produced better comprehension of text in that domain.

For a reader working in a second language there is a condition attached. Background knowledge pays only once enough of the words are known, a point the research literature calls the linguistic threshold. Work on second-language listening finds that learners below a certain vocabulary level get no benefit at all from knowing the topic. The usable version of that finding is the coverage figure attributed to Hu and Nation: about ninety-eight per cent of the running words need to be known for unassisted reading, and about ninety-five per cent for minimally acceptable comprehension.

So this book has to do two things at once. It has to raise knowledge, and it has to stay inside the threshold while doing it. Ninety-eight per cent of three hundred words is where the arithmetic lands: at most six words per passage may fall outside the level's assumed vocabulary, and at the first two levels every one of them is explained where it stands.

The figures above reached this book through secondary sources. Clarke, Hu and Nation, and Recht and Leslie should be checked at source before any of them is quoted in a classroom.

How the book is built

Five fields, ten strands in each, four levels in each strand. A strand is one thread of content followed through all four levels, so each strand receives twelve hundred words of treatment and can be read downward as well as across.

The four levels do four different jobs, in the same order every time. This is what makes the coverage graded rather than merely longer.

Level 1, Foundation — phenomenon. One concrete case, named and dated: what happens and what it looks like. Sentences average 13 to 18 words; vocabulary is assumed to the top 16,000 most frequent word forms.

Level 2, Developing — mechanism. How it works and how much: cause, comparison, magnitude. Sentences average 16 to 21 words; vocabulary is assumed to the top 22,000 most frequent word forms.

Level 3, Target — evidence. How anyone knows: the study, the document, the measurement, and its limits. Sentences average 19 to 24 words; vocabulary is assumed to the top 30,000 most frequent word forms.

Level 4, Stretch — dispute. What is contested: two readings of one body of evidence, and the writer's stance. Sentences average 21 to 28 words; vocabulary is assumed to the top 40,000 most frequent word forms.

So one strand in biology runs: a reindeer herd that reached six thousand and then forty-two (phenomenon), the growth curves and the lag that let a population pass its own ceiling (mechanism), the Hudson's Bay fur returns and what a trapping ledger can and cannot show (evidence), and the experiment that found food and predators acting together rather than separately (dispute).

Every one of the two hundred target words from Words in Context appears somewhere in these passages, in the same field and at the same level, used in running prose. A student can meet a word here and meet it again as a question.

How to use it

Read across or read down. Reading across means taking all ten passages of one field at one level, which gives a survey of the field in three thousand words.

Reading down means taking one strand through its four levels, which shows the same subject treated four ways and is the better use of the book for anyone with time.

Each passage ends with three lines headed what this passage is for. Read them after the passage, not before. If none of the three is a surprise, the passage has done its work and can be left. If one of them is, the passage is worth a second reading, and the sentence it came from is worth finding.

Do not look up every unfamiliar word. The passages are built so that no more than six words in three hundred fall outside what a reader at that level is assumed to know, and the technical ones are explained in the sentence where they appear. A reader who stops at every name will never find out that the names do not matter.

Keep a list of the field terms rather than of the hard words. Appendix B collects every term the book explains, with its explanation and the passage it appears in, and that list is the actual content of the book.

What this book is not

These passages are not past papers and they are not a syllabus. The College Board publishes no list of subjects and no reading list, so the five fields and the fifty strands are a curricular judgment, informed by what the test is described as covering. The passages were written from general knowledge and checked for standard figures; a reviewer who intends to assert any particular fact in a classroom should verify it first.

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Level 3 — Target · evidence

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The fifty strands

Appendix A — strand index

Appendix B — the field terms explained

Appendix C — where the vocabulary words appear

Appendix D — reading log

The fifty strands

Each line is one strand. The four entries under it are the four levels, in order: phenomenon, mechanism, evidence, dispute.

History and Civics

S01 The Declaration and the claim of natural rights

July 1776 in Philadelphia: what the document is, who wrote it, what it lists.
 The argument's shape: self-evident truths, consent, and the right to alter government.
 What the drafts show: Jefferson's deleted passage on the slave trade, and what the edit tells us.
 Whether the Declaration promised equality or only independence: Douglass and Calhoun read the same sentence.

S02 The Constitution's machinery

Three branches, named, with one power each that the others can block.
 Why the machinery was built that way: faction, and Madison's remedy in Federalist 10.
 Hamilton on a single executive in Federalist 70, and what the ratification debates record.
 Whether the checks protect liberty or produce paralysis: two readings of the same design.

S03 Who counts as a citizen

Seneca Falls, 1848: a convention, a declaration, and a list of grievances.
 How the franchise actually widened: property, race, sex, age, and the amendments that did it.
 Reading the suffrage record: petitions, poll books, and what a signature count can show.
 Whether rights are won in courts or in the street: the suffrage movement's own quarrel.

S04 Abolition and the argument against slavery

Frederick Douglass at Rochester, 1852: the Fourth of July speech and its question.
 The argument from natural rights, and the proslavery answer from property and order.
 The evidence abolitionists used: narratives, ledgers, ship manifests, and their reliability.
 Whether moral suasion or political power ended slavery: Garrison against the Free Soilers.

S05 Reconstruction and the retreat from it

1865 to 1877: three amendments, the Freedmen's Bureau, and what was built.
 How the retreat worked: violence, the courts, the 1876 bargain, and disenfranchisement.
 What the sources say: Congressional testimony, registration rolls, and the gap between them.
 Whether Reconstruction failed or was defeated: Du Bois against the Dunning school.

S06 Industry, labor and the regulating state

The Triangle fire, 1911: a locked door, 146 deaths, and what followed.
 How regulation spread: inspection, liability, hours laws, and the cost of each.
 The evidence behind factory law: commission reports, mortality tables, and their gaps.
 Whether regulation raised wages or cost jobs: the same statistics, two conclusions.

S07 Civil rights: courts, statutes and the street

Montgomery, 1955: a bus, a boycott, and a year of walking.
 How the law changed: Brown, the 1964 Act, the 1965 Act, and the mechanism of each.

What the record holds: court transcripts, registration figures, and FBI files.
 Whether the movement's gains came from law or from pressure: King's own argument with himself.

S08 Empire and independence

Salt at Dandi, 1930: a march, a handful of mud, and sixty thousand arrests.
 How empires were run and how they ended: revenue, railways, soldiers, and debt.
 Reading colonial records: tax rolls, famine reports, and whose numbers they are.
 Whether independence was granted or taken: Attlee's account against Nehru's.

S09 War, emergency and civil liberty

Executive Order 9066, 1942: an order, a map, and 120,000 people moved.
 How emergency powers work: suspension, detention, censorship, and the usual arguments.
 The evidence at the time: the Ringle report, the Munson report, and what was withheld.
 Whether security ever justifies the suspension of rights: Korematsu and its dissents.

S10 The press, the vote and public opinion

The penny press, 1833: a cheap paper, advertising, and a new kind of reader.
 How a story becomes news: sources, deadlines, competition, and what gets left out.
 Measuring opinion: the 1936 Literary Digest poll, its two million replies, and its failure.
 Whether the press leads opinion or follows it: two readings of the same campaign.

Biology and Earth Science

S01 Natural selection and variation

Peppered moths on Manchester bark: two forms, soot, and a changing count.
 The mechanism in four conditions: variation, heredity, differential survival, time.
 The Galapagos finch record: thirty years of beak measurements through drought and flood.
 Whether selection or chance shapes most change: the neutral theory argument.

S02 Inheritance, from pea plants to genomes

Mendel's garden at Brno: tall and short peas, and a ratio that kept appearing.
 How the ratio arises: two copies, dominance, and the shuffling at each generation.
 From chromosome maps to sequencing: what each method can and cannot resolve.
 Whether genes explain traits or only correlate with them: heritability misread.

S03 The cell and its energy

A yeast cell in a warm sugar solution: what goes in, what comes out, what rises.
 Respiration and fermentation compared: oxygen, yield, and the cost of going without.
 How the energy path was worked out: labeled atoms, poisons, and isolated mitochondria.
 Whether the cell is a machine or an ecology: two ways to read the same organelles.

S04 Ecosystems: who eats whom, and how much

Yellowstone, 1995: wolves returned, and the willows along the creeks grew back.
 Trophic levels and the ten per cent rule: why there are few wolves and many voles.
 Exclosure experiments and isotope ratios: how a food web is actually traced.
 Whether predators or rivers rebuilt the willows: the trophic cascade dispute.

S05 Populations: growth, limit and collapse

Reindeer on St. Matthew Island: 29 animals, then 6,000, then 42.
 Exponential and logistic growth: the ceiling, the overshoot, and the lag.
 The Hudson's Bay fur returns: a ten-year cycle across a century, and what a ledger can show.
 Whether food or predators drive the hare cycle: fifty years of disagreement.

S06 Microbes and disease

Broad Street, 1854: a pump, a map, and a handle removed.
 How transmission works: dose, route, incubation, and the reproduction number.
 Koch's postulates, and the organisms that refuse to satisfy them.
 Whether the germ or the ground matters more: Snow, Pettenkofer, and modern echoes.

S07 Plants, soil and the nitrogen cycle

A clover field and a wheat field in rotation: why the wheat does better after clover.
 Fixation, nitrification, uptake and loss: the four steps and their rates.
 The Rothamsted plots: 180 years of the same treatments on the same soil.
 Whether fertilizer feeds the world or exhausts it: two readings of the yield curve.

S08 Carbon, climate and the record of it

Mauna Loa, 1958 to now: one instrument, one graph, a sawtooth that climbs.
 The mechanism: absorption bands, residence time, and the ocean's slow uptake.
 Ice cores, tree rings and corals: three proxies, three error bars, one picture.
 Whether the record settles attribution: how a signal is separated from noise.

S09 The restless Earth: plates, rock and deep time

Iceland's rift at Thingvellir: two walls of rock moving apart an inch a year.
 Plate motion and its consequences: ridges, trenches, faults and mountains.
 How the age of the Earth was fixed: lead ratios, half-lives, and Holmes's meteorites.
 Whether catastrophe or slow change built the landscape: the old quarrel, revived.

S10 Water, ocean and ice

The Thwaites glacier front: a cliff of ice, a warm current, and a retreating line.
 Why warm water does the work: latent heat, salinity, and the shape of the sea floor.
 Satellite gravimetry and tide gauges: two ways to weigh an ice sheet, and their limits.
 Whether the retreat is reversible: the marine ice-cliff argument.

Physical Science**S01 Measurement, error and uncertainty**

Weighing the same stone eleven times and getting eleven answers.
 Random error against bias: what averaging fixes and what it cannot.
 Calibration chains: how a laboratory balance is tied to a national standard.
 Whether a measurement can ever be separated from its theory.

S02 Forces and motion

A dropped hammer and a dropped feather on the Moon, 1971.
 The three laws at work in one braking truck: mass, friction and distance.
 How gravity's constant was measured: Cavendish's torsion balance and its error.

Whether Newton was wrong or incomplete: what Mercury's orbit decided.

S03 Energy: stored, moved and lost

A reservoir above a town: water, a pipe, a turbine, and lit houses.

Forms and conversions: how much is lost at each step, and where it goes.

Measuring efficiency: input, output, and the hard problem of the boundary.

Whether efficiency saves energy: the rebound argument.

S04 Heat, gases and the second law

A bicycle pump that grows hot in the hand.

Pressure, volume and temperature: the gas law and what it predicts.

Joule's paddle wheel: how the mechanical equivalent of heat was pinned down.

Whether the second law is a law of nature or of bookkeeping.

S05 Electricity and the circuit

A house fuse that blows, and the kettle that blew it.

Current, voltage, resistance and power: why the thin wire gets hot.

How a national grid is kept in balance, second by second, and how it is measured.

Whether the grid should be large or local: two readings of the same blackout.

S06 Waves, light and sound

A straw that looks bent in a glass of water.

Refraction, reflection and diffraction: one rule, three effects.

Measuring the speed of light: from Fizeau's wheel to a defined meter.

Whether light is wave or particle: what the two-slit result forced.

S07 Matter: atoms, bonds and reactions

Steel wool that burns, and a steel bar that does not.

Bonds, energy barriers and rates: why surface area and heat change everything.

The periodic table as prediction: gallium, germanium, and Mendeleev's gaps.

Whether chemistry reduces to physics: the argument over emergence.

S08 Materials: strength, failure and choice

The Tay Bridge, 1879: cast iron, a gale, and seventy-five deaths.

Stress, strain, yield and fracture: four numbers that decide a design.

Testing to destruction: how a safety factor is actually arrived at.

Whether failures are technical or organizational: two inquiries, one bridge.

S09 The nucleus, radiation and dating

A radium dial painter's jaw, 1925, and what the X-rays showed.

Half-life, dose and shielding: the three numbers that govern exposure.

Carbon-14 dating: the method, the calibration curve, and what it cannot date.

Whether low doses are safe: the linear no-threshold dispute.

S10 Distance, light and the scale of the sky

Why the Moon takes 1.3 seconds to reach us and the Sun eight minutes.

Parallax, brightness and the ladder of distance.

Cepheid variables: how Leavitt's law turned brightness into a ruler.

Whether the expansion rate is known: the tension between two good measurements.

Humanities

S01 Who is telling this: narrator and point of view

Two openings of one scene: the maid's account, then the mistress's.
 First person, third limited, omniscient: what each can and cannot report.
 How critics establish an unreliable narrator: the textual evidence they cite.
 Whether a narrator's judgment is the author's: the quarrel over irony.

S02 Character, motive, and the gap between them

A man who gives away his coat and cannot say why.
 How writers build motive: action, speech, gesture, and what is withheld.
 Reading a character from the manuscript: what the deleted paragraph shows.
 Whether characters have psychology or only functions: two schools reading one novel.

S03 Image, figure and the physical word

A poem that is four objects and no statement.
 Metaphor, simile and metonymy: three moves and the work each does.
 How an image is traced through a text: concordances, repetition, and pattern.
 Whether imagery carries meaning or only mood: the argument over symbolism.

S04 Form: the line, the stanza, the turn

A sonnet counted out: fourteen lines, a rhyme scheme, a hinge at line nine.
 How form creates expectation, and how breaking it produces meaning.
 Scansion as evidence: what a metrical analysis can actually prove.
 Whether free verse has form: Whitman's defenders and his critics.

S05 The stage as a machine

A trapdoor, a balcony and a bare platform: the Globe and what it could do.
 How a playhouse shapes a play: entrances, asides, daylight, and the crowd.
 Reading the promptbook: cuts, business, and what a production actually did.
 Whether a play belongs to its author or its stage: the directors' quarrel.

S06 Genre and what a reader expects

A locked room, a body and a detective: the shape of a kind of story.
 How conventions work, and what an author gains by refusing them.
 Tracing a genre's history: first appearances, borrowings, and bibliographies.
 Whether genre limits a writer or enables one: two critical positions.

S07 Painting and the trained eye

One room of a Dutch house, painted in 1658, and everything in it.
 How a picture directs the eye: line, light, scale and the diagonal.
 What conservation reveals: X-rays, pigment analysis, and repainted hands.
 Whether a painting means what its age meant by it: the iconography argument.

S08 Music: pattern in time

A tune of eight bars, repeated, altered, and repeated again.
 How tension is built and released: key, cadence, rhythm and expectation.
 What a manuscript shows: revisions, performance marks, and the editor's choices.

Whether a score is the work or a set of instructions: the performance debate.

S09 Architecture and the shaped city

A nineteenth-century arcade: glass, iron, shops and shelter from rain.

How a street is made to work: frontage, width, sightlines and use.

Measuring a city: counts of footfall, surveys, and what each misses.

Whether the block or the tower serves a city better: Jacobs against Le Corbusier.

S10 Criticism: how arguments about art are made

Two reviews of one first night, printed the same morning.

What a critical argument needs: a claim, evidence from the work, and a standard.

How a judgment is tested: counter-examples, context and the record of reception.

Whether judgments of art can be true: taste, standards and the long argument.

Social Science

S01 Asking people things: surveys and their traps

Two wordings of one question, and the twenty-point gap between the answers.

Sampling, response rate and question order: three ways a survey goes wrong.

How a modern poll is actually built: frames, weights and margins of error.

Whether polls can be fixed or should be replaced: after 2016 and 2020.

S02 Experiment, control and the drawing of lots

A school that gave half its pupils breakfast and measured the other half.

Why lots matter: what randomization does that matching cannot.

Field experiments: the Oregon lottery, the Perry preschool, and what each shows.

Whether randomized trials are the gold standard: the external validity objection.

S03 Correlation, cause and confounding

Towns with more bookshops have fewer burglaries, and why that proves nothing.

Confounding, selection and reverse causation: three reasons a pattern misleads.

Natural experiments: twins, lotteries, border effects, and what they buy.

Whether observational data can ever establish cause: smoking and the long argument.

S04 Samples, averages and spread

A street where the mean income is high and nobody is rich but one.

Mean, median, spread and skew: four numbers and what each hides.

Confidence intervals: what a margin of error actually says.

Whether the average is a useful object at all: the distributional objection.

S05 Incentives and how people answer them

A city that made the first hour of parking free and emptied its curbs.

Price, salience and substitution: why a small fee can beat a large campaign.

Measuring a response: before-and-after, control cities, and the elasticity estimate.

Whether incentives crowd out motives: the blood-donation argument.

S06 Norms, groups and conformity

An office where nobody sits at the window table, and no rule says so.

How a norm is enforced: watching, gossip, exclusion, and the cost of breaking it.

The conformity experiments: Asch's lines, their replications, and their limits.
Whether norms are chosen or inherited: two accounts of the same village.

S07 Cities, movement and where people settle

A tram line opened in 1890, and the houses that followed it out of town.
Rent, commuting time and the trade-off that decides where people live.
Reading a census: boundaries, undercounts, and the problem of comparison.
Whether building more houses lowers rents: two readings of the same city.

S08 Work, wages and the labor market

A dockside where men waited each morning to be picked for a day's work.
Supply, demand, bargaining power, and what sets a wage.
The minimum wage studies: New Jersey, Pennsylvania, and the method that started a fight.
Whether a minimum wage costs jobs: thirty years of contested estimates.

S09 Inequality and how it is measured

Two neighborhoods three miles apart with an eleven-year gap in life expectancy.
Income, wealth, mobility: three different inequalities and their different causes.
The Gini coefficient and its rivals: what each measure is blind to.
Whether inequality or poverty is the problem: the argument over the top one per cent.

S10 Judgment under uncertainty

A doctor, a test that is ninety-nine per cent accurate, and a rare disease.
Base rates, anchoring and availability: three predictable errors.
How the heuristics were demonstrated: the original problems and their replications.
Whether people are irrational or merely economical: the two camps.

LEVEL 1

Foundation

phenomenon

One concrete case, named and dated: what happens and what it looks like.

Passage length — 288 to 312 words

Sentences — averaging 13 to 18 words

Vocabulary assumed — the top 16,000 most frequent word forms

Words above that — at most 6, each explained in place

Clue to the reader — Read for the case itself. Every passage names one thing that happened, or one thing that is so, and holds to it. If the reader can say afterward what the passage was about in a single sentence, the passage has done its work.

History and Civics

Passages 1 to 10

1

HIS · S01 · phenomenon

Four Days in Philadelphia

The Second Continental Congress met in Philadelphia through the summer of 1776. On July 2 the delegates voted to separate from Britain. Two days later they approved the text that explained why. The Declaration of Independence was printed that night in a shop a few streets away. Copies were read out in public squares within the week, and in some towns the reading ended with a bonfire. Fifty-six men signed the parchment copy, most of them on August 2, by which time the war had been running for more than a year.

Thomas Jefferson wrote the first draft in about two weeks. A committee of five was named to do the work, but the others left most of the writing to him, and Congress then cut about a quarter of what he produced. The document has three parts. The preamble, which is the opening section that states a purpose, makes a general claim about where any government gets its power. The middle section is a list of grievances, meaning formal complaints, against George III. He had shut down elected assemblies, kept soldiers in the colonies in time of peace, and closed off their trade with other countries.

The grievances take up most of the page. That surprises readers who know only the opening lines. Congress was building a legal case, and a legal case needs particulars. Each complaint named something the king had actually done, on a date, in a place. Together they argued that he had broken the agreement by which he ruled, and that the colonies therefore owed him nothing further. Congress had no army fund and no treasury of its own that July; it established both within the year. News of the vote took six weeks to cross the Atlantic. The parchment copy is now kept in Washington under filtered light.

WHAT THIS PASSAGE IS FOR

- The vote for independence and the text explaining it were two separate acts, two days apart.
 - Most of the document is a list of particular complaints, not a statement of principle.
 - The argument is legal: the king broke an agreement, so obedience ended.
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Questions 1.1 to 1.10

on passage 1, Four Days in Philadelphia

1.1

CENTRAL IDEA · EASY · INFORMATION AND IDEAS

Which choice best states the main idea of the text?

- A) Copies of the Declaration were printed and read aloud across the colonies within a week of its approval
- B) Most of the Declaration is a list of particular complaints, because Congress was building a legal case against the king
- C) Thomas Jefferson wrote the Declaration alone, and Congress approved his draft without altering it
- D) The Declaration proves that every government everywhere draws its power from the people it rules

1.2

DETAIL · EASY · INFORMATION AND IDEAS

According to the text, what did Congress do with the draft that Jefferson produced?

- A) It approved the draft without making changes to it
- B) It returned the draft to a committee of five to be rewritten
- C) It added the list of grievances to the draft itself
- D) It cut about a quarter of what he had written

1.3

EVIDENCE · MEDIUM · INFORMATION AND IDEAS

Which quotation from the text most strongly supports the claim that the Declaration was written as a legal argument rather than as a statement of principle?

- A) "Congress was building a legal case, and a legal case needs particulars"
- B) "Thomas Jefferson wrote the first draft in about two weeks"
- C) "Copies were read out in public squares within the week"
- D) "He had shut down elected assemblies, kept soldiers in the colonies in time of peace"

1.4

INFERENCE · MEDIUM · INFORMATION AND IDEAS

The Declaration was approved on July 4, 1776, but Congress had no army fund and no treasury of its own that July. It established both within the year, which suggests that the vote to separate ____

Which choice most logically completes the text?

- A) was delayed until the institutions it required already existed
- B) had little effect on how the Congress afterward spent its time
- C) created obligations the Congress was not yet equipped to meet
- D) was carried out by Thomas Jefferson without the other delegates

1.5

WORDS IN CONTEXT · EASY · CRAFT AND STRUCTURE

As used in the text, what does the word "established" most nearly mean?

- A) demonstrated as true
- B) brought into being
- C) made peace with
- D) took control of

1.6

STRUCTURE · MEDIUM · CRAFT AND STRUCTURE

Which choice best describes the function of the third paragraph in the text as a whole?

- A) It explains why the bulk of the document is taken up by complaints
- B) It lists the grievances that the second paragraph had only named
- C) It describes how copies of the document were distributed and read
- D) It corrects the account of the drafting given in the paragraph before

1.7

CROSS TEXT · HARD · CRAFT AND STRUCTURE

Text 2 is passage 51 of this book. It argues that the second paragraph of the Declaration moves from premises to a conclusion in order, and that the grievances are offered as evidence for a factual claim rather than as argument about principle.

Text 1 reports that the grievances take up most of the document. Based on Text 2, how would its author most likely explain that proportion?

- A) The grievances were the only part of the document Congress could agree on
- B) The opening principles were added after the complaints had been drafted
- C) The document was shortened by a quarter, which left mostly complaints
- D) The premises needed no proof, so the space went to the factual claim

1.8

TRANSITIONS · EASY · EXPRESSION OF IDEAS

Congress had no army fund and no treasury of its own that July. ___ it established both within the year.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) Likewise,
- C) Nevertheless,
- D) In other words,

1.9

BOUNDARIES · EASY · STANDARD ENGLISH CONVENTIONS

The Declaration of Independence was printed in a shop a few streets from the room where Congress had voted ___ copies were read out in public squares within the week.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) voted; copies
- B) voted, copies
- C) voted copies
- D) voted: copies

1.10

SYNTHESIS · MEDIUM · EXPRESSION OF IDEAS

While researching a topic, a student has taken the following notes.

- The grievances take up most of the page.
- Congress was building a legal case, and a legal case needs particulars.
- Each complaint named something the king had actually done, on a date, in a place.
- Readers who know only the opening lines are surprised by the proportion.

The student wants to explain to an audience unfamiliar with the subject why the document contains so many complaints. Which choice most effectively uses relevant information from the notes to accomplish that goal?

- A) The Declaration has three parts, and the middle one is a list of grievances against the king
- B) Readers who know only the opening lines are surprised, although the grievances take up most of the page

- C) The Declaration gives most of its space to grievances because Congress was making a legal case, which needs named particulars
- D) Congress named each complaint on a date and in a place, and the war had already run for more than a year

ملخص المقطع 1 بالعربية

خلاصة: اجتمع المؤتمر القاري الثاني في فيلادلفيا صيف عام ألف وسبعمئة وستة وسبعين، وصوّت المندوبون في الثاني من تموز على الانفصال عن بريطانيا، ثم أقرّوا بعد يومين النص الذي يشرح الأسباب. كتب توماس جيفرسون المسودة الأولى، وحذف المؤتمر نحو ربعها. وتتألف الوثيقة من ثلاثة أقسام: مقدمة تقرّر مصدر سلطة أي حكومة، وقسم أوسط يسرد المظالم على الملك جورج الثالث، ثم الخاتمة.

المعنى: المعنى الأساسي أن الوثيقة بُنيت مرافعة قانونية لا بيان مبادئ. فالمظالم تشغل معظم الصفحة لأن القضية القانونية تحتاج تفاصيل محدّدة: فعل بعينه، في تاريخ، وفي مكان. وهذا يفسّر ما يفاجئ من يقرأ السطور الأولى وحدها.

الأهمية في المجال: في مجال التاريخ والنظام المدني تمثّل هذه المادة مستوى الظاهرة: واقعة مؤرّخة ومسمّاة يُبنى عليها لاحقاً فهم الآلية والدليل والخلاف. ومن دون معرفة ما تقوله الوثيقة فعلاً يصعب تقييم أي حجة تُبنى عليها.

الصلة بالاختبار: في اختبار سات تتكرّر النصوص التأسيسية كثيراً، ويُسأل عنها عادةً في الفكرة المركزية والدليل النصي وبنية المقطع. والفجّ الشائع اختبار تفصيل صحيح لا يجيب عن السؤال، كذكر طباعة النسخ أو قراءتها في الساحات بدل سبب طول قائمة المظالم. ويقترن هذا المقطع بالمقطع الحادي والخمسين ليكونا نصين متقابلين.

Three Branches and One Lock

The Constitution written at Philadelphia in 1787 divides the work of government among three branches, and gives each of them something the other two cannot do. Congress makes law. The President carries it out. The courts decide what the law means in particular cases. Each branch also holds at least one way of stopping the others, and the men who drew up the plan put those blocks in deliberately. The first printing of the finished text ran to four pages.

Congress holds the money. No federal dollar may be spent unless Congress has passed an appropriation, which is a vote that releases funds, so a President with a plan and no appropriation has nothing. The President may veto a bill, meaning return it to Congress without signing, and Congress can then pass it anyway only with two thirds of each chamber. The Senate must agree to treaties and to senior appointments, judges included. The courts may hold that a law conflicts with the Constitution and refuse to apply it, a power called judicial review. Federal judges are appointed for life and cannot be removed for rulings that prove unpopular.

None of this was an accident of drafting. The men at the convention had watched a king rule without consent, and had then watched state assemblies do much as they liked. They wanted authority to be real but answerable, so they gave each branch a lock on part of the work of the others. The result is slow by design. A measure that two branches want and the third does not will usually fail, while a measure all three want can move in a week. Readers meeting the system for the first time often expect a chain of command. What the document sets up instead is a standing argument among equals.

WHAT THIS PASSAGE IS FOR

- The three branches are defined by function: making, executing and interpreting law.
 - Each branch holds at least one power that can stop another.
 - The slowness is deliberate, not a defect of drafting.
-

A Convention at Seneca Falls

In July 1848 about three hundred people came to a chapel in Seneca Falls, a small town in western New York. The meeting had been announced a few days earlier in a county newspaper, and no one had time to travel far. Elizabeth Cady Stanton and Lucretia Mott had called it to discuss the condition of women. Forty of those who came were men, and one of them was Frederick Douglass, who ran an abolitionist newspaper, that is, a paper campaigning for the ending of slavery, in Rochester.

The convention produced a document called the Declaration of Sentiments. Its form was borrowed on purpose. It opened with the words of the Declaration of Independence, changed to read that all men and women are created equal, and then set out a list of complaints in the same manner as 1776. Married women could not own property in their own names. They could not keep the wages they earned. They had almost no standing in court, little access to higher education, and no vote. Sixty-eight women and thirty-two men put their names to it, and some later asked for their names to be removed.

Eleven resolutions, that is, formal proposals put to a vote, came before the meeting. Ten passed easily. The one that asked for the vote was argued over and nearly lost. Several of those present thought that a demand to expand the franchise, meaning the right to vote, would make the rest of the list look ridiculous. Douglass spoke for it and it carried by a narrow margin. Newspapers treated the whole affair as a joke for weeks. The vote arrived seventy-two years later, in 1920, by which time only one of those who signed was still alive to use it.

WHAT THIS PASSAGE IS FOR

- Seneca Falls borrowed the form of the Declaration of Independence deliberately.
 - The demand for the vote was the one resolution that nearly failed.
 - The gap between the demand and the Nineteenth Amendment was seventy-two years.
-

A Fourth of July Speech

On July 5, 1852, Frederick Douglass spoke in Corinthian Hall in Rochester, New York. The invitation came from the Ladies Anti-Slavery Society, an abolitionist group, meaning one working for the ending of slavery. About six hundred people paid to hear him. Douglass had been born into slavery in Maryland and had escaped fourteen years earlier. He taught himself to read as a boy, bought his own freedom with money raised in Britain, and by 1852 owned and edited a newspaper.

He began mildly. For twenty minutes he praised the men of 1776, called them brave, and said their work deserved the respect of their children. Then he turned. The holiday, he told the hall, was theirs and not his. He asked what the Fourth of July meant to an American slave, and answered that it revealed to such a man, more than any other day in the year, the wrong done to him. The speech ran for nearly two hours, and the audience is reported to have stood at the end.

Douglass did not reject the founding documents. That is the part readers miss. He argued the opposite case, that the Declaration said exactly what he wanted said, and that the country was failing its own words rather than keeping them. Near the end he turned to the churches, the press and the political parties in turn, and named the share of each one in the failure. The address was printed as a pamphlet, which is a short work sold cheaply, within a few weeks, and it sold widely in the North. It is now usually published under a title that Douglass himself never gave it. The hall where he spoke was pulled down in 1898. His newspaper, the North Star, had about four thousand readers at its height, many of them in Britain. Douglass lived another forty-three years and advised two presidents.

WHAT THIS PASSAGE IS FOR

- Douglass praised 1776 for twenty minutes before turning on his audience.
 - His argument was that the country failed its own founding words, not that the words were wrong.
 - The speech named churches, press and parties as sharing the failure.
-

What Was Built After 1865

The war ended in April 1865. Four million people who had been held as property were free, with no land, no schooling and no standing in law. The twelve years that followed are called Reconstruction. A great deal was built in them, in a hurry, by a Congress that had just won a war and did not expect to keep its majority for long.

Three amendments were added to the Constitution. The Thirteenth ended slavery. The Fourteenth made anyone born in the country a citizen, and required every state to give equal protection of the laws. The Fifteenth banned the denial of the vote on grounds of race. A federal agency, the Freedmen's Bureau, set up schools, wrote labor contracts and ran courts where state courts would not sit. Black men voted in large numbers across the South. More than fifteen hundred held public office, including sixteen seats in Congress and two in the Senate. South Carolina had a legislature with a Black majority. The first public school systems in several southern states date from these years.

None of it lasted in the form it was built. Northern interest declined through the 1870s as other matters took the front page and as the cost of keeping soldiers in the South rose. The last federal troops were pulled out in 1877. Within thirty years nearly every state in the region had written new voting rules that removed most Black voters from the rolls without mentioning race at all. Among them were poll taxes, which are charges that had to be paid before a person could vote, and reading tests set by local officials. The three amendments stayed in the Constitution, never repealed and largely not enforced, for another sixty years. Mississippi wrote the first of the new constitutions in 1890. Black turnout in Louisiana fell from about ninety-five percent to under two percent in eight years.

WHAT THIS PASSAGE IS FOR

- Three amendments ended slavery, created birthright citizenship and barred racial denial of the vote.
 - Reconstruction put more than fifteen hundred Black officeholders into office.
 - The amendments survived while their enforcement did not.
-

A Locked Door on Washington Place

The Triangle Shirtwaist factory filled the top three floors of a ten-floor building off Washington Square in New York. On Saturday, March 25, 1911, a few minutes before the end of the shift, a fire started in a bin of cloth scraps on the eighth floor. Within eighteen minutes it was over. One hundred and forty-six workers were dead, most of them women, many of them under twenty, and some of them fourteen.

The building was not especially old and the stairs were not especially narrow. What killed people was a set of ordinary decisions. One of the two stair doors was kept locked, because the owners wanted to check bags at a single exit to prevent theft. The fire escape was a light iron structure that gave way and threw about twenty people into the yard below. The fire department ladders reached the sixth floor. There were no sprinklers, which are ceiling pipes that release water in a fire, and the company had turned down an inspection offered two years earlier.

Two hundred thousand people walked in the funeral march. The state appointed a commission, meaning a body set up to investigate and report, and it spent two years visiting factories across New York. The reform that followed was unusually quick. Between 1911 and 1914 the legislature passed more than thirty laws. They required fire drills, unlocked exits, water pipes in tall buildings and limits on hours for women and children. They also set up an inspection service with the power to enter without warning. Several of the people who ran that commission spent the next thirty years in national office. The New York rules became the pattern for federal law in the 1930s. The factory building itself still stands and belongs to a university.

WHAT THIS PASSAGE IS FOR

- A locked exit door, kept locked to check bags for theft, was central to the death toll.
 - The fire killed 146 workers in about eighteen minutes.
 - More than thirty state laws followed within three years.
-

A Year of Walking in Montgomery

On December 1, 1955, Rosa Parks was arrested in Montgomery, Alabama, for refusing to give up her seat on a city bus. She sewed clothes for a living, and she was also the secretary of the local branch of the National Association for the Advancement of Colored People. She had spent part of the previous summer at a training school for organizers in Tennessee. Three other women had been arrested on the same ground that year.

The response was ready within three days. Printed sheets went out calling for a one-day boycott, meaning an organized refusal to ride, on the Monday. About forty thousand Black residents stayed off the buses, which was close to every one of them, and they made up about three quarters of all riders. That evening a meeting at the Holt Street church voted to carry on. A minister of twenty-six named Martin Luther King Jr. was chosen to speak for the campaign, partly because he was new in town and had made no enemies in it.

The boycott lasted three hundred and eighty-one days. Carpools, which were shared cars run from church basements, moved people to work and back. Drivers were arrested for small traffic offenses, insurance companies canceled their policies, and the house where King lived was bombed. What finally ended segregation on the buses was not the boycott itself but a lawsuit, *Browder v. Gayle*, which the Supreme Court upheld in November 1956. The campaign had done something no court could do. It had shown that a whole city could unite behind one demand and hold together for a year. The bus company lost about two thirds of its income in that year. Parks lost her job at the department store within a month of the arrest, and left Alabama in 1957.

WHAT THIS PASSAGE IS FOR

- Parks was an experienced NAACP officer, not an accidental protester.
 - The boycott ran 381 days, with carpools run from church basements moving people to work.
 - Segregation on the buses ended by court ruling, not by the boycott alone.
-

A Handful of Mud at Dandi

Under British rule in India it was against the law for an Indian to make salt. The government held a monopoly, which is the sole right to produce and sell, and it taxed what it sold. Salt is not a luxury in a hot country where most people work outdoors. The tax therefore reached every household, including the poorest, and it had been collected in one form or another for most of a century.

On March 12, 1930, Mohandas Gandhi left his settlement near Ahmedabad with seventy-eight followers and walked toward the coast. The walk took twenty-four days and covered about two hundred and forty miles. Reporters went with him. The crowds grew at every village along the road, and by the end they were enormous. On the morning of April 6 he waded into the sea at Dandi, picked up a handful of salt-crusted mud, and held it up. The act was useless as trade and perfect as argument. He had broken a law that nobody could defend in public without looking foolish.

All along the coast people began boiling sea water in pans. Something like sixty thousand were arrested in the months that followed, Gandhi among them. The tax was not lifted, and the march won nothing at the time. What it did was force a choice. A government that arrests a man of sixty for picking up mud on a beach has rejected its own claim to rule by consent. Photographs of those arrests appeared in newspapers from London to New York within the week. Independence came seventeen years later, in 1947, after a world war had emptied the British treasury. The salt tax itself was abolished in 1946. Gandhi walked about twelve miles a day and was sixty years old at the time.

WHAT THIS PASSAGE IS FOR

- The salt monopoly taxed something nobody in a hot country can do without.
 - The march was designed as an argument, not as a way of getting salt.
 - Roughly sixty thousand arrests followed, and no immediate concession.
-

An Order, a Map and a Suitcase

On February 19, 1942, ten weeks after the attack on Pearl Harbor, President Franklin Roosevelt signed Executive Order 9066, which is a written instruction from a president that carries the force of law. The order named no group at all. It gave the army power to declare military areas and to restrict who could live in them. Within weeks the whole Pacific coast had been declared such an area, and the people removed from it were almost entirely of Japanese descent.

About one hundred and twenty thousand men, women and children were given between six days and two weeks to settle their affairs. Two thirds of them were American citizens by birth. They could take only what they could carry. Farms, shops, boats and houses were sold for whatever price could be got in two weeks, or left with neighbors, or simply abandoned. A piano went for five dollars in Seattle. The families were held first at race tracks and fair halls, then moved to ten camps built inland, in deserts and on wet ground, with thin wooden barracks, shared washrooms and wire fences.

Nobody of Japanese descent in the United States was ever charged with spying or sabotage during the war. The records of the time include two reports to the government saying as much, and both were set aside. Meanwhile a regiment of Japanese American soldiers, the 442nd, fought in Italy and France and became the most decorated unit of its size in the army. Some of its men had volunteered from inside the camps where their parents were still held. The last camp closed in 1946. Congress apologized in 1988 and paid twenty thousand dollars to each surviving person who had been held. About eighty-two thousand people received payments. One of the camps, Manzanar, is now a national historic site.

WHAT THIS PASSAGE IS FOR

- The order named no group; the army's area declarations did the work.
 - Two thirds of those removed were American citizens by birth.
 - No person of Japanese descent in the United States was charged with espionage during the war.
-

10

HIS · S10 · phenomenon

A Paper for One Cent

Before 1833 a New York newspaper cost six cents and arrived by subscription, meaning payment in advance for a year of issues. That was most of a day's pay for a laborer. The papers were four pages of shipping lists, prices, political notices and long letters, and they were written for merchants and party men. Perhaps one household in fifty took one, and a single copy was often read by a dozen people in a coffee house.

In September 1833 a printer named Benjamin Day began selling the New York Sun on the street for one cent. He hired boys to shout the headlines. They were paid in bundles of papers, which they then sold at a profit. He filled the sheet with police court reports, fires, accidents, crimes and conflict of every ordinary kind. Within two years the Sun was selling about fifteen thousand copies a day, more than any paper in the city had ever sold, and three rivals had copied the formula.

The penny papers made their money from advertising rather than from readers, and that changed what they printed. A paper paid for by the people who take it answers to them. A paper paid for by advertising needs the largest crowd it can gather. Crowds are gathered by stories that anyone can follow. The new papers also hired reporters, which the old ones had barely done. They sent them out to courts, docks and police stations to look for material. Within one generation the idea that a newspaper should carry news, gathered by its own employees, had replaced the older idea that it should carry opinion, supplied by friends. The Sun also printed a series of invented reports about life on the moon in 1835, and sales rose.

WHAT THIS PASSAGE IS FOR

- A six-cent subscription paper was priced beyond most wage earners.
 - Advertising, not subscription, became the source of revenue, which changed the content.
 - Paid reporters sent out to gather news were a new institution.
-

Biology and Earth Science

Passages 11 to 20

11

BIO · S01 · phenomenon

Moths on a Sooty Wall

The peppered moth rests on tree bark by day with its wings open. Most of these moths are pale gray with small dark spots. A collector near Manchester recorded a nearly black one in 1848, and at the time it was a curiosity worth writing down. Fifty years later the black form made up more than nine in ten of the moths caught in that city. Nothing about the moths had changed in any obvious way. What had changed was the bark.

Coal smoke had killed the pale lichens, which are the crusty growths that cover healthy bark, and had left the trunks black with soot, which is the black powder left by burning coal. On a clean trunk a pale moth is almost invisible and a black one stands out. On a sooty trunk the reverse is true. Birds hunt these moths by sight, and they take whichever form they can see. The counts in rural Dorset, where the air stayed clean, stayed overwhelmingly pale through the same years.

Then the direction reversed. Britain passed a clean air law in 1956, coal smoke fell away over the next thirty years, the lichens came back, and the black moths became rare again. By the 1990s they were below five percent around Manchester. The same reversal happened near Detroit, which gives two records rather than one. A population that has adapted to one kind of background will swing back when the background changes, and the moths did it twice within living memory. The whole story can be read off a series of ordinary counts made with a light trap and a notebook. Collectors had been keeping such records since the 1830s, for reasons of their own, and the numbers were there when biologists came looking for them. Soot turned out to be the quiet agent in all of it.

WHAT THIS PASSAGE IS FOR

- A single moth population shifted from pale to black and back within 150 years.
 - The moths did not change; the bark they rest on did.
 - Clean air reversed the change, which gives two records instead of one.
-

Peas in a Monastery Garden

Gregor Mendel was a monk and a science teacher in Brno, a city now in the Czech Republic. Between 1856 and 1863 he grew peas in the monastery garden and counted them. He chose seven features that came in two clear forms and nothing in between. Seeds were either round or wrinkled, and pods were either green or yellow. Plants were tall or short, with no middling heights. Over seven years he grew and counted something near twenty-eight thousand plants, by hand, in a plot about the size of a tennis court.

His method was simple and exact. He began with lines that bred true, meaning that tall plants from those lines always gave tall offspring. Then he crossed a tall line with a short one. Every plant in the first generation was tall, and yet the short form had not been diluted or lost. When he let that generation breed among itself, the short plants came back, and they came back in a steady proportion. For every three tall plants there was about one short plant. He got a ratio close to three to one for all seven features he had chosen.

Mendel published the work in 1866 in the journal of a local natural history society. Almost nobody read it. He sent copies to well-known scientists and received little in reply, became an abbot, and did no more breeding work. The paper sat unused for thirty-four years. In 1900 three separate researchers, working on their own plants, found the same ratios and then found his paper. What any inherited feature looks like in the next generation depends on rules of this kind. Mendel had those rules written down while other naturalists were still arguing about whether offspring simply blend their two parents together. He also chose his plant well. Peas pollinate themselves unless a breeder interferes, so every cross was controlled.

WHAT THIS PASSAGE IS FOR

- Mendel chose features with two clear forms and no middle ground.
 - The short form vanished in one generation and returned in a three to one ratio.
 - The work was published in 1866 and ignored for thirty-four years.
-

What a Yeast Cell Does

A teaspoon of dried yeast holds several billion cells, each one a single rounded cell about five thousandths of a millimeter across. Stirred into warm water with a little sugar, the powder begins to foam within ten minutes. The foam is gas. The cells are taking in sugar and giving back two things, carbon dioxide and alcohol, and a baker and a brewer want different halves of the same process.

In bread the gas does the work. Each bubble of carbon dioxide pushes the stretchy dough outward, and a loaf left in a warm kitchen roughly doubles in ninety minutes. The alcohol boils off in the oven. In beer and wine the alcohol is the point and the gas is allowed to escape, except in bottles that are sealed early on purpose. The same organism, *Saccharomyces cerevisiae*, has been used for both for at least five thousand years, long before anyone knew that a living thing was involved.

Yeast can work with air or without it. With oxygen available it burns sugar all the way down and gets far more energy from each molecule. Without oxygen it takes a short cut, called fermentation, and must release alcohol as a waste product because the chemistry stops halfway. The short cut yields less than a tenth as much energy, which is why yeast grows slowly in a sealed vessel and quickly in an open one. Louis Pasteur showed this difference in the 1860s by measuring how much sugar disappeared and how much yeast was produced in each case. The cell has no choice in the matter. The chemistry available to it is decided by what is dissolved in the liquid around it. A brewer controls that liquid deliberately. Air is kept out of the vessel, the temperature is held near twenty degrees, and the sugar is allowed to run down over several days.

WHAT THIS PASSAGE IS FOR

- Yeast takes in sugar and gives out carbon dioxide and alcohol.
 - Bakers use the gas and brewers use the alcohol from the same process.
 - Without oxygen the cell gets less than a tenth as much energy from the same sugar.
-

Wolves and Willows in Yellowstone

Wolves were shot out of Yellowstone National Park by the 1920s. The last known pair was killed in 1926. For the next seventy years the park had no wolves, and the elk, which are the large deer that wolves hunt there, lived in large numbers on the valley floors. Along the creeks the willows and young aspens were eaten down to stumps every year, and the beaver, which need willow, almost disappeared from the northern part of the park.

In 1995 and 1996 the park service brought thirty-one wolves from Canada and released them in the Lamar Valley. The animals were held in pens for ten weeks first, then let out. They bred quickly. Within ten years there were well over a hundred wolves in the park. Elk numbers on the northern range had fallen by more than half. Part of that was hunting by wolves, and part was that the elk now moved more and spent less time standing in open valleys.

Then the plants changed. Willows along some creeks that had been browsed flat for decades grew to two or three meters. Cottonwood seedlings survived their first years for the first time in living memory. Beaver colonies on the northern range rose from one in the early 1990s to nine by 2015. Each new dam held water back and made still more willow possible. Songbirds that nest in streamside bushes recovered along with them. The park service has photographs of the same stretches of creek taken in 1996 and in 2015, and the difference is not subtle. What is in dispute is not the change but its cause, and that argument is taken up later in this strand. The park keeps counts of wolves, elk and beaver every year, and has done so since the release. The willow measurements are taken at fixed points along the same creeks each summer.

WHAT THIS PASSAGE IS FOR

- Yellowstone had no wolves for about seventy years.
 - Thirty-one wolves were released in 1995 and 1996 and elk numbers later fell by more than half.
 - Willows, cottonwoods and beaver colonies all increased afterward.
-

Reindeer on St Matthew Island

St Matthew Island lies in the Bering Sea, about two hundred miles from the nearest other land. It is thirty-two miles long, treeless, and covered in moss, low grass and lichen, which is the slow crusty growth that reindeer eat. Nobody lives there. In 1944 the United States Coast Guard put twenty-nine reindeer ashore as an emergency meat supply for nineteen men stationed on the island. The war ended, the station closed, and the men left. The reindeer stayed.

They had no predators of any kind. No wolves, no bears, no hunters. The lichen on the island had been growing undisturbed for thousands of years and lay thick on the ground. A biologist counted the herd in 1957 and found about thirteen hundred animals in good condition. He counted again in 1963 and found six thousand. That is roughly two hundred times the number put ashore nineteen years earlier, on an island of fixed size.

The next count was in 1966. There were forty-two reindeer left, forty-one of them female and one male, and no young of that year. The winter of 1963 had been severe, with deep snow, but the important fact was underneath the snow. The lichen was gone, grazed down to bare rock and soil across most of the island, and lichen grows back at a rate measured in decades rather than years. The herd had eaten its own future. By the 1980s there were no reindeer on St Matthew Island at all. The whole episode took less than forty years from start to finish, and every number in it comes from counts made on the ground. A population can survive on borrowed capital for a while, and the bill falls due in one bad winter. Nothing about the island was unusual except the absence of anything that eats a reindeer. Mainland herds, hunted by wolves and by people, had held steady for centuries.

WHAT THIS PASSAGE IS FOR

- Twenty-nine reindeer were put on a remote island in 1944 with no predators.
 - The herd reached about six thousand by 1963 and forty-two by 1966.
 - The cause was the loss of the lichen, which regrows over decades.
-

The Pump on Broad Street

In late August 1854 people in Soho, a crowded district of London, began to die of cholera. The disease killed quickly, often within a day. Within a week more than five hundred were dead in a few streets. Most doctors believed that cholera traveled in bad air rising from drains and rotting matter. John Snow, a local physician, did not believe it, and he went out to look.

Snow collected the address of every death he could trace and marked each one on a street map as a small black bar. The bars piled up around one place: a public water pump at the corner of Broad Street. Houses a short walk away, but closer to a different pump, had almost none. He also found exceptions and chased them down. Workers at a brewery on the same street drank beer and water from the brewery's own well, and none of them fell ill. A widow in Hampstead, three miles off, died after a cart brought her a bottle of Broad Street water because she liked the taste.

On September 8 Snow persuaded the parish board to take the handle off the pump. Deaths were already falling, because many residents had fled, so the removal did not end the outbreak and Snow said so himself. What the map did was show that the disease had spread from a single source that people drank from, not from the air they breathed. Years later an inspection found a leaking cesspit, which is a pit for household waste, less than a meter from the well. Cloths from a sick baby had been washed into it. The water board did not accept the argument for another decade. Snow died in 1858 without seeing it accepted. The map itself is still reprinted in textbooks, and the pump site is marked with a replica on the street now called Broadwick.

WHAT THIS PASSAGE IS FOR

- Snow mapped each death as a bar on a street plan.
 - The brewery workers and the Hampstead widow were the exceptions that tested the case.
 - Removing the handle did not end the outbreak, and Snow said so.
-

Why Wheat Follows Clover

Two fields on the same farm can be given the same seed, the same rain and the same plowing and still produce very different crops of wheat. One useful predictor is what grew in each field the year before. Wheat sown after clover, which is a low plant of the pea family grown for hay, will usually beat wheat sown after wheat. The difference can be a third or more. English farmers knew this long before anyone could say why.

In the eighteenth century a rotation became standard across light soils in Norfolk. The farmer divided a field into four parts and moved the crops around them on a four-year cycle: turnips, then barley, then clover, then wheat. The turnips were a root crop lifted for winter feed. The clover was cut for hay or eaten by sheep, whose droppings stayed on the land, and the roots were then turned into the soil. Yields on such farms roughly doubled over the century. The system spread across northern Europe as the Norfolk four-course rotation.

The explanation is in the roots. Clover and other members of the pea family carry small swellings called nodules on their roots. Inside these nodules live bacteria that take nitrogen from the air and turn it into a form that plants can use. Wheat cannot do this and must consume what it finds in the soil. Clover leaves the soil richer than it found it; wheat leaves it poorer. A farmer who alternates the two runs a slow cycle of deposit and withdrawal. One who grows wheat on wheat is making withdrawals only, and the yields show it by the third or fourth year. The practice spread because it paid, not because the chemistry was understood. Nitrogen itself was not identified as the active element until the 1830s, two generations after the rotation had become normal.

WHAT THIS PASSAGE IS FOR

- Wheat after clover outyields wheat after wheat, often by a third.
 - The Norfolk four-course rotation moved turnips, barley, clover and wheat around one field.
 - Bacteria in clover root nodules take nitrogen from the air.
-

One Instrument on a Volcano

Near the top of Mauna Loa, a volcano on the island of Hawaii, an instrument has been drawing air through a tube and measuring the carbon dioxide in it every hour since 1958. The site was chosen because it is far from cities, forests and factories, eleven thousand feet up, in air that has crossed thousands of miles of ocean. Charles David Keeling set up the first measuring system there and ran it for forty-seven years.

The record is one of the most familiar graphs in science. It does two things at once. It wobbles up and down each year, and it climbs. The yearly wobble has a sharp shape, like the teeth of a saw. Carbon dioxide falls through the northern spring and summer, when the forests and crops of the northern half of the world are growing and taking it in. It rises again through the fall and winter as leaves drop and rot. The swing is about six parts per million from peak to trough.

The climb is the other half. The first readings in 1958 were near three hundred and fifteen parts per million, which is the unit in which the measurement is reported. The figure passed four hundred in 2013 and has gone on rising since. The yearly wobble has not changed much, but the level around which it wobbles has shifted upward every single year of the record without exception. That is the part no one disputes, because it is a direct reading from an instrument rather than an estimate. Other stations, from Alaska to the South Pole, now run the same measurement and give the same climb with a slightly different wobble. The South Pole record shows almost no yearly wobble, because the southern half of the world has little plant life to drive one. Keeling nearly lost his funding twice in the 1960s.

WHAT THIS PASSAGE IS FOR

- One station on Mauna Loa has measured carbon dioxide hourly since 1958.
 - The yearly sawtooth comes from northern plant growth taking carbon dioxide in and releasing it.
 - The level has risen every year of the record, from about 315 to over 400 parts per million.
-

A Rift You Can Walk Into

At Thingvellir in southwest Iceland there is a valley with a wall of dark rock on either side. The two walls are the torn edges of a crack in the crust, which is the outer rock layer of the Earth. A visitor can walk along the floor between them. The ground there is pulling apart at about an inch a year, which is roughly the rate at which a fingernail grows. Over a thousand years that is thirty feet, and the valley floor has dropped as the sides have moved away from each other.

The crack is part of a line that runs the length of the Atlantic Ocean, some ten thousand miles from north of Iceland to the far south. Almost all of it is two miles under water. Iceland is one of the few places where the line comes ashore, because an unusually hot region beneath the island has built up enough rock to stand above the sea. New ground is made along the crack: molten rock rises, cools and sticks to both edges, so the two sides grow outward from the middle.

Thingvellir has a second kind of history. Iceland's assembly met in this valley from the year 930 until 1798. That makes it one of the oldest meeting places in Europe. The chieftains who gathered there had no idea that they stood between two plates. The rock they used as a speaking platform is basalt, which is rock that was once liquid. Earthquakes are common and small. When the land moves suddenly rather than slowly, farms and roads are threatened, and Iceland's building rules reflect that. The slow pull continues whether anyone is watching or not. Surveyors have measured the width of the valley with satellite instruments since the 1990s. Those figures agree with the long-term average taken from the age of the rock on each side.

WHAT THIS PASSAGE IS FOR

- The valley walls at Thingvellir are the torn edges of a crack in the crust.
 - The ground is pulling apart at about an inch a year.
 - New ground forms along the crack as molten rock rises, cools and sticks to both edges.
-

A Cliff of Ice in West Antarctica

Thwaites Glacier sits in West Antarctica and reaches the sea along a front about eighty miles wide. The ice behind that front is more than a mile thick in places. If all of it melted, the water added to the ocean would raise sea level by roughly two feet. The glacier also acts as a plug for a much larger body of ice behind it. For that reason more field money has been spent on it in the last ten years than on any other glacier.

The front is not a cliff standing on rock. For the last part of its journey the ice floats, and the line where it lifts off the sea bed is called the grounding line. That line has moved inland. Radar and satellite measurements put the retreat at about nine miles between 1992 and 2011 in one part of the glacier. The floating ice in front, which gives the glacier behind it some shelter from the open sea, has thinned and cracked, and large pieces have broken away.

The reason is not warm air. Air temperatures over that part of Antarctica stay below freezing for most of the year. The reason is water. A current of relatively warm sea water reaches the underside of the floating ice. It is about two degrees above the freezing point of salt water. That is enough to melt the ice from below at tens of meters a year in places. In 2019 a team drilled through six hundred meters of ice and lowered a robot into the cavity, meaning the space under the floating ice. That is how the melting rate at the grounding line was first measured directly. The hole took five days to make with a hot-water drill. The robot sent back pictures of a staircase of ice overhead.

WHAT THIS PASSAGE IS FOR

- Thwaites holds enough ice to raise sea level by roughly two feet and plugs more behind it.
 - The grounding line, where ice lifts off the sea bed, retreated about nine miles in twenty years.
 - Warm sea water, not warm air, does the melting.
-

Physical Science

Passages 21 to 30

21

PHY · S01 · phenomenon

Eleven Weighings of One Stone

A smooth stone from the bed of the river Aire is placed on a laboratory balance, which is an instrument that compares an unknown weight against a known one. The reading is 48.213 grams. It is lifted off and put back, and the reading is now 48.209. Eleven weighings in a row give eleven numbers. They differ in the third decimal place, and sometimes in the second. Nothing has been done to the stone between weighings. It has not been chipped, wetted or warmed. The stone has one mass, and the instrument has given eleven answers.

Some of the spread has obvious sources. A door opens and moves the air. The bench carries a slight vibration from a corridor. A fingerprint adds a film of oil that weighs something. The room is warming through the morning, and the metal parts of the balance expand as it does. None of these is a mistake in the ordinary sense. Nobody wrote the number down wrong. They are the limits of the arrangement being used.

What a careful worker does next is not to pick the reading that looks best. The eleven numbers are kept, and their average is taken, which here is 48.211 grams. The spread of the eleven is also recorded, because that spread is the honest statement of how well this balance can measure this stone in this room. A result given as 48.211 grams, plus or minus 0.002, says two things at once: a best figure and a width. A result given as 48.213 grams on its own looks more confident and tells the reader less. Every quantity in a science report carries a width of that kind, whether or not it is printed, and the habit of asking how wide is the beginning of reading such a report properly. Working laboratories record how many repeats were made.

WHAT THIS PASSAGE IS FOR

- One stone weighed eleven times gives eleven different readings.
 - The spread comes from the arrangement, not from mistakes.
 - An honest result states a best figure and a width.
-

A Hammer and a Feather

On August 2, 1971, at the end of the last walk of the Apollo 15 mission, David Scott stood in front of a television camera on the surface of the Moon. In his right hand he held a geologist's hammer that weighed about one and a half pounds on Earth. In his left he held a falcon feather. He let go of both at the same moment. They fell together and landed together, and the film runs for about one and a third seconds.

On Earth the same demonstration fails every time. A feather dropped beside a hammer drifts and turns and arrives much later. The reason is not weight. The reason is air. A feather has a large surface for its small mass, so the air it must push aside slows it a great deal, while the same air barely affects the hammer. Remove the air and the difference goes with it. Galileo had argued for this result in the 1630s without being able to produce it cleanly, because he had no way to remove the air from a room.

The Moon has no air to speak of. Scott had suggested the demonstration himself, and the flight planners agreed to it partly because it would be watched by schoolchildren. The fall also shows a second thing. Both objects reached the ground in about one and a third seconds from a height of about five feet, which is slower than the same drop on Earth. The Moon pulls less hard. A sensitive instrument can detect that difference without any dropping at all, but a hammer, a feather and a camera make it visible to anyone. The pull at the surface of the Moon is about one sixth of the pull at the surface of the Earth. Scott reported afterward that he had been slightly worried the feather would stick to his glove.

WHAT THIS PASSAGE IS FOR

- A hammer and a feather dropped on the Moon fell together.
 - On Earth the difference is caused by air, not by weight.
 - The fall is also slower than on Earth, because the Moon pulls less hard.
-

Water, a Pipe and a Lit House

Above the town of Foyers in the Scottish Highlands there is a lake, and below it there is a power station. The lake stands about one hundred and seventy meters higher than the turbine hall. Water runs down a steel pipe under its own weight, strikes a wheel at the bottom, and turns it. The turning wheel drives a generator, and the generator sends electricity along wires to houses, some of them forty miles away. The whole arrangement has been working since 1896, which makes it one of the oldest such stations in Britain.

Nothing in that chain creates energy. Each step takes energy in one form and gives most of it back in another. Water high up has energy because of its position. Falling water has energy because of its motion. The spinning generator has electrical energy, and the bulb in a kitchen converts most of that into light and heat. At every step some of what went in does not come out in the useful form. Water sliding down a pipe warms the pipe slightly. Bearings warm. Wires warm. The air around the transformer warms.

Those losses are not large in a modern station. A good turbine and generator together deliver about nine tenths of the energy that the falling water carries. The old bulb was the weak link in the chain for a century, because a filament lamp turned only about five percent of its electricity into light and the rest into heat. That is why replacing the bulbs in a town saves more energy than improving the turbines. The chain is only as efficient as its worst step, and the worst step is usually the one closest to the user. A modern light-emitting diode turns about forty percent of its electricity into light.

WHAT THIS PASSAGE IS FOR

- Each step in the chain changes energy from one form to another.
 - Some energy leaves as heat at every step.
 - The chain is only as efficient as its worst step.
-

A Pump That Grows Hot

Robert Boyle described the behavior of squeezed air in 1662, and anyone with a bicycle can repeat the core of it. Pump up a tire from flat and the barrel of the pump becomes hot to hold within about thirty strokes. The base of the pump, nearest the valve, is the hottest part. Nothing is burning and nothing is rubbing hard enough to explain it. A thermometer taped to the barrel will show a rise of twenty degrees or more on a cool day. The heat comes from the air inside.

Air is mostly empty space. The molecules in it are far apart and moving fast, and the push that a tire feels from the inside is the sum of countless small collisions against the rubber. A pump stroke squeezes a fixed quantity of air into a smaller volume. The piston moving down does work on the air, and that work goes into the motion of the molecules, so they move faster. Faster molecules are what a thermometer reports as a higher temperature. The air has been heated without a flame anywhere near it.

The same relation runs the other way, which is easier to feel than to explain. Let the air out of a full tire quickly and the escaping stream is noticeably cold, because the air expands as it leaves and does work pushing the outside air aside. A refrigerator uses exactly this. It squeezes a gas until the gas is hot and lets that heat transfer to the kitchen through a radiator at the back. The gas is then allowed to expand inside the cabinet, where it is now colder than the food. Pressure, volume and temperature cannot be changed one at a time. Fix any two and the third is decided.

WHAT THIS PASSAGE IS FOR

- A bicycle pump grows hot because work is done on the air inside it.
 - Temperature is a report on how fast the molecules are moving.
 - A refrigerator runs the same relation in reverse.
-

The Fuse and the Kettle

A kettle, a toaster and an iron are switched on in the same kitchen, and the lights go out. In the cupboard under the stairs a small switch has moved to the off position. Nothing is broken. The switch is a circuit breaker, which is a device that opens the circuit when too much current flows through it, and it has done what it was installed to do.

Electricity in a house behaves in a way that a water pipe makes easy to picture. The voltage at the socket is like the pressure in a main. The current is like the flow in gallons a minute. The resistance of whatever is plugged in decides how much flows for a given pressure. A kettle has low resistance because it is supposed to turn electricity into heat quickly, so it draws about thirteen amps in a British house and about twelve in an American one. A table lamp draws less than half an amp. Adding the kettle, the toaster and the iron on one circuit asks for more current than the wire in the wall was designed to carry.

The danger is not the appliance. The danger is the wire buried in the plaster, which cannot be inspected and cannot shed its heat. Push too much current through it and it warms, the insulation around it ages, and in the worst case it sets fire to the joists beside it. The breaker is sized to protect that wire and not the kettle. It is set below the current at which the cable would overheat, so the circuit is broken long before the plaster is. A householder who fits a larger breaker to stop the nuisance has removed the only thing standing between the kitchen and the timber.

WHAT THIS PASSAGE IS FOR

- The breaker opens the circuit when the current is too high.
 - Resistance decides how much current flows for a given voltage.
 - The breaker protects the buried cable, not the appliance.
-

The Bent Straw in a Glass

Ptolemy wrote about this effect in Alexandria in the second century. Stand a straw in a glass of water and look at it from the side. The straw appears to break at the surface. The part under water seems to be at a different angle from the part above, and if the glass is moved the break moves with it. The straw is perfectly straight. A ruler laid against it in the air, and then in the water, demonstrates that no physical bending has occurred.

Light travels more slowly through water than through air, by about a quarter. When a beam crosses the surface at an angle, the side of the beam that enters first is slowed first, and the whole beam swings. The technical word is refraction, meaning the bending of light as it passes from one material into another. Everything seen through the water is therefore seen in a shifted position. The eye has no mechanism for detecting that the light changed direction along the way. It reports the straw where the light appears to have come from.

The same rule explains a long list of ordinary things. A swimming pool looks shallower than it is, which is why bathers misjudge the deep end. A spoon in water looks broken. A spoon in oil looks more broken, because oil slows light more than water does. A lens in a pair of spectacles is a piece of glass ground to a curve, so that the bending happens in a chosen direction and brings light to a focus on the retina. Some light is also reflected at the surface, which is why a window shows a faint image of the room. Some is absorbed by the water itself, which is why the deep sea is dark. One surface does three things to a beam at once.

WHAT THIS PASSAGE IS FOR

- The straw looks broken at the surface although it is straight.
 - Light bends because it travels more slowly in water than in air.
 - A surface reflects, bends and absorbs light at the same time.
-

Steel Wool and a Steel Bar

Antoine Lavoisier established in the 1770s that burning is a reaction with oxygen. Hold a match to a steel bar and nothing happens. The bar warms near the flame and stays a bar. Hold the same match to a pad of fine steel wool and the wool catches, glows orange, and travels across the pad in a slow front that leaves a gray ash behind it. A nine-volt battery held across the same pad will start it just as well. The two objects are the same metal, from the same kind of ingot, and one of them burns.

What differs is surface. Steel reacts with oxygen in the air to make iron oxide, which is rust. That reaction can happen only where metal and air meet. A solid bar presents a few square centimeters of surface for every hundred grams. The same hundred grams, drawn into threads a hundredth of a millimeter thick, presents several square meters. Every thread is also thin enough to be heated right through. The reaction then releases enough heat in one spot to light the thread beside it.

Rust on a gate is the same reaction running at a crawl. It takes years because the surface is small, the heat escapes, and a layer of oxide shields what is underneath. The chemistry does not change with the shape of the metal; only the rate does. That is why a flour mill can explode while a bag of flour on a shelf is perfectly stable. Fire crews treat a grain silo and a sack of grain as different problems. A substance has no single level of danger. It has a level for each state it is divided into. Powdered sugar and powdered coal have both destroyed buildings where the solid form had been handled safely for years.

WHAT THIS PASSAGE IS FOR

- The same metal burns as wool and not as a bar.
 - Reaction can happen only where metal and air meet.
 - Shape changes the rate of a reaction, not the chemistry.
-

The Bridge at Dundee

The first Tay Bridge carried a single railway track two miles across the water at Dundee in Scotland. It opened in 1878 and was the longest bridge in the world at the time. Queen Victoria crossed it in June 1879 and knighted the engineer who had designed it. On the night of December 28 of that same year, in a winter gale, the thirteen central spans went into the river while a train was crossing. Everyone aboard died. The usual figure given is seventy-five.

The inquiry found a chain of causes rather than one. The high girders were carried on columns of cast iron, a material that is strong when squeezed and weak when pulled or struck. Some of the castings had holes in them from the foundry and had been filled with a paste of iron filings and wax, which looked sound and carried nothing. The bolt holes had been cast rather than drilled, which left them rough. No allowance had been made in the design for the sideways push of a gale on the side of a train, and the measured winds that night were near eighty miles an hour.

The engineer had never calculated what the structure must resist from the wind. That was not unusual in 1879. It became unusual immediately afterward. The second Tay Bridge, built beside the first and still in use, used wrought iron and steel, had doubled columns, and was designed against a wind load of fifty-six pounds per square foot. The stumps of the old piers still stand in the water beside it, and every train from Edinburgh to Aberdeen passes them. The inquiry also asked who should have noticed. Workmen had reported loose bolts and rattling joints for months, and the reports had gone no further than the foreman.

WHAT THIS PASSAGE IS FOR

- The bridge fell in a gale fourteen months after it opened.
 - Cast iron is strong in compression and weak in tension.
 - No wind load had been calculated in the design.
-

The Women Who Painted Dials

In 1917 a factory in Orange, New Jersey, began painting watch and clock dials with a paint that glowed in the dark. The glow came from radium, an element discovered by Marie and Pierre Curie in 1898. About seventy young women did the painting. The brushes were fine camel hair, and to keep a point on them the women were taught to shape each brush with their lips between strokes. Some of them painted their nails and teeth with the paint for a party trick.

The first deaths were put down to other causes. A dial painter named Mollie Maggia lost teeth through 1922 and then pieces of her jaw. She died in September of that year, with the cause recorded as a disease of the mouth. By 1924 a dentist in the town had seen enough cases to write to the state. Radium behaves chemically much like calcium, so a body builds it into bone, where it sits for the rest of a life giving out radiation at close range. The jaw and the hip were the usual sites of damage.

Five of the women sued in 1927, in a case the newspapers followed closely. All five were too ill to raise their arms to take the oath. The settlement was small and they were already dying, but the case established that a company could be held liable for a disease contracted at work years earlier. The factories were made to stop the lip-pointing, to issue washable aprons and screens, and to reduce the time any one worker spent at the bench. Later work on those same patients produced the first serious figures on what a given amount of radium in the body does to bone. Those figures were still being used to set safety limits fifty years afterward.

WHAT THIS PASSAGE IS FOR

- Dial painters shaped their brushes with their lips and swallowed radium.
 - Radium behaves like calcium, so the body stores it in bone.
 - The 1927 case established liability for a disease contracted years earlier.
-

Eight Minutes of Sunlight

Light moves at a fixed speed, about three hundred thousand kilometers in a second. Nothing in the universe carries information faster than that. Albert Michelson measured it to within a fraction of one percent in 1879. That single fact turns every distance in the sky into a length of time. The times are easier to hold in the head than the distances. A distance tells a reader how far away a thing is. A time tells the reader how old the view of it is.

The Moon is about four hundred thousand kilometers away, so its light reaches an observer in a little over one second. Radio instructions to the Apollo crews took the same time, which is why those recordings have an odd pause in every exchange of speech. The Sun is four hundred times further off. Sunlight takes eight minutes and twenty seconds to arrive, so the Sun that anyone sees above the horizon left the surface before the viewer sat down. Jupiter is between thirty-three and fifty-three minutes away, depending on where the two planets stand. Light from the edge of the solar system takes several hours.

Beyond that the numbers change scale. The nearest star beyond the Sun, Proxima Centauri, is about four and a quarter light years off. The light arriving tonight set out when a child now at school was born. The nearest large galaxy, Andromeda, lies two and a half million light years away. It is still close enough to see with the naked eye on a dark night, as a faint smudge about the width of a thumb. Every view of the night sky must therefore contain many different dates at once. A telescope is a kind of record, and the further it is pointed the older the record is. Nobody has ever seen any part of the universe as it is now.

WHAT THIS PASSAGE IS FOR

- Light speed turns every distance in the sky into a time.
 - Sunlight takes eight minutes and twenty seconds to arrive.
 - A single view of the sky contains many different dates.
-

Humanities

Passages 31 to 40

31

HUM · S01 · phenomenon

Two Proposals, Two Narrators

Charlotte Bronte published *Jane Eyre* in 1847. Jane tells her own story, and when Rochester proposes to her the reader is inside her head. The reader learns that her heart is beating and that she suspects a trick. Nothing is reported that Jane does not know. Rochester's thoughts are a closed door. When he lies to her, the reader is deceived at the same moment she is, and finds out when she finds out.

Jane Austen published *Pride and Prejudice* in 1813. Darcy proposes to Elizabeth in a room at Hunsford, and the teller stands outside both of them. The reader is told what Elizabeth says and how Darcy colors as he listens. The voice also comments. It can say that Darcy spoke with little attempt at civility, which is a judgment Elizabeth would not have phrased that way. Austen uses that freedom constantly, and a reader who misses it takes the narrator's dry remarks for plain fact.

Neither choice is better. Each buys something and gives something up. The first person keeps the reader locked in one mind, which creates suspense and sympathy and makes deception easy to arrange. The outside voice can move between rooms and years, can tell a reader what two people are each thinking, and can judge them both. What it cannot do is surprise the reader by not knowing something. Every novel reveals its world through a particular position, and the first useful question about any passage is who is standing where. The answer is usually settled in the opening sentences and holds for the rest of the book. Bronte made her choice part of the title page, which announced the novel as an autobiography edited by Currer Bell. Austen left her name off her books entirely in her lifetime. Both facts tell a reader something about the voice before a word of the story is read.

WHAT THIS PASSAGE IS FOR

- In *Jane Eyre* nothing is reported that Jane does not know.
 - Austen can report two minds and comment on both.
 - The first question about a passage is who is telling it.
-

The Clerk Who Would Prefer Not To

Herman Melville published *Bartleby, the Scrivener* in 1853. A lawyer on Wall Street hires a copyist named Bartleby, who works steadily for a few days. Then the lawyer asks him to check a document against the original. Bartleby answers that he would prefer not to. He is not angry and he does not explain. Asked again, he gives the same answer in the same mild voice, and over the following weeks he stops doing anything at all.

The story never says why. The lawyer tries every explanation available to him. He wonders whether Bartleby is ill, or mad, or ruined by earlier work in a dead letter office. That last guess is offered in the final paragraph as a rumor rather than a fact. He offers money. He moves his own offices rather than remove the man. The reader is given exactly what the lawyer is given, which is a sentence of five words repeated, and no access to the mind behind it.

That withholding is the design of the story and not a gap in it. Melville portrays a man entirely through what he refuses to do, and the pressure of the tale falls on the narrator instead. What the reader learns a great deal about is the lawyer: his comfort, his fear of scenes, his wish to be thought kind, and the exact point at which his patience runs out. A character can be built out of an absence. When a writer keeps a motive back, the question worth asking is what the silence does to everyone else on the page. Melville had published *Moby-Dick* two years earlier to poor sales. He was working as a customs inspector in New York within a few years of writing this story, and he published almost no prose for the next thirty.

WHAT THIS PASSAGE IS FOR

- Bartleby answers requests with one mild refusal and gives no reason.
 - The story offers the dead letter office only as a rumor.
 - Withholding the motive turns the attention onto the narrator.
-

A Poem Made of Four Objects

William Carlos Williams published a poem of sixteen words in 1923. It names a red wheelbarrow, rain water, white chickens, and the fact that a great deal depends on them. There is no story. Nobody speaks. No feeling is named at any point. The poem does not say that the scene is beautiful or sad or important, and the only abstract word in it is the claim that so much depends.

Williams was a doctor in Rutherford, New Jersey, and wrote between patients. He argued for years that poetry should work through things rather than through statements about things. He put the principle in a line now quoted more often than any of his poems: no ideas but in things. The wheelbarrow is the argument in practice. A reader is handed three objects, told that something rests on them, and left to supply the rest.

The effect is easy to test. Replace the objects with abstractions and the poem dies at once. A line about a feeling of rural simplicity is not a poem at all, and it says less rather than more. The concrete version works because rain water on red paint can be seen, and because the reader's own eye does the work of connecting the glaze to the chickens. Williams did not invent the method. He stated it plainly enough to teach it, and a generation of poets took the lesson. A thing on the page can express what a statement about it cannot. Williams published the poem in a collection called *Spring and All*, among prose passages that argued the same case at length. The poem has outlived the argument around it. It is now one of the most anthologized short poems in English, and it is still sixteen words.

WHAT THIS PASSAGE IS FOR

- The poem names objects and no feelings.
 - Williams argued for poetry that works through things.
 - Replacing the objects with abstractions destroys the poem.
-

Fourteen Lines and a Hinge

A sonnet has fourteen lines. The form reached England from Italy in the sixteenth century, and Shakespeare composed a hundred and fifty-four of them. The English version rhymes in a fixed pattern: four lines, then four more on new rhymes, then four more, then a pair at the end that rhyme with each other. Each line runs to ten syllables with the stress falling on every second one, a pattern called iambic pentameter.

The interesting part is not the counting but the hinge. An Italian sonnet divides eight lines against six, and the turn comes after line eight. The reader is given a situation, a problem or a picture, and then the poem changes direction. The English form keeps a turn near line nine as well. It then adds a second one, because the closing pair can confirm or undercut what came before. Sonnet 130 spends twelve lines listing everything the speaker's mistress is not. The closing pair then says that he loves her all the same. The joke only lands because the reader has been counting.

A reader who can count to fourteen therefore has a map. Finding the turn is usually enough to find the argument, and the argument is what examiners ask about. The form is also old enough to carry its own weight, so a modern poet who writes a sonnet about a bus station is borrowing four centuries of love poems and knows it. Wordsworth, Keats, Hopkins, Frost, Heaney and Terrance Hayes have all used the shape. The rules are simple, written down, and still being argued with. Hayes published a book of seventy sonnets in 2018, every one of them titled the same way. The form has survived four hundred years of poets trying to break it.

WHAT THIS PASSAGE IS FOR

- A sonnet is fourteen lines with a fixed rhyme pattern.
 - The turn, not the counting, carries the argument.
 - The English form adds a closing couplet that can undercut what came before.
-

A Platform, a Balcony, a Trap

The Globe was built on the south bank of the Thames in 1599 from the timbers of an older playhouse. It was a ring of galleries about a hundred feet across, open to the sky in the middle, with a platform stage pushed out into the yard. Two thousand people could get in. The cheapest paid a penny and stood in the open around three sides of the stage. Plays began in the early afternoon because there was no artificial light.

The building offered a company three useful things. The platform had a trap door, so a grave could be dug or a devil could rise. Above the stage was a gallery, which could serve as a balcony, a city wall or the deck of a ship. At the back were two doors and a curtained space, which let an actor be hidden and then discovered. There was no front curtain and almost no scenery, so a scene ended when the actors walked off and the next began when others walked on.

Those arrangements shaped the writing. A speech can be addressed to two thousand people standing in daylight, which is why asides and soliloquies work there and would be strange in a dark room. Place is established by being named, because nothing on stage can show it. Scenes run straight into one another without pause, which is why the plays move so quickly between Rome and Egypt or between a court and a field. The theater burned down in 1613 when a cannon used in a performance set the roof alight. A modern copy was built two hundred yards from the original site in 1997, and it plays in daylight without microphones. Audiences there stand in the yard in the rain, which is the one part of the experience that no indoor theater can reproduce.

WHAT THIS PASSAGE IS FOR

- The Globe held about two thousand people and played in daylight.
 - A trap, a gallery and a curtained space were the whole machinery.
 - Place had to be named because nothing on stage could show it.
-

The First Locked Room

Edgar Allan Poe published *The Murders in the Rue Morgue* in 1841. Two women are killed in a room on the fourth floor of a Paris house. The door is locked from the inside and the key is in the lock. The windows appear to be nailed shut. A crowd of neighbors hears voices on the stairs but cannot agree on what language the second voice was speaking. The police arrest the wrong man almost at once.

The story introduced a set of parts that are still in use. There is a brilliant private investigator, Dupin, who is not a policeman and who solves the case by reasoning from physical detail. There is a slower companion who tells the story and asks the questions a reader would ask. There is an official force that is competent and wrong. There is a solution that is strange but mechanical. Here it turns on a nail broken so that a window appears fastened when it is not. The explanation is given in the last pages, and every clue needed for it has already appeared.

Arthur Conan Doyle read Poe closely and did not pretend otherwise. Sherlock Holmes arrived forty-six years later with the same arrangement of parts. Doyle even has Holmes dismiss Dupin by name in the first novel, which is what a writer does when the debt is obvious. Later writers did not simply imitate the pattern. They counted on a reader knowing it, which is a different and more useful thing, and it is where the pleasure of the modern form begins. Agatha Christie built a career on breaking the rules the pattern had established, and her readers could only be shocked because they knew them. Poe himself wrote only three Dupin stories and thought little of them.

WHAT THIS PASSAGE IS FOR

- The room was locked from the inside with the key still in the lock.
 - Poe assembled the parts the genre still uses.
 - Later writers counted on the reader knowing the pattern.
-

A Dutch Kitchen in 1658

Johannes Vermeer painted *The Milkmaid* in Delft at about the end of the 1650s. The picture is small, roughly eighteen inches high. A servant stands at a table pouring milk from an earthenware jug into a bowl. That is the whole event. Light comes through a window on the left, and the wall behind her is plain plaster with a nail and a nail hole in it.

The things in the room are painted with great care. There is bread on the table, broken and rough. There is a blue cloth, a brass pot, a basket and a polished jug hanging on the wall. Low down on the right there is a foot warmer, a small wooden box for hot coals, and a line of blue tiles along the base of the wall. The milk is the only moving thing in the painting, and Vermeer painted it as a thin white thread with a thickened edge where it meets the bowl.

Every surface reflects light differently and the painter has recorded each one. The glaze on the jug carries a hard highlight. The bread crust holds small dots of thick paint that catch light like real crumbs. The whitewashed wall is warm where the sun falls on it and cool in the shadow. Nothing is happening, and yet the picture has held visitors in front of it for three hundred years. A later passage in this strand asks how a painter arranges such a scene so that the eye goes where it is sent. The first step is noticing how much is actually there. Vermeer left about thirty-five paintings in all. He worked slowly, borrowed money, and was largely forgotten for nearly two centuries after his death in 1675. *The Milkmaid* is now the property of the Dutch state.

WHAT THIS PASSAGE IS FOR

- The painting records one ordinary action in a plain room.
 - Each surface is given its own kind of highlight.
 - The milk is the only thing in motion.
-

One Tune, Twelve Ways

Mozart wrote a set of twelve variations for keyboard on a French song that English speakers know as the tune of Twinkle, Twinkle, Little Star. The tune itself is eight bars long and uses only a handful of notes. He states it plainly first, so that every listener knows exactly what will be worked on. Then he changes it twelve times without ever quite losing it.

The methods are easy to hear. In one variation the right hand runs in fast even notes while the tune hides inside the run. In another the tune moves to the left hand underneath. One variation turns the key from major to minor, which darkens the whole character while the shape of the melody stays where it was. One slows almost to a halt. The last speeds up and changes the count of beats in a bar. In every case a listener can still follow the original, and the pleasure comes from hearing it survive.

The form is old and was never a Mozart's invention. A composer takes something already known, often a popular song or a dance, and demonstrates what can be done to it. Bach, Beethoven, Brahms and Rachmaninoff all wrote large sets of this kind, and Beethoven was inspired to write thirty-three variations on a waltz by a publisher he thought little of. Jazz musicians work the same way nightly. The theme is stated, and then it is taken apart in front of the audience. That only works if everyone in the room remembers the theme. Mozart wrote his set in his twenties, probably in Paris, and it was published in 1785. The song itself was already old and nobody knows who wrote it. Several hundred sets of variations on borrowed tunes were published in Europe in that decade alone.

WHAT THIS PASSAGE IS FOR

- The tune is stated plainly before it is altered.
 - Each variation changes one thing while leaving the tune recognizable.
 - The form depends on the audience remembering the theme.
-

Shopping Under Glass

Burlington Arcade opened off Piccadilly in London in 1819. It is a straight covered lane, about six hundred feet long, lined on both sides with small shops and roofed in glass. It was built on the edge of a private garden, partly to stop passers-by from throwing oyster shells over the wall. Beadles, meaning uniformed officers employed to keep order, patrolled it in top hats. Running, singing and whistling were forbidden, and still are.

The idea spread quickly because it solved several problems at once. London rain made street shopping miserable, and pavements were narrow, filthy and shared with carriages. Under glass a shopper could walk the whole length dry, look in forty windows, and meet people. The shops were small and the rents were high, which suited trades selling expensive light goods: gloves, jewelry, prints, watches. Gas lighting, and then electric lighting, let the arcade stay open into the evening.

Bigger versions followed across Europe. Milan built the Galleria Vittorio Emanuele II between 1865 and 1877, with a glass dome a hundred and fifty feet high over a crossing of two covered streets. Cleveland, Brussels, Naples and Moscow each built their own version within fifty years of the Milan galleria. These were admired as engineering as much as architecture. A long glass roof carried on iron was a new thing to be able to make. The arcade is also the direct ancestor of the shopping mall. One difference matters. An arcade is a public street with a roof on it, and it has doors at both ends that open onto the city. A mall has one owner, one set of opening hours and a car park. Burlington Arcade still has its beadles, and the rules against whistling were enforced into the present century.

WHAT THIS PASSAGE IS FOR

- An arcade is a covered lane lined with small shops.
 - It solved rain, dirt and narrow pavements at once.
 - An arcade is a public street with a roof, open at both ends.
-

Two Reviews of One Riot

The Rite of Spring had its first performance in Paris on May 29, 1913. Igor Stravinsky wrote the music and Vaslav Nijinsky arranged the dancing. The audience began to object within the first minutes. There was laughing, then shouting, then fighting in the aisles, and the noise grew loud enough that the dancers could not hear the orchestra. Nijinsky stood in the wings calling out the counts.

The reviews the next morning did not agree about what had happened. One Paris critic called the work a laborious and puerile barbarity and said the theater should be ashamed. Another praised it as a work of rare strength and wrote that the hissing came from people defending their own habits. A third spent most of his column on the clothes in the audience. All three had sat in the same room for the same thirty-three minutes.

That disagreement is the ordinary condition of criticism rather than a scandal. Each of the three had a standard in mind and the standards were different. The first wanted melody and grace and found neither. The second wanted a work that did something new and found one. The third wanted a social occasion. A reader who wants to use any review must therefore find the standard behind it, which is often unstated. Within a decade the piece was being played in concert halls to respectful audiences, and Stravinsky lived another fifty-eight years as the most famous composer alive. The riot is now part of the music's reputation, which none of the three critics could have predicted. Stravinsky himself gave several accounts of the evening in later life, and they do not agree with one another either. The theater that staged it had opened only six weeks before, on a new avenue in a fashionable district. Its manager said afterward that he had expected trouble and had sold the seats anyway.

WHAT THIS PASSAGE IS FOR

- The first performance ended in shouting and fighting.
 - The reviews disagreed because the standards behind them differed.
 - A reader of a review must find the standard behind it.
-

Social Science

Passages 41 to 50

41

SOC · S01 · phenomenon

Forbid, Allow and Twenty Points

In 1940 an American researcher named Donald Rugg ran a simple test on question wording. He had two versions of one question about free speech. One asked whether the United States should forbid public speeches against democracy. The other asked whether the United States should allow such speeches. Logically these are the same question. A person who wants to forbid something does not want to allow it. The two versions went to two matched groups of people in the same survey.

The answers did not match. About fifty-four percent said yes to forbidding. About seventy-five percent said no to allowing. Those two figures describe different publics, and the gap is roughly twenty points. The study has been repeated many times since, in several countries, with the same result. People are more willing to refuse permission than to impose a ban, even when the practical effect is identical. The word forbid is heavier than the phrase not allow, and respondents, meaning the people who answer a survey, hear the weight.

That is one of a family of effects that pollsters have to manage. The order of the items changes answers. A question about national problems placed after a question about crime produces more mentions of crime. Offering a middle option pulls perhaps a fifth of respondents out of the two sides. Asking about aid to the poor gets warmer answers than asking about welfare, though the programs named are the same ones. None of this means that surveys are useless. It means that a number from a survey is a number about a particular form of words, and that the wording influences the result as much as the subject does. A reader who is given a percentage without the question is not being given much. Good polling organizations publish the exact wording alongside the figure for this reason.

WHAT THIS PASSAGE IS FOR

- Two logically identical wordings gave answers twenty points apart.
 - People refuse permission more readily than they impose a ban.
 - A survey figure is a figure about one particular form of words.
-

Thirteen Children or Twenty-Five

Tennessee ran an unusual experiment in its schools between 1985 and 1989. It is still known by the name Project STAR. The state wanted to know whether small classes help young children. Earlier studies had compared schools that happened to have small classes with schools that did not, and those comparisons were hard to trust. Small classes are common in rich districts and in schools with spare rooms. Any difference in results might come from the wealth or the building rather than the class size.

So the state assigned children by lot. About eleven thousand six hundred pupils in seventy-nine schools entered the study in kindergarten. Within each school, children were put at random into a small class of thirteen to seventeen, or into a regular class of twenty-two to twenty-five, or into a regular class with a teacher's aide. Teachers were assigned by lot as well. The children stayed in their assigned kind of class for up to four years and were tested each year. The design is called a randomized trial.

The small classes did better. The gap was clearest in the first two years and in reading, and it was larger for children from poorer families. The effect did not vanish when the children returned to ordinary classes, and some of it could still be found years later in high school test taking. The experiment also cost a great deal, because small classes need more teachers and more rooms, and that cost is the reason the policy argument continued after the result was in. Tennessee published the data, and economists including Alan Krueger reanalyzed it for years afterward. What the lot made possible was a fair comparison. Nothing except the class size distinguished the two groups at the start.

WHAT THIS PASSAGE IS FOR

- Children and teachers were assigned to class sizes by lot.
 - About 11,600 pupils in 79 schools took part from kindergarten.
 - The lot is what made the comparison fair.
-

Chocolate and the Nobel Prize

In 2012 a Swiss physician named Franz Messerli published a short paper in a well-known medical journal. He had plotted two figures for twenty-three countries. On one axis was the amount of chocolate eaten per person each year. On the other was the number of Nobel prizes won per million people. The points lined up remarkably well. Switzerland sat at the top on both measures. China sat near the bottom on both. The relationship was strong enough that a statistician would call it a tight fit.

Messerli was making a joke with a serious point, and he said so in the paper. Nobody thinks that eating chocolate produces scientists. Several other explanations are available at once. Rich countries eat more chocolate and also fund more laboratories, so national wealth could produce both figures without either touching the other. The measures are also crude. Prizes are awarded to people who did their work decades earlier, often in a different country from the one that claims them. A country of two million people needs only one laureate, meaning one person awarded the prize, to lead the table.

The paper is now used in teaching, which was not quite its purpose. It is a clean case of a pattern that is real, strong, measurable and useless as a guide to action. Two figures can rise together because one causes the other, because the second causes the first, because a third thing causes both, or because a small sample has fallen out that way. A reader who sees two lines moving together and does not ask which of those four is in play will assume the first. That reader will be wrong most of the time. The habit of asking is most of what a course in statistics is for.

WHAT THIS PASSAGE IS FOR

- A real published paper found a tight fit between chocolate and Nobel prizes.
 - National wealth could produce both figures without either touching the other.
 - Two figures can move together for four quite different reasons.
-

The Street With One Rich Man

Nine households on a street each have an income of forty thousand dollars a year. The tenth household has an income of four million. The mean income on the street is four hundred and four thousand dollars. That figure is correct and it describes nobody. Nine of the ten families are nowhere near it, and the tenth is ten times above it. The middle household, which is the median, meaning the middle value when the cases are put in order, earns forty thousand dollars like its eight neighbors.

Real national figures behave in the same way for the same reason. In the United States the mean household income runs well above the median, by something like a third, because incomes have no upper limit and do have a lower one. A small number of very large incomes pull the mean up and cannot pull the median anywhere, since the median only asks which household stands in the middle of the line. Any measure that sums and divides is sensitive to the largest values in the set. Any measure that counts positions is not.

Which figure a writer chooses therefore decides what the sentence says. A report that uses the mean can describe a decade in which almost every household lost ground as a decade of rising incomes, provided the top of the distribution rose enough. A report that uses the median will show the opposite trend from the same raw data, and neither report is lying. The responsible practice is to give both, along with some measure of the spread. When only one is offered, the useful question is which one, and the second useful question is why that one was chosen. Government statistical offices publish both as a matter of routine. Newspapers usually print whichever supports the sentence already written.

WHAT THIS PASSAGE IS FOR

- A mean can be correct and describe nobody in the group.
 - The median asks which household stands in the middle.
 - Mean and median can show opposite trends from the same data.
-

Parking by the Hour in San Francisco

San Francisco began an experiment with parking meters in 2011. The problem was not a shortage of spaces. It was that drivers circled looking for one, which filled the streets with slow traffic. Studies in several cities had found that a substantial share of downtown traffic at some hours was made up of cars hunting for a space rather than going anywhere. The city decided to attack the hunting rather than build more parking.

The method was price. Meters in the test districts were fitted with sensors, which are devices that reported whether the space was occupied. Rates were then adjusted block by block and hour by hour, with a target of leaving about one space free on every block. Where a block was always full, the price went up by twenty-five cents an hour at the next review. Where a block was often empty, the price came down. Rates ended up ranging from twenty-five cents to six dollars an hour, and on many blocks at many hours they fell rather than rose.

The city reported that the time spent searching for a space dropped by about forty percent in the test areas. Average rates in those areas went down, which surprised people who had expected a revenue grab. The policy works by paying attention to the one thing a driver cannot ignore, and by letting the price reward whoever is willing to walk two blocks. A campaign asking drivers to circle less would have cost less and done nothing. Several cities have since copied the design. The main obstacle everywhere has been political rather than technical. A price that moves is easy to attack and hard to defend in one sentence. Los Angeles, Seattle and Calgary now run versions of the same scheme.

WHAT THIS PASSAGE IS FOR

- Much downtown traffic was cars hunting for a space.
 - Rates were adjusted block by block toward one free space per block.
 - Search time fell by about forty percent and average rates fell too.
-

The Card in the Hotel Bathroom

Hotels put cards in bathrooms asking guests to use their towels more than once. In 2008 three researchers, led by Noah Goldstein, tested what those cards should say. They ran the test in a real hotel, across several hundred rooms, over several months. Housekeeping staff recorded whether the towels in each room had been hung up for reuse. The guests did not know that anything was being measured. That is what makes the study a field experiment, meaning a trial run in a real setting rather than a laboratory.

One version of the card made the usual environmental appeal and asked guests to help save the environment. Another version reported a plain fact about other guests, saying that most people who stay in the hotel reuse their towels. A third named the room: most guests who have stayed in this room have reused their towels. The cards were otherwise identical and were assigned to rooms at random. Reuse rose by roughly a quarter with the second card compared with the first. The version naming the room did better still.

The finding has been reproduced in other settings, including electricity use and tax letters, and it has limits that later passages in this strand will take up. What it shows is modest and useful. A statement about what other people actually do is a stronger lever than an appeal to a value, and it costs the same to print. People behave partly by reading the room, and a sentence that tells them what the room is doing will be acted on by some of them. Nobody in the hotel was arguing about the environment. They were deciding, in about four seconds, what a normal guest does with a towel. The hotel saved laundry costs, which is why it agreed to the study.

WHAT THIS PASSAGE IS FOR

- Housekeeping recorded towel reuse without the guests knowing.
 - A card reporting what other guests do beat an environmental appeal.
 - A statement about what others do is a stronger lever than an appeal to a value.
-

Houses That Followed the Streetcar

Before 1850 most people in Boston lived within walking distance of their work. The built-up city was about two miles across. A clerk who had to be at a counting house by eight walked there, and a city therefore could not be much wider than an hour on foot. Rich and poor lived close together, often in the same street, because nobody could afford the time to live far out.

Then came the omnibus, which is a large horse-drawn carriage running a fixed route, and the steam railway, and from 1889 the electric streetcar. A streetcar held forty people and ran at about eight miles an hour, which put land four or five miles out within reach of a downtown office. Builders bought fields along the new lines. They put up houses in rows. Between 1870 and 1900 Boston spread from a two-mile city to a ten-mile one. The historian Sam Bass Warner counted twenty-three thousand houses built in three suburbs in those thirty years, almost all of them by small builders putting up a handful at a time.

Who moved out was decided by the fare. A ride cost five cents each way. That was real money for a laborer earning a dollar and a half a day. It was nothing to a clerk earning three. Those who could pay it moved to new wooden houses with gardens, and those who could not stayed in the center. The new districts sorted themselves by income along each line. A pattern of this kind is not a matter of taste alone. People prefer space when they can buy the travel time, and the price of the travel decides who is allowed to prefer it. The same sorting happened later with the automobile, at a larger scale and a higher price.

WHAT THIS PASSAGE IS FOR

- A walking city could not be much wider than an hour on foot.
 - The electric streetcar from 1889 put land four or five miles out in reach.
 - The five-cent fare decided who could move out.
-

Hired by the Half Day

At the London docks in the 1880s men were hired twice a day. A foreman came to a gate at about half past seven, and again after noon, and chose from the crowd that had gathered. The crowd was always larger than the number of jobs. Work lasted four hours and paid by the hour. A man who was not picked went home with nothing and came back for the afternoon call. Those who were picked might work two days that week or six, depending on how many ships were in.

The system suited the dock companies exactly. Ship arrivals were irregular, and a company that employed a steady workforce would be paying men to stand about on quiet days. Casual hiring, meaning taking men on by the half day with no contract, moved that risk onto the workers. It also gave the foreman very great power, since he chose faces each morning and could be bought a drink the night before. The men who did best were not the strongest but the best known to the man at the gate.

Hiring of that kind was normal across the whole port. In August 1889 the dockers struck for a guaranteed minimum call of four hours and a rate of sixpence an hour. The strike lasted five weeks, drew financial support from as far away as Australia, and largely won. The deeper arrangement took much longer to change. Casual hiring at the docks was not ended in Britain until 1967. The same practice ran in New York under the name of the shape-up into the 1950s. A market can be organized so that one side carries nearly all the uncertainty, and the side that benefits will usually describe the arrangement as flexibility. The word has not changed much since. Delivery work booked by the hour raises the same question in a new form.

WHAT THIS PASSAGE IS FOR

- Men were chosen from a crowd at the gate twice a day.
 - Casual hiring moved the risk of irregular shipping onto the workers.
 - The 1889 strike won the four-hour call but casual hiring lasted much longer.
-

Thirty Years Across One City

Streeterville and Englewood are both neighborhoods of Chicago. They lie about nine miles apart and are served by the same city government, the same state laws and the same hospitals in principle. Life expectancy at birth means the average number of years a person born now is expected to live. In figures published in 2019 it was about ninety years in Streeterville. In Englewood it was about sixty. The gap is roughly thirty years, inside one city, between two places a short train ride apart.

Nothing in those figures is about individual choices in any simple sense. The two areas differ in income and in the share of adults with work. They differ in the quality of housing and the number of grocery stores. They differ in air quality near the expressways and in the presence of violence. They differ in how far a resident must travel for unhurried medical care. They also differ in history. The boundaries follow lines drawn by lending maps and by covenants, which were clauses in deeds restricting who could buy a house. These things arrive together and are hard to separate.

Figures of this kind are now published for most large American cities and for many European ones, usually by small area. Glasgow produced a famous example. The reported gap there was more than twenty years between two districts eight miles apart. The exact numbers move when the boundaries or the years change. Some early figures were later revised downward, which is a reason to read any single pair of numbers carefully. The pattern does not move. A postal code will predict the length of a life better than most facts about the person living there. Later passages in this strand ask what such a measure is actually measuring.

WHAT THIS PASSAGE IS FOR

- Two Chicago neighborhoods nine miles apart differ by about thirty years.
 - The differences arrive together and are hard to separate.
 - A postal code predicts length of life better than most personal facts.
-

A Test That Is Usually Wrong

In 1978 three researchers put a short problem to sixty doctors and medical students at Harvard. A disease affects one person in a thousand. A test for it never misses a case. It does give a false positive, which is a positive result in a person who does not have the condition, in five percent of healthy people. A patient has tested positive and nothing else is known about him. The doctors were asked for the chance that he has the disease.

Most of the sixty answered ninety-five percent. Only about one in six gave an answer near the right one. The right answer is about two percent. The arithmetic is short enough to do on a napkin. Take a thousand people, of whom exactly one has the disease, and that one tests positive as the test never misses a case. Of the nine hundred and ninety-nine who do not have it, about fifty also test positive. So fifty-one people in the thousand test positive and only one of them is actually ill, and one in fifty-one is about two percent.

What goes wrong is not the arithmetic but the starting point. The five percent figure is vivid and the one in a thousand figure is not, so the second gets dropped. The rate at which a condition occurs before any test is run is called the base rate. Ignoring it is among the most reliable errors in human judgment. It matters wherever a rare thing is screened for in a large population. That covers cancer screening, airport security, drug testing and fraud detection. A positive result on a good test for a rare condition is usually a false alarm. How a doctor or an official should respond to one depends entirely on the number that most people forget. Screening programs are designed around this arithmetic.

WHAT THIS PASSAGE IS FOR

- Most of the sixty doctors answered ninety-five percent.
 - The right answer is about two percent.
 - Ignoring the base rate is one of the most reliable errors in judgment.
-

LEVEL 2

Developing

mechanism

How it works and how much: cause, comparison, magnitude.

Passage length — 288 to 312 words

Sentences — averaging 16 to 21 words

Vocabulary assumed — the top 22,000 most frequent word forms

Words above that — at most 6, each explained in place

Clue to the reader — Read for the working. These passages do not stop at what happens; they say why it happens and how much. Watch for the sentence that carries the cause, and for the number that gives the size of it.

History and Civics

Passages 51 to 60

51

HIS · S01 · mechanism

The Shape of the Argument

The second paragraph of the Declaration is a piece of reasoning laid out in order, and the order matters more than the famous phrases. It begins with premises, meaning statements offered without proof because the writer expects agreement. They are that all men are created equal, that they hold certain rights which cannot be given away, and that governments exist to secure those rights. From those premises a conclusion follows, and Jefferson draws it immediately. A government that destroys those rights may be altered or abolished by the people who set it up. Nothing in the paragraph is decoration.

Everything after that is evidence for a factual claim rather than argument about principle. The twenty-seven grievances are offered to show that this particular government has in fact destroyed those rights, in named ways, over a stated period. The document also closes off the obvious objection. It states that prudence requires long-established governments not be changed for light and transient causes, and that the colonists had petitioned repeatedly and been answered with repeated injury. The structure is a legal brief: here is the standard, here is the record, here is the remedy.

That shape explains what the document does and does not do. It makes no promise about who shall vote, how the new states shall be governed, or what equality requires in practice, because none of those questions bears on the narrow case being argued. Readers who come to it expecting a constitution find a brief for separation instead. The actual machinery was built eleven years later and ratified in 1788, by men who had the Declaration in front of them and treated it as settled ground rather than as a plan to be carried out. Jefferson owned more than a hundred people at the time of writing, which later readers have had to reckon with.

WHAT THIS PASSAGE IS FOR

- The second paragraph moves from premises to a conclusion in order.
- The grievances are evidence for a factual claim, not argument about principle.
- The document is a brief for separation, not a plan of government.

Madison on Faction

The men who wrote the Constitution were not afraid of kings only. They were at least as worried about majorities, and Federalist 10 is James Madison explaining why. His term for the danger is faction, meaning a group united by a common interest or passion that is adverse to the rights of others or to the general good. Factions, he argues, cannot be prevented, because their causes are sown in the nature of man. Any attempt to remove them would require removing liberty or enforcing a single opinion on everyone.

If the cause cannot be removed, the effects must be controlled, and Madison offers two mechanisms. The first is size. In a small republic one interest can easily become a majority, but across a large and various country a faction must assemble so many different groups that its demands get moderated in the assembling. The second is representation. Power is delegated to elected members rather than exercised directly, which puts a deliberating body between a public mood and a statute. Madison expected election to filter upward toward people of some judgment, and he may have been wrong about that.

The design that follows from this reasoning is the one in the Constitution. Two chambers chosen on different principles, by different electorates, for different terms. A president elected separately from either. Judges appointed rather than elected and holding office for life. Each of these slows a majority down and forces it to persist through more than one season of opinion. Whether the result protects minorities or simply protects property has been argued since the ratification debates, and that argument is taken up at the end of this strand. Federalist 10 was published in a New York newspaper in November 1787 under the name Publius, and was one of eighty-five such essays.

WHAT THIS PASSAGE IS FOR

- Madison treats faction as unpreventable and aims at its effects.
 - His two mechanisms are the size of the republic and representation.
 - Each device forces a majority to persist through more than one season.
-

Four Barriers and Four Amendments

Voting in the United States was never restricted by one rule. It was restricted by four, and each came down separately. Property requirements were the first to go, removed state by state between about 1790 and 1850, so that by the Civil War most white men could vote without owning land. Race was next in law and last in practice: the Fifteenth Amendment of 1870 forbade denial of the vote on racial grounds, and was then evaded for ninety years by devices that did not mention race at all.

Sex was removed by the Nineteenth Amendment in 1920, after campaigners had petitioned, marched and picketed for seventy years. Age came last. The Twenty-Sixth Amendment of 1971 set the voting age at eighteen, and passed in under four months during a war in which eighteen-year-olds were being drafted. A fifth barrier was the poll tax, which was a charge that had to be paid before a person could vote. It was abolished for federal elections by the Twenty-Fourth Amendment in 1964. The Supreme Court struck it down for state elections two years after that.

The pattern in all of this is that a constitutional amendment settles the rule and settles nothing else. Each of these changes required a second round of enforcement machinery, usually decades later, because a state that wishes to exclude voters has many instruments and only one of them is an explicit rule. Literacy tests administered by local officials, residence requirements, registration hours, the location of polling places and the drawing of district lines all survive the removal of the barrier they replaced. The history of the franchise is therefore not a straight line but a series of rules followed by workarounds. Each new amendment also had to be ratified by three quarters of the states, which meant persuading legislatures that had no obligation to listen.

WHAT THIS PASSAGE IS FOR

- Four separate barriers to voting came down at different times.
 - The Fifteenth Amendment was evaded for ninety years by devices that avoided race.
 - An amendment settles the rule and nothing else.
-

Two Arguments About Property

The case against slavery that Douglass and others advocated rested on a single premise and a single move. The premise is that a human being is not a kind of property, because the capacity to reason and to choose puts a person in a different category from a horse or a field. The move is to apply the founding documents literally. If rights are unalienable, meaning incapable of being transferred or surrendered, then no bill of sale can transfer them and no statute can make the transfer valid. Slavery is therefore not merely cruel but void.

The defense of slavery did not answer that argument on its own ground. It shifted the ground. John Calhoun and others argued from order rather than from rights. They held that every settled society rests on some arrangement of unequal labor, and that the southern arrangement was more humane than the wage labor of northern factories. They added that abolition would destroy property worth more than the whole industrial capital of the country. A second line argued from history, claiming that the Constitution had recognized slavery in three clauses and therefore protected it.

Each side was arguing toward a different kind of conclusion. One asked what is owed to a person. The other asked what will hold a society together and what the law already permits. Arguments of that shape cannot be settled by better evidence, because the parties disagree about what counts as evidence in the first place. That is why the dispute moved from the pulpit and the pamphlet to the territories and then to the battlefield, and why the amendment of 1865 was framed as a flat prohibition rather than as a compromise. Douglass had read Calhoun and quoted him, which is why his speeches so often answer a specific sentence rather than a general position.

WHAT THIS PASSAGE IS FOR

- The antislavery case turns on the claim that a person is not property.
 - The defense argued from order and from the existing Constitution.
 - The two sides disagreed about what counts as evidence.
-

How the Retreat Was Managed

The reversal of Reconstruction was not one event, and it was not mainly achieved by changing the law. Four instruments did the work, and they reinforced one another. The first and most immediate instrument was organized violence on a large scale. Armed groups attacked officeholders, teachers and voters through the years around 1870, and federal prosecutions under the enforcement acts fell away almost entirely once the troops had left the region. The second instrument was the courts, which in a series of decisions from 1873 onward read the Fourteenth Amendment narrowly, holding in 1883 that Congress could not reach discrimination by private individuals.

The third instrument was national politics. The disputed presidential election of 1876 was settled by an arrangement under which the remaining federal troops were withdrawn, and after that no administration was willing to spend political capital on enforcement. The fourth was state law, and it proved the most durable. Between 1890 and 1908 southern states enacted new constitutions and statutes imposing literacy tests, poll taxes, property requirements, all-white primaries, which were party contests from which Black voters were excluded, and complicated registration rules.

The combination worked because each part covered a weakness in the others. Violence created the conditions in which the new rules could be written, narrow court rulings removed the federal remedy, and national indifference removed the political cost of doing nothing. None of the new state rules mentioned race, which is what allowed them to survive judicial review for decades. The result can be measured. Black registration in Louisiana fell from over ninety percent of eligible men in 1896 to about one percent by 1904. Nothing in the Constitution had changed in those eight years, and what changed instead was everything around it. The architects of the new constitutions said in public what they were doing, and the convention debates were printed.

WHAT THIS PASSAGE IS FOR

- Four instruments reversed Reconstruction and reinforced each other.
 - The new state rules avoided mentioning race and so survived review.
 - Black registration in Louisiana fell from over ninety percent to about one.
-

Four Ways to Make a Factory Safer

A state that wants fewer factory deaths has four instruments, and they work in quite different ways. The first is a rule with an inspector behind it. A statute imposed a duty, such as keeping exits unlocked, and an inspection service with the right to enter unannounced makes that duty real. The weakness is cost and coverage. New York had about fifty inspectors for tens of thousands of workplaces in 1912, so the chance of a visit in any given year was small.

The second instrument is liability, meaning legal responsibility for the cost of an injury. If an injured worker can sue and win, the employer carries the cost of danger and has a reason to spend on prevention. Nineteenth-century courts blocked this with three doctrines. The first was that a worker who knew of a risk had accepted it. The second was that an employer was not answerable for the carelessness of a fellow servant. The third was that any fault by the worker barred recovery altogether. Workers' compensation laws, which are schedules of automatic payments replacing the lawsuit, were passed by most states between 1910 and 1920.

The third instrument is insurance pricing. Once compensation payments became automatic, insurers had a direct interest in safer plants, and they began to inspect, grade and charge accordingly. The fourth is the hours law, which reduces exposure rather than hazard, since a worker in the building for ten hours instead of fourteen is at risk for less time and is less tired while there. The record suggests that no single instrument did much on its own, and that the sharp falls in industrial death rates came where three or four of them were operating at once. Britain had reached much the same conclusion fifty years earlier, and its factory inspectorate dated from 1833.

WHAT THIS PASSAGE IS FOR

- Rules need an inspection service with the right to enter.
 - Three legal doctrines had blocked injured workers from suing.
 - Automatic compensation gave insurers a direct interest in safer plants.
-

What the 1965 Act Actually Did

Brown v. Board of Education in 1954 declared segregated schooling unconstitutional and provided no machinery at all. Enforcement depended on lawsuits brought district by district, and ten years later fewer than two percent of Black children in the states of the old Confederacy attended schools with white children. A ruling that must be enforced by litigation moves at the speed of litigation, and that speed is set by how many lawyers exist.

The Civil Rights Act of 1964 took a different route. Rather than rest on the Fourteenth Amendment, which reaches only state action, Congress used its power over interstate commerce to reach private businesses, and attached conditions to federal funding for school districts. A district that discriminated could lose its money without anyone filing suit. The Voting Rights Act of 1965 went further and is the clearest case of machinery in the whole period. It suspended literacy tests in covered areas, sent federal examiners, who were officials able to register voters directly where local registrars refused, and required those areas to obtain federal approval before changing any voting rule.

The results separate sharply. Black registration in Mississippi rose from under seven percent in 1964 to about sixty percent in 1967. School desegregation in the same states, driven by funding conditions rather than by courts alone, moved further between 1964 and 1972 than in the whole previous decade. The pattern is consistent. Where a statute created an administrative mechanism that operated without a plaintiff, change prevailed quickly. Where it created a right that had to be claimed in court, change was slow and uneven. The approval requirement in the 1965 Act was narrowed by the Supreme Court in 2013, and several covered states changed their voting rules within weeks. The two statutes are therefore studied together as a lesson about remedies: a right without an administrator is a right that waits.

WHAT THIS PASSAGE IS FOR

- Brown provided no enforcement machinery and depended on lawsuits.
 - The 1964 Act used commerce power and funding conditions instead.
 - Mississippi Black registration rose from under seven percent to about sixty.
-

The Arithmetic of Empire

An empire is an administrative problem before it is anything else. British India was governed by a civil service of about a thousand British officials over a population that passed three hundred million. That was possible only because almost all the work was done by Indian employees and by local rulers who kept their thrones. The army was similar. Most soldiers were Indian, paid out of Indian revenue, and the same revenue paid for the railways, the telegraph and the salaries of the officials.

That arrangement had a condition attached. The colony had to produce a surplus. Land revenue, which was the tax on agricultural land that funded most of the administration, together with the salt tax and customs duties, had to cover the cost of government and leave something over. Two things broke the arithmetic in the twentieth century. The first was the cost of consent. Each concession to nationalist demand, from the councils of 1909 to the provincial governments conceded in 1935, moved spending decisions toward people with no interest in remitting money abroad. The second and decisive pressure was the cost of two world wars fought on borrowed money.

Britain finished the Second World War owing India about thirteen hundred million pounds. The debt was recorded as sterling balances, meaning sums owed in British currency and held in London. By 1945 the colony had become a creditor of the country that governed it. At the same time the British army was stretched across four continents, the Indian officer corps had been largely Indianized, and the police could no longer be relied on in a general strike. Independence in 1947 was neither a gift nor simply a victory. It was the point at which the cost of holding on exceeded any plausible return. France, the Netherlands and Portugal each reached a version of the same reckoning within twenty years.

WHAT THIS PASSAGE IS FOR

- About a thousand British officials governed a population over three hundred million.
 - The arrangement required the colony to produce a surplus.
 - By 1945 Britain owed India about thirteen hundred million pounds.
-

The Machinery of Emergency

Emergency powers in a constitutional state follow a recognizable pattern, and the removal of 1942 used three of its standard instruments. The first is delegation, meaning a grant of broad authority from a legislature to an executive to act in a defined emergency, usually without specifying what will be done, because the point of the grant is speed. Executive Order 9066 handed the army power to define areas and exclude persons from them, and Congress then made violation of such an order a crime.

The second instrument is the suspension of ordinary procedure. Detention without charge, military courts, and the removal of a right to challenge detention before a civilian judge all shorten the distance between a decision and its execution. The third is control of information, and it operates in two directions at once. One is censorship of what the public learns. The other is classification, which is the withholding of the government's own evidence from examination, so that the basis for a measure cannot be tested while the measure is in force.

Those three together produce a characteristic difficulty. The executive holds the facts, the courts defer to the executive on military necessity, and dissent is therefore conducted in the dark. In the Japanese American cases the Supreme Court was told that military necessity required removal, and was not shown the reports already in the government's possession that said otherwise. Powers of this kind are rarely repealed outright. They are allowed to lapse, or narrowed quietly, or left on the books unused, which is why each emergency begins with a stock of instruments inherited from the last one. The order itself was not formally revoked until 1976, thirty-four years after it was signed. Lawyers for the detained men discovered the suppressed reports in government archives in the early 1980s.

WHAT THIS PASSAGE IS FOR

- Emergency power begins with a broad delegation to the executive.
 - Control of information works through censorship and classification together.
 - The courts deferred while the executive held the facts.
-

How a Story Gets Into the Paper

News is not a selection from everything that happened. It is a selection from whatever reached a newsroom in a usable form before a deadline, and that difference explains a great deal. A reporter has a beat, meaning an assigned area such as the courts, the city hall or the police, and the beat delivers a steady supply of material from official sources who are easy to reach and willing to talk. Events that occur where no reporter is assigned arrive late or never.

Three pressures then shape what survives. The deadline rewards whatever can be confirmed quickly, which favors documents and official statements over slow inquiry. Competition rewards whatever rivals do not have, which favors novelty over importance, and a new development in a long-running story over the story itself. The need for a narrative rewards events with actors, a beginning and a visible change, which is why a vote is covered and a trend is not. Nothing in this requires any journalist to be dishonest.

The consequences are measurable. Coverage of crime rises and falls with police practice as much as with crime itself. Famines, slow epidemics and gradual improvements are systematically underreported because they have no event structure. When a government department is abolished there is a story on the day and no story at all about the work that quietly stops. A reader who understands the production process can therefore predict the shape of the gaps, and that is a more useful skill than detecting bias in any single article. Research on American local news finds that the closing of a paper is followed by measurable falls in turnout and in local borrowing costs. The same pressures now operate on a faster clock, because a story that takes six hours to confirm has already been overtaken.

WHAT THIS PASSAGE IS FOR

- News is selected from what reached a newsroom usable and in time.
 - Deadline, competition and narrative each favor a particular kind of event.
 - Slow processes are underreported because they have no event structure.
-

Biology and Earth Science

Passages 61 to 70

61

BIO · S01 · mechanism

Four Conditions and a Clock

Natural selection is not a force and not a tendency. It is an outcome that follows whenever four conditions hold at the same time. There must be variation, meaning that the members of a population differ from one another, and some of that variation must be heritable, meaning passed from parent to offspring through the genes. The variants must then differ in how many offspring they leave, and there must be time, counted in generations rather than in years.

Each condition can be checked, and that is what makes the mechanism testable rather than a story. In the peppered moth the variation was a single difference in color, and it was heritable, because dark moths bred dark offspring. Survival differed because birds hunting by sight took whichever form stood out against the bark. The generation time is one year, so fifty years allowed fifty rounds. Change the bark and the same four conditions produce the opposite result, which is exactly what the counts recorded after the clean air laws.

Two common misreadings are worth naming. The first is that individuals adapt. They do not, because a pale moth cannot darken and nothing in its lifetime changes its color. What changes across generations is the proportion of each form in the population. The second is that selection makes things better in general. It makes a population fit the conditions it is actually in, which is why a variant that lets a population thrive under soot becomes a liability once the soot is gone. The mechanism has no memory and no aim. It is simply arithmetic applied to who leaves offspring. Breeders had been using the same arithmetic deliberately for centuries, which is where Darwin got his clearest evidence. He devoted the opening chapter of his book to pigeons and cabbages rather than to wild species.

WHAT THIS PASSAGE IS FOR

- Selection follows whenever four conditions hold together.
 - Individuals do not adapt; proportions in a population change.
 - Selection fits a population to current conditions only.
-

Two Copies and a Shuffle

The three to one ratio that Mendel counted follows from three facts, none of which he could see. First, a pea plant carries two copies of each gene, one from each parent. Second, the two copies may be different versions, and each version is called an allele, meaning one of the alternative versions of a gene. Third, when the plant makes pollen or egg cells, the two copies separate and each cell receives one of them at random.

Follow a cross with that in hand. A true-breeding tall plant carries two tall alleles and a short plant carries two short. Their offspring must receive one of each, so every plant in the first generation carries one of both. All of them are tall, because the tall allele is dominant, meaning the version that shows whenever it is present. The short allele is still there and still intact. When that generation breeds among itself, each parent contributes either a tall or a short allele with equal chance, which gives four equally likely combinations. Three of the four contain at least one tall allele.

That is the whole derivation, and it predicts more than the ratio. It predicts that about one third of the tall plants in the second generation will breed true and two thirds will not, which Mendel also tested and confirmed. It predicts that two features carried on different chromosomes will emerge in all four combinations in fixed proportions. It fails, usefully, for features carried close together on the same chromosome, and that failure is how the first gene maps were later drawn. Mendel had no word for a gene and no knowledge of chromosomes, both of which arrived decades later. What he had was a counting habit and seven features that behaved simply.

WHAT THIS PASSAGE IS FOR

- Each plant carries two copies and passes on one at random.
 - The first generation is uniform because one allele is dominant.
 - The derivation predicts more than the ratio and fails usefully.
-

With Oxygen and Without

A cell does not use sugar directly. It converts the energy in sugar into a molecule called ATP, which is the working currency of energy inside every living thing, and then spends the ATP on whatever needs doing. The question for any cell is how many units of that currency a single sugar molecule will yield, and the answer depends entirely on whether oxygen is available.

The first stage is the same either way. Glycolysis, which is the step that splits the sugar in half, happens in the main body of the cell and nets two units of ATP. Without oxygen that is where the accounting stops. The half-molecules must be disposed of, and yeast disposes of them as alcohol while a working human muscle disposes of them as lactate. Either way the cell keeps its two units and throws away most of the chemical energy it was handed.

With oxygen the halves are taken into the mitochondria and dismantled completely to carbon dioxide and water. That path yields something in the region of thirty units rather than two. The difference explains a great deal of biology. It explains why a sprinter can run flat out for twenty seconds and not for twenty minutes, since the fast path cannot sustain the pace. It explains why yeast in a sealed vessel grows slowly. It also explains why oxygen-breathing life could support large active animals at all, because fifteen times the yield from the same food is the margin that pays for muscle, nerve and a warm body. The figure of about thirty units is an estimate rather than a constant, and textbooks have revised it downward over the years. It depends on how much energy the cell spends moving materials across the mitochondrial membrane.

WHAT THIS PASSAGE IS FOR

- A cell spends ATP, not sugar.
 - The first stage nets two units whether or not oxygen is present.
 - The oxygen path yields roughly fifteen times as much.
-

Why Wolves Are Rare

Every food web has a shape, and the shape follows from one arithmetical fact. Energy is lost at every transfer. A plant captures a small fraction of the sunlight falling on it. An animal that eats the plant uses most of what it takes in on moving, breathing and staying warm, and stores only a fraction as its own tissue. The rough figure used in textbooks is ten percent, meaning that about a tenth of the energy at one trophic level, which is a step in the chain of who eats whom, becomes available at the next.

Run that forward and the consequences are strict. If a hundred units of plant material supports ten units of grazing animal, then ten units of grazer supports one unit of predator. The biomass, meaning the total weight of living material, therefore falls by roughly a factor of ten at each step. That is why grass is abundant, deer are common, wolves are rare, and animals that eat wolves effectively do not exist. Four or five steps is the practical limit for a food chain anywhere on Earth.

The same arithmetic explains the Yellowstone observations without any appeal to design. A small number of wolves can affect a large quantity of willow, because each wolf represents a great deal of elk, and each elk represents a great deal of plant tissue. The leverage runs downward through the chain. It also explains why top predators are the first species lost when a habitat shrinks, since they need the largest area to support the smallest number of animals. The ten percent figure is a convenience rather than a law, and measured transfer efficiencies range from about one percent to over thirty. Cold-blooded animals do better than warm-blooded ones, because they spend nothing on holding a temperature.

WHAT THIS PASSAGE IS FOR

- Energy is lost at every transfer in a food web.
 - Biomass falls by roughly a factor of ten at each step.
 - Four or five steps is the practical limit for a food chain.
-

The Ceiling and the Overshoot

Two curves describe almost all population growth. The first is exponential growth, meaning growth in which the increase is proportional to the number already present. Thirty reindeer producing offspring at a steady rate give sixty, then a hundred and twenty, and the absolute increase gets larger every year although the rate per animal has not changed. Plotted on paper the line curves upward and keeps curving.

Nothing grows that way for long, because something eventually runs short. The second curve, the logistic, bends over and flattens at a level called the carrying capacity, which is the number the habitat can support indefinitely. As food gets scarce, births fall or deaths rise, and the population settles near that level. This is the textbook picture, and the textbook picture has a flaw: it assumes the population feels the shortage immediately.

Real populations respond late, and the delay is what killed the reindeer on St Matthew Island. A reindeer born in a good year is large, healthy and fertile, and goes on breeding for several years afterward on the strength of that start. The herd therefore keeps growing after the lichen has begun to fail, passes its own ceiling, and eats into the stock that the ceiling depended on. The crash then falls below where the population would have settled, because the resource itself has been damaged. Populations with delays of this kind do not settle. They fluctuate, and the longer the delay, the wilder the swing. The same arithmetic applies to any renewable resource harvested by anything, which is why fishery managers argue about lags. A quota set from last season's catch data is a quota set on a delay. Fishery scientists call the resulting pattern a boom and bust, and they can produce it in a tank with water fleas.

WHAT THIS PASSAGE IS FOR

- Exponential growth increases in proportion to the number present.
 - The logistic curve flattens at the carrying capacity.
 - Delay lets a population overshoot and damage its own ceiling.
-

Dose, Route and the Number R

An outbreak is governed by a small number of quantities, and public health work consists largely of measuring them and then changing one. The first is the route. Cholera travels in water, influenza in air and on hands, malaria in mosquitoes, and each route suggests a different intervention: a pump handle, a mask, a net. The second is dose, since most infections require more than one organism to establish, which is why dilution matters and why a leaking cesspit beside a well is worse than one at a distance.

The third quantity is the reproduction number, meaning the average number of new cases that one case produces in a population with no immunity. If that number is above one the outbreak grows, and if it is below one the outbreak dies out. Nothing else about the disease changes that threshold. Measles has a reproduction number in the range of twelve to eighteen, which is why it needs very high vaccination coverage, while influenza is usually between one and two.

The fourth is the incubation period, meaning the gap between infection and the first symptom, and it decides whether contact tracing can work at all. A long incubation gives time to find and isolate contacts before they infect others. A short one, or an infection that spreads before symptoms appear, lets cases disperse through a population faster than any tracing team can follow. John Snow could act on a map in 1854 because cholera is waterborne and the source was fixed in one place. An airborne infection with a two-day incubation offers no equivalent handle. Each of these numbers is estimated from data collected during an outbreak, which means the early estimates are the worst ones. They are also local: a reproduction number is a property of a disease in a particular population, not of the organism alone.

WHAT THIS PASSAGE IS FOR

- Each route of transmission suggests a different intervention.
 - An outbreak grows if the reproduction number is above one.
 - A short incubation period defeats contact tracing.
-

Four Steps in the Nitrogen Cycle

Nitrogen is four fifths of the air and almost useless in that form. The two atoms in a nitrogen molecule are held together so tightly that very few living things can break them apart. Fixation, which is the conversion of nitrogen gas into a form that living things can use, is therefore the gate through which all biological nitrogen must pass. In nature it is done by bacteria, some free in the soil and some in the root nodules of clover and its relatives, and in small amounts by lightning.

Once fixed, the nitrogen moves. Plants take it up through their roots as nitrate or ammonium and build it into protein. Animals eat the plants. Both eventually die, and other bacteria break the remains back down, releasing nitrogen into the soil again. A third group of bacteria completes the circle by turning soil nitrogen back into gas, which returns to the atmosphere. The cycle is closed, but a field is not a closed system.

Three leaks matter to anyone growing a crop. The harvest removes nitrogen in the grain and takes it away in a truck. Leaching, which is the washing of dissolved nitrogen down below the root zone by rain, removes more, and nitrate is highly soluble. Gas losses take a further share. A field cropped year after year without return is therefore depleted at a rate that can be estimated, and the Norfolk rotation worked because clover and grazing animals put back roughly what the wheat took out. Since 1913 the Haber process has fixed nitrogen industrially, using high pressure, high temperature and a great deal of gas. Something close to half the nitrogen in a person alive today passed through that process. Soil tests now measure what is there before a field is dressed.

WHAT THIS PASSAGE IS FOR

- Nitrogen gas must be fixed before anything living can use it.
 - Harvest, leaching and gas losses are the three leaks from a field.
 - The rotation worked because clover replaced what wheat removed.
-

Why the Gas Holds Heat

Sunlight arrives mostly as visible light, passes through the atmosphere with little absorption, and warms the ground. The ground then radiates that energy back upward, but at much longer wavelengths, in the infrared, which is the band at which any warm surface radiates. The atmosphere is not transparent to those wavelengths. Water vapor and carbon dioxide absorb strongly in parts of the infrared band, take up the energy and re-emit it in all directions, including downward.

The mechanism is therefore not a blanket and not a trap. It is a delay. Energy still leaves the planet, and must, or the temperature would rise without limit. What changes when the concentration of an absorbing gas rises is the height in the atmosphere from which the escaping radiation finally gets away, and therefore the surface temperature needed to balance the books. The effect of each added unit is smaller than the last, because the strongest parts of the band are already saturated, and the useful absorption happens at the edges.

Two further quantities decide how much this matters. The first is residence time, meaning how long a molecule may inhabit the air before being taken up elsewhere. Methane breaks down in about a decade. Carbon dioxide is partly absorbed by the ocean within years, but a substantial fraction stays in circulation for centuries, which is why the Mauna Loa curve has no downward step after any recession. The second is feedback: a warmer atmosphere holds more water vapor, and water vapor is itself a strong absorber in the infrared. John Tyndall measured the absorption of several gases in a laboratory in 1859 and identified water vapor and carbon dioxide as the important ones. His apparatus was a brass tube, a heat source and a sensitive thermopile.

WHAT THIS PASSAGE IS FOR

- The ground radiates in the infrared, where the atmosphere is not transparent.
 - The mechanism is a delay in escape, not a trap.
 - A substantial fraction of carbon dioxide stays in circulation for centuries.
-

What Moving Plates Produce

The surface of the Earth is broken into about a dozen large plates and several dozen small ones, and every one of them is moving, at a speed of between one and ten centimeters a year. Since the plates cannot overlap and cannot leave gaps, everything interesting happens at their edges, and there are only three kinds of edge.

Where two plates move apart, as at Thingvellir, magma, which is molten rock from below, rises into the gap and freezes onto both sides. That builds new crust and produces ridges, rift valleys and shallow earthquakes. Where two plates move toward each other, one of them usually goes down beneath the other, a process called subduction. The descending plate drags the sea floor into a deep trench, melts at depth, and feeds a line of volcanoes above it, which is the origin of the Andes, Japan and the Cascades. Where two plates slide past one another, as in California, nothing is made or destroyed and the motion is released in earthquakes along a fault.

The three kinds of edge account for the distribution of nearly every mountain range, volcano and major earthquake on the planet. They also set up a race. Collision lifts rock upward at a few millimeters a year, while rain, ice and rivers erode it downward, and the height of a range is the balance between the two rates. The Himalaya is still rising because the collision has not finished. The Appalachians, whose collision ended long ago, are the worn remains of something that once stood as high. Satellite positioning now measures plate motion directly. The precision is about a millimeter a year. Before that it was inferred from magnetic stripes on the sea floor and from the ages of volcanic islands.

WHAT THIS PASSAGE IS FOR

- There are only three kinds of plate edge.
 - Subduction feeds a line of volcanoes above the descending plate.
 - The height of a range is a balance between uplift and erosion.
-

Why Warm Water Does the Work

Water is a far better carrier of heat than air, which is why a swimmer feels cold at a temperature that feels mild on land. The same comparison explains why Antarctic ice is being lost from below rather than from above. Melting ice also absorbs a great deal of energy without changing temperature at all, a quantity called latent heat, so a current only two degrees above freezing can remove an enormous amount of ice if it keeps arriving.

Around West Antarctica a mass of relatively warm salty water sits at depth, below a colder and fresher surface layer. Wind patterns determine how much of it is pushed up onto the continental shelf and into the cavities beneath the floating ice. Once there it melts the underside, and the fresh melt water, being lighter, rises along the sloping ice and draws more warm water in behind it. The circulation is self-sustaining while the warm water continues to migrate shoreward.

The sea bed decides how this ends. Under Thwaites the rock surface deepens as it goes inland, which is called a retrograde slope. On such a slope every retreat of the grounding line puts the ice front into deeper water, where the ice is thicker, so more ice is exposed to melting and the retreat accelerates. On a slope of the opposite sense the retreat would slow itself and stop. That single geometrical fact is why Thwaites is studied more closely than glaciers that are losing ice faster today. The shape of the bed under the ice was mapped by aircraft carrying ice-penetrating radar, flown in grids over several seasons. Gaps in that map are still the largest source of uncertainty in the projections. The grounding line under Thwaites has about eighty miles of such slope ahead of it.

WHAT THIS PASSAGE IS FOR

- Water carries heat far better than air.
 - Melt water rising along the ice draws more warm water in.
 - A bed that deepens inland makes retreat accelerate.
-

Physical Science

Passages 71 to 80

71

PHY · S01 · mechanism

Scatter and Bias

The spread in eleven weighings comes from two quite different sources, and confusing them is the commonest mistake in handling data. Random error is scatter that falls on both sides of the true value, caused by air movement, vibration and the limits of the instrument. Systematic error is an error that pushes every reading the same way, caused by a balance that reads high, a ruler that has shrunk, or a stopwatch started late by the same half second every time.

Averaging repairs the first and is helpless against the second. Take enough readings and the random part shrinks in proportion to the square root of the number of readings, so a hundred weighings are ten times better than one. Nothing about repetition touches a balance that reads two milligrams high, because that error is present and identical in every reading, including the average. A thousand careful measurements of the wrong thing give a very precise wrong answer.

The two therefore require different remedies. Random error is reduced by repeating, by steadying the apparatus, and by using a finer instrument. Systematic error is found only by comparison with something independent, such as a second balance, a mass certified by the National Institute of Standards and Technology, or a different method altogether. Experimenters refine a result by attacking both, and they report them separately, which is why a careful paper distinguishes a statistical uncertainty from a systematic one. When two laboratories disagree by more than their stated uncertainties, at least one of them has a systematic error it has not found. The distinction has a famous example. Measurements of the speed of light published between 1900 and 1950 drifted steadily, each new value sitting inside the error bars of the last and all of them wrong in the same direction. Later workers concluded that experimenters had unconsciously weighted their results toward the accepted figure.

WHAT THIS PASSAGE IS FOR

- Random error scatters both ways; systematic error pushes one way.
 - Averaging shrinks random error with the square root of the number of readings.
 - Systematic error is found only by independent comparison.
-

Why a Truck Takes Longer to Stop

A loaded truck and a small car moving at the same speed on the same road will not stop in the same distance. The reason is a chain of three steps rather than a single fact, and Isaac Newton supplied all three of them. The first step is that stopping requires a force, and the only force available is friction, which is the force between the tires and the road surface. That force is roughly proportional to the weight pressing down, so a heavy vehicle does get more braking force than a light one.

The second step undoes the first. A heavier vehicle also carries more kinetic energy, which is the energy an object has because it is moving, and that energy must all be removed as heat in the brakes and tires. Mass appears on both sides of the problem and largely cancels. What does not cancel is speed, because kinetic energy rises with the square of the speed. Double the speed and the stopping distance roughly quadruples, which is why the braking distance at seventy miles an hour is about four times that at thirty-five.

The third step is the one drivers forget. Before any braking happens the vehicle travels for the time it takes to notice and react, usually near three quarters of a second. In that time it covers a distance proportional to the speed, not to its square. The two parts add. A truck loses further because of load transfer, brake heating on long descents, and tires that cannot accelerate sideways while already working hard to slow down. Stopping distance tables in driving manuals are built from exactly this arithmetic and then rounded. They assume a dry road and a fit driver, and both assumptions fail regularly.

WHAT THIS PASSAGE IS FOR

- Friction is the only force available to stop a vehicle.
 - Mass appears on both sides of the problem and largely cancels.
 - Stopping distance rises with the square of the speed.
-

Where the Energy Goes

Energy accounting rests on one rule, the conservation of energy, which states that energy is never created or destroyed but only moved or converted. That rule makes every machine auditable. Whatever goes in must come out somewhere, and the only question is how much of it comes out in the form that was wanted. The fraction that does is the efficiency, meaning the share of the energy put in that emerges in the useful form.

Multiply the steps and the losses compound. In a coal power station the chemical energy of the fuel becomes heat. The heat becomes steam, the steam turns a turbine, and the turbine drives a generator. Each stage has its own efficiency, and the biggest loss is unavoidable. A heat engine working between a boiler and a cooling tower cannot convert more than a limited share of its heat into motion, and that limit depends only on the two temperatures. Modern coal plants yield about forty percent of the fuel energy as electricity and discard the rest as warm water and gas.

Transmission loses a few percent more, and then the appliance has its own efficiency. The whole chain therefore multiplies rather than adds. An electric heater is close to perfect at its own step, since all the electricity becomes heat. But if that electricity came from a coal station, it represents more than twice the fuel that burning the coal in the room would have needed. Comparing two technologies by their last step alone is the standard way of getting this wrong. The theoretical limit on a heat engine was worked out by Sadi Carnot in 1824, before anyone had a clear idea of what heat was. His result still sets the ceiling for every power station built since.

WHAT THIS PASSAGE IS FOR

- Conservation of energy makes every machine auditable.
 - A heat engine is limited by its two temperatures alone.
 - Efficiencies multiply along a chain rather than adding.
-

Three Quantities, One Relation

The behavior of a gas is governed by a single relation among three quantities. Pressure multiplied by volume is proportional to the absolute temperature, which is temperature counted from the coldest possible point rather than from the freezing point of water. Fix any two of the three and the third follows. That one sentence predicts a surprising amount, provided the gas is not close to becoming a liquid.

Work through the cases. Hold the temperature steady and compress the gas into half the volume, and the pressure doubles: that is a bicycle pump with the valve shut. Hold the volume steady and warm the gas, and the pressure rises: that is why a sealed container in a fire bursts, and why a tire gauge reads higher after a long drive. Hold the pressure steady and warm the gas, and it expands: that is a hot air balloon, where warming the air inside by about eighty degrees makes it light enough to lift a basket.

The fourth case is the useful one for weather. Air pushed up a mountainside arrives at a height where the surrounding pressure is lower, expands, does work on the air around it, and cools without losing any heat to anything. The rate is about ten degrees for every thousand meters of dry ascent. That is why mountain tops are cold, why cloud forms on the windward slope, and why the air coming down the far side arrives warm and dry. The relation was assembled from three separate laws found by Boyle, Charles and Gay-Lussac over about a century and a half. It holds well for air at ordinary pressures and fails badly for steam near its boiling point. Weather forecasters apply the same relation several million times a day inside their models.

WHAT THIS PASSAGE IS FOR

- Pressure times volume is proportional to absolute temperature.
 - A sealed container in a fire bursts because volume is fixed.
 - Rising air expands and cools without losing heat to anything.
-

Why the Thin Wire Gets Hot

Four quantities describe any circuit, and two relations connect them. Voltage is the push. Current is the rate of flow. Resistance is the opposition a conductor offers to that flow, and it depends on the material, the length and the thickness of the wire. The first relation says that current equals voltage divided by resistance. The second says that power, which is the rate at which energy is delivered or lost, equals current multiplied by voltage.

Combining them gives the fact that matters in a wall. The power lost as heat in a wire is the current squared multiplied by the resistance of that wire. Current squared is the harsh term. Double the current and the heating in the cable goes up fourfold, which is why an overloaded circuit does not warm gently but runs away. A thin wire makes this worse twice over, because halving the cross-section roughly doubles the resistance.

The same arithmetic explains why electricity is sent across country at high voltage. To deliver a given amount of power, a higher voltage needs a smaller current, and a smaller current loses far less in the line. A transmission line at four hundred thousand volts carries the output of a power station with losses of a few percent, where the same power at household voltage would be gone within a mile. Thick copper conducts well, but no practical thickness of copper can beat the arithmetic of raising the voltage and lowering the current. Transformers make the voltage changes possible, and they work only with alternating current. That is the technical reason the alternating system won the argument in the 1890s, against direct current and against Edison. Household wiring is therefore sized by the current it must carry, and the table of sizes is set by law.

WHAT THIS PASSAGE IS FOR

- Current equals voltage divided by resistance.
 - Heating in a wire rises with the square of the current.
 - High voltage means low current and small losses.
-

One Rule, Three Effects

Reflection, refraction and diffraction look like three subjects and are one. Each follows from the behavior of a wavefront, which is the moving line along which a wave has the same phase. One simple rule does the work: every point on such a line acts as a source of the next ripple. Work out where the ripples overlap and the three effects fall out of the same calculation.

Where a wavefront meets a flat barrier the ripples add up along a path that leaves at the same angle it arrived, which is reflection. Where the front crosses into a material in which the wave moves more slowly, the side that enters first falls behind, so the whole front swings. That is refraction. Where the front meets an edge or a narrow opening, the ripples from the surviving points spread into the space behind, which is diffraction, meaning the spreading of a wave after it passes an edge or a gap.

The amount of spreading depends on the size of the gap compared with the wavelength, and that comparison explains a familiar asymmetry. Sound has wavelengths of about a meter, so it bends round a doorway and a voice can be heard from the next room. Visible light has wavelengths under a thousandth of a millimeter, so it casts sharp shadows instead. A telescope mirror cannot resolve detail finer than diffraction allows, and that limit, not the quality of the glass, is what eventually distorts the image of a distant star into a small disc. Christiaan Huygens published the rule about every point acting as a source in 1690. It was ignored for a century in favor of a particle account, and then revived when interference was demonstrated. The same limit applies to a camera lens and to the human eye.

WHAT THIS PASSAGE IS FOR

- All three effects follow from one rule about wavefronts.
 - Diffraction depends on the gap compared with the wavelength.
 - Diffraction, not glass quality, sets the limit on a telescope.
-

Barriers, Rates and Surfaces

A reaction that releases energy overall does not necessarily happen. Paper in air at room temperature sits next to a reaction that would release a great deal of heat, and it sits there for centuries. Svante Arrhenius put the matter on a quantitative footing in 1889. The reason is the activation energy, meaning the push a reaction needs before it will start even when the overall balance is favorable. Bonds must be broken before new ones can form, and breaking them costs energy that has to be supplied first.

That single idea explains the whole set of levers available for changing a rate. Heat supplies the push, so raising the temperature increases the proportion of molecules carrying enough energy to get over the barrier, which is why a rough rule of thumb has reaction rates roughly doubling for every ten degrees. Concentration raises the number of collisions. Surface area raises the number of places where collisions can happen, which is the whole difference between steel wool and a steel bar. A catalyst, which is a substance that lowers the barrier without being consumed, provides an easier route over it.

Dissolving a solid is a useful case to think through, because it is governed by the same accounting. A sugar cube and the same sugar ground to powder dissolve at very different rates and reach the same end point. Stirring removes the saturated layer at the surface and lets fresh water reach the solid. Hot water supplies more energy per collision. None of these change how much sugar will eventually dissolve, which is set by temperature alone. Rate and extent are separate questions, and confusing them is the usual error. Catalysts are the reason modern chemical industry exists at all, since most useful reactions are far too slow unaided.

WHAT THIS PASSAGE IS FOR

- A favorable reaction can still be slow because of a barrier.
 - Heat, concentration, surface and catalysts all act on the same barrier.
 - Rate and extent are separate questions.
-

Four Numbers That Decide a Design

An engineer choosing a material needs four numbers, and the Tay Bridge failed because two of them were not considered. The first is stress, which is the force carried by each unit of area, and it is the reason a thin rod and a thick one made of the same steel behave differently under the same load. The second is the yield point, meaning the load beyond which the material keeps the shape it is bent into rather than springing back.

The third is the ultimate strength, the stress at which the material finally parts. The gap between the yield point and that final figure is where the safety of a structure actually lives. A material with a wide gap bends, sags visibly, and gives warning before it breaks. A material with almost no gap, such as cast iron, holds its shape until the moment it shatters, and offers no warning at all. The fourth number is toughness, which measures how much energy a material absorbs before fracture, and it falls sharply in some steels as the temperature drops.

Those four explain most failures. A structure that bends and stays bent has yielded and can often be repaired. A structure of brittle material under a sudden sideways load can collapse in seconds from a crack that nobody could have seen. Cold makes some steels brittle, which is why several wartime cargo ships broke in half in the North Atlantic while sister ships in warm water did not. The choice of material is a choice about what kind of failure is acceptable. Modern practice therefore prefers materials that give warning, and designs load paths with more than one route. A bridge with a single critical member is a bridge with no margin, whatever the arithmetic says.

WHAT THIS PASSAGE IS FOR

- Stress is force divided by area, not force alone.
 - The gap between yield and fracture is where safety lives.
 - Brittle material gives no warning before it shatters.
-

Half-Life, Dose and Distance

Radiation exposure is governed by three numbers, and the women at the Orange factory were unlucky in all three. The first is half-life, meaning the time in which half of a quantity of a radioactive element decays away. Radium has a half-life of about sixteen hundred years, so a quantity of it in a body will emit radiation at a nearly constant rate for the whole of a human life. Elements with short half-lives are intensely active and gone quickly, which is a different and often more manageable problem.

The second is the kind of radiation. Alpha particles are heavy and stopped by a sheet of paper or the outer layer of skin, so an alpha emitter outside the body is almost harmless. Inside the body, with no barrier at all between the source and living tissue, the same particles deposit all their energy within a fraction of a millimeter. Radium is an alpha emitter. Paint on a brush that is shaped with the lips crosses from the harmless case to the worst one.

The third is dose, meaning the energy absorbed per unit of body mass, and it falls off with distance and with shielding. A technician standing two meters from a source receives a quarter of the dose received at one meter, because the intensity falls with the square of the distance. Lead absorbs most of what reaches it. None of these three remedies is available for a source that the body has built into bone, which is why internal emitters are treated as a separate class of hazard with its own limits. The modern unit of dose is the sievert, and a typical annual exposure from natural sources is about two or three thousandths of one. Occupational limits are set at twenty thousandths a year, averaged over five years.

WHAT THIS PASSAGE IS FOR

- Radium has a half-life of about sixteen hundred years.
 - An alpha emitter is nearly harmless outside the body and dangerous inside it.
 - Dose falls with the square of the distance.
-

The Ladder of Distance

Nobody has ever measured the distance to a star directly, by any means at all. No tape reaches that far. Astronomical distances come instead from a ladder of methods, each rung calibrated by the one below it. The structure is only as sound as its lowest steps, and those steps are plain geometry of a kind any surveyor would recognize at once. Every figure printed in an astronomy textbook rests on that chain.

The first rung is geometry. Parallax, which is the shift in a star's position as the Earth moves from one side of its orbit to the other, gives a distance by triangulation, exactly as a surveyor uses a baseline. The baseline is the width of the Earth's orbit, about three hundred million kilometers, and even so the angles are tiny. The nearest star shifts by less than one second of arc, which is the width of a coin seen from three miles. Ground telescopes manage parallax out to a few hundred light years, and a satellite has now pushed it to tens of thousands.

Beyond that, brightness takes over. If an object's true brightness is known, its apparent dimness gives its distance, because light spreads out with the square of distance. Such an object is called a standard candle. Certain pulsating stars and certain exploding stars both serve. Their true brightness is fixed by finding examples close enough to have a parallax, or in clusters densely enough populated to be measured another way. Each rung is tied to the one below, so an error low on the ladder moves every distance in the universe above it. The ladder now reaches billions of light years. Edwin Hubble used that chain in 1924 to establish that other galaxies lie outside our own.

WHAT THIS PASSAGE IS FOR

- Astronomical distances come from a ladder of methods.
 - Parallax is triangulation from the width of the Earth's orbit.
 - An error low on the ladder moves every distance above it.
-

Humanities

Passages 81 to 90

81

HUM · S01 · mechanism

What Each Position Can Report

A narrator is defined by access rather than by personality, and there are three standard positions. A first-person narrator reports only what one character perceives, remembers or guesses. A third-person limited narrator stands outside but stays with one mind at a time, so the reader learns what that character notices and nothing more. An omniscient narrator, meaning one with access to every mind in the book, may report what two people think at the same moment and may also comment on both.

Each position decides what kind of surprise is available. Deception works easily in the first person, because the reader is inside the deceived mind. Irony works easily in the omniscient position, because the voice can show a character being wrong in a sentence the character would endorse. Suspense of the simplest kind, in which the reader fears something the character does not suspect, is almost impossible in strict first person and routine in the others.

Nineteenth-century novelists developed a fourth option which is now the commonest of all. In free indirect style, meaning a third-person voice that slides into a character's way of thinking without announcing it, the sentence carries the character's vocabulary and judgments while staying grammatically outside. Jane Austen uses it constantly, and so does almost every modern novel. It lets a writer convey a mind from the inside and withdraw to judge it in the next clause. Recognizing the shift is a skill worth practicing, because the whole effect depends on the reader not quite noticing where the voice changed. An examiner asking about tone is usually asking about exactly that movement. The test is simple: ask whether the sentence could be spoken by the character, and whether it could be endorsed by the author.

WHAT THIS PASSAGE IS FOR

- A narrator is defined by access rather than by personality.
 - Each position decides what kind of surprise is possible.
 - Free indirect style carries a character's judgments from grammatically outside.
-

Four Ways to Build a Motive

A reader infers motive from four kinds of evidence, and a writer chooses how much of each to supply. The first is exposition, meaning direct statement of background by the narrator: this man lost a brother at sea and has not spoken of it since.

Exposition is efficient and expensive, because a reader who is told why a character acts has nothing left to work out. The second is action, including the small choices that nobody in the story remarks on, such as which of two chairs a visitor takes.

The third is speech, and it is less reliable than it looks. Characters lie, boast, and tell themselves stories, so a stated motive is evidence about the speaker rather than about the cause. The fourth is interiority, which is direct access to what a character is thinking, available only in positions that allow it. Each kind can also be withheld, and withholding is itself a device. Melville gives *Bartleby* action and speech and no interiority at all.

The effects differ in a way worth naming. Motive supplied by exposition is settled and inert. Motive built from action and silence stays open, and readers argue about it for a century, which is a kind of value an author may prefer to certainty. The risk is obvious: a character whose motive is too subtle to recover reads as arbitrary rather than deep. Writers manage that risk with pattern, by repeating a gesture or an avoidance often enough that a reader can see a shape without being handed a reason. Henry James described the technique as a matter of what to leave in shadow. Chekhov said that an author should not solve problems but state them correctly. A reader who cannot say what a character wants is usually reading a book that has decided not to say.

WHAT THIS PASSAGE IS FOR

- Motive comes from exposition, action, speech and interiority.
 - A stated motive is evidence about the speaker.
 - Pattern is what keeps a withheld motive from reading as arbitrary.
-

Three Moves With Words

Figurative language is usually taught as a list of names, which hides the fact that the figures do different work. A simile, meaning a comparison that announces itself with the word like or as, keeps the two things separate: the reader is told to hold a hand and a leaf side by side and notice a resemblance. A metaphor removes the announcement and asserts the identity, so that the hand is the leaf, and the reader must do the work of deciding in what respect.

Metonymy is a third operation and a different one. It names a thing by something associated with it rather than something like it, as when the press means journalists, or the crown means a monarchy, or Washington means a government. Nothing is being compared. The reader is being pointed at a whole through one of its parts or attachments, which is why metonymy feels like plain speech while metaphor feels like writing. Most newspaper prose is thick with metonymy and nobody calls it poetic.

What all three share is economy. An image that is vivid makes the reader supply the detail, so four words do the work of a paragraph. That is also where they fail. An image that points in two directions at once confuses rather than compresses, and a metaphor repeated until nobody notices it, as in the heart of the matter, has stopped doing any work at all. Such expressions are dead metaphors, and the usual advice is to use a live one or a plain word, not something halfway. Orwell made the test practical in 1946: can the writer see the image. A writer who says a thing is a hotbed of unrest is not seeing a bed, and the sentence has gone to sleep.

WHAT THIS PASSAGE IS FOR

- A simile keeps the two things separate; a metaphor asserts identity.
 - Metonymy names a thing by association, not by resemblance.
 - All three work by economy and fail by confusion.
-

Expectation and the Broken Line

A fixed form works by setting up an expectation that can then be used. Once a reader has heard three lines of ten syllables, the fourth is predicted, and a poet who delivers eleven has done something the reader feels before being able to name it. The effect does not require the reader to be counting consciously, any more than a listener counts the beats in a dance tune.

Two devices exploit the line directly. Enjambment, meaning the running of a sentence past the end of a line without a pause, sets the grammar against the rhythm, so that the eye arrives at the line break while the sense keeps going. That produces a small suspension, and poets use it to hang a word in the air. A caesura, which is a strong break inside a line, does the opposite, stopping the voice where the meter says to continue. Both only work because the underlying pattern is regular enough to be felt.

This is why metrical substitution carries weight. A line that is iambic throughout and then reverses one foot puts a stress where none was expected, and the reversed foot is almost always on a word the poet wants struck. Shakespeare opens a sonnet with *Shall I compare thee*, which is regular, and elsewhere opens with a reversed foot that lands like a hand on a table. A lyrical effect of this kind is produced by arithmetic. The reader hears the feeling and the poet counted. Free verse does not escape the mechanism, because a reader brings expectations to any line. What it does is make the poet establish the pattern inside the poem rather than inherit one. Reading a line aloud twice, once flat and once with the stresses, is the quickest way to hear it.

WHAT THIS PASSAGE IS FOR

- A fixed form works by creating an expectation that can be used.
 - Enjambment sets the grammar against the rhythm.
 - A reversed foot lands on a word the poet wants struck.
-

How a Building Writes a Play

Several habits of Elizabethan playwriting are consequences of the building rather than choices of taste. The audience stood in daylight on three sides of a platform, so an actor could speak directly to them without breaking anything. That is why the aside, meaning a line spoken to the audience and not to the other characters, is a standard device rather than a trick. A dark auditorium with a lit stage makes the same line awkward, which is why the aside nearly disappeared in the nineteenth century and returned when playhouses went back to open staging.

Because there was no scenery, place had to be established in the dialogue, and Shakespeare does it in the first few lines of almost every scene. Because there was no front curtain, a scene could not end with a held tableau and had to end with an exit. That is why so many scenes finish on a rhyming couplet that tells the audience the stage is clearing. Because the stage had a discovery space, which is a curtained opening at the back, a body or a sleeper or a statue could be revealed rather than carried on.

The conditions also set the length of a speech. An actor in the open air with two thousand people standing has to hold attention by voice alone, and the verse renders that possible by giving the voice a pattern to lean on. Prose in these plays is used for different registers and rarely sustains a long solo turn. None of this makes the plays less artful. It locates the art in a particular room. Modern productions indoors have to solve problems that the original staging did not have. Cutting an aside, or lighting a soliloquy as private thought, changes the relationship the line was written for.

WHAT THIS PASSAGE IS FOR

- Daylight on three sides made the aside a standard device.
 - With no scenery, place had to be established in dialogue.
 - Verse gave the voice a pattern to lean on in the open air.
-

The Uses of a Rule

A convention, meaning a rule that a kind of story is expected to follow, does two jobs at once. It tells a reader how to read. A body in the first chapter signals that the rest of the book is a search for a cause, and the reader begins sorting detail into clues without being asked. It also creates the possibility of significant deviation, because a rule that everybody knows can be broken on purpose and the breaking will be noticed.

Detective fiction wrote its conventions down. In the 1920s members of a London club drew up a list of rules. One was the requirement of fair play, meaning that every clue must be shown to the reader before the solution is given. Agatha Christie then built her most famous book on an apparent breach of one of them, and the argument about whether she had cheated is still going. The scandal only existed because the rule existed. A reader with no expectations cannot be surprised.

The same mechanism works through allusion, which lets a later work lean on an earlier one. A western in which the hero walks down a street at noon is borrowing a hundred earlier walks. A film that shows the hero arriving to find the street empty comments on all of them in one shot. None of this requires the audience to have studied the history. Recognition can be entirely unconscious, and the writer only needs the pattern to be present, not to be named. Genres also drift, because every successful breach becomes the next convention. The unreliable narrator was a shock in 1926 and is now a standard item on a publisher's list. The useful question about any genre work is therefore which rule it is leaning on and which it is pushing against.

WHAT THIS PASSAGE IS FOR

- A convention tells a reader how to read.
 - Fair play requires every clue to be shown before the solution.
 - A reader with no expectations cannot be surprised.
-

Where the Eye Is Sent

A painting is read in an order, and the painter decides that order. Composition, meaning the arrangement of shapes and spaces within the frame, is the machinery for doing it, and three devices carry most of the load. The first is tonal contrast, which is the difference between the lightest and the darkest areas. The eye goes to the strongest contrast first, which is why Vermeer puts his brightest white on the milk and the cap and leaves the rest of the wall in a narrow range.

The second device is line. Edges that run diagonally lead the eye along them, and painters arrange table edges, floor tiles, arms and gazes to point where they want attention to travel. The third is scale and position. A figure placed on a diagonal from a corner reads differently from one on the center line. An object cut by the frame reads as continuing beyond it, which pulls the viewer into imagining a room rather than a panel.

What follows is a working method for looking. Ask where the eye went first and why. Follow the path it took afterward and find the edge or the gaze that sent it. Notice what has been left in low contrast, because that is what the painter decided could wait. A crowded and ornate picture is not necessarily badly composed, since a painter may want the eye to wander, but a composition that fails will leave the eye drifting without arriving. The test is whether the order of looking can be described at all. Conservators can sometimes show that an artist moved a figure to improve this order. X-ray images of the Milkmaid reveal a map on the back wall and a basket that were painted out.

WHAT THIS PASSAGE IS FOR

- The eye goes to the strongest tonal contrast first.
 - Diagonal edges lead the eye along them.
 - A failed composition leaves the eye drifting without arriving.
-

Tension, Delay and Arrival

Music works on expectation in a way that can be described without technical training. A piece in a key establishes a home note, the tonic, and a listener who has heard a few bars can hum it without knowing its name. Moving away from that home creates tension. Returning to it releases the tension. Almost everything else is management of the distance and the delay between those two events.

A cadence, meaning a closing formula that signals an ending, is the standard machinery of arrival. Composers learned early that an expected cadence can be interrupted, and the interruption is one of the strongest effects available. The music arrives at the moment of resolution and goes somewhere else, often to a chord that is close by and wrong, and the listener feels the swerve as a physical disappointment. Haydn built jokes on it and Wagner built whole acts by refusing to arrive for an hour at a time.

Rhythm does the same job in time rather than in pitch. A pattern once established can be stretched, interrupted or contradicted, and a strong beat left silent is louder than any note. What this means for a listener is practical. The question to ask of a passage is not whether it is beautiful but what it has set up and whether it has paid. A piece that evokes grief usually does so by establishing something and then withholding it, and the emotion sits in the gap rather than in the notes. Film composers work almost entirely with this mechanism and often with very simple material. A single unresolved chord held under dialogue is doing the work of a page of script. That is why a description of a piece as sad is less useful than a note of where it was expected to resolve and did not.

WHAT THIS PASSAGE IS FOR

- Moving away from the home note creates tension and returning releases it.
 - An interrupted cadence is felt as a physical disappointment.
 - A strong beat left silent is louder than any note.
-

What Makes a Street Work

A street can be described by numbers, and a few of them predict most of what it feels like to walk down. The first is frontage, meaning the length of building face that meets the pavement, and more exactly active frontage, which is frontage with doors and windows that people actually use. A hundred meters with twelve doorways behaves quite differently from a hundred meters with one, even if the buildings are the same height and the same age.

The second number is width, and the proportion of width to building height decides whether a street reads as a room or as a gap. Older European streets commonly run between one to one and one to two, and above about one to three the space stops holding together. The third is permeability, counted as the number of ways out of a block. A fine grid gives walkers' choices and spreads traffic; a single loop road forces everyone onto one route and makes walking pointless.

These numbers conflict, which is the whole difficulty. Wide roads move vehicles and break walking routes. Large blocks are cheaper to build and produce long blank walls. A tower set back in open ground may be austere and handsome from a distance and leave the pavement beside it dead, because nothing at ground level opens onto it. Jane Jacobs made the argument from observation in 1961, counting uses and watching sidewalks, and most of what she claimed has since been measured by people who walked the same streets with clipboards. Transport departments now count footfall, crossings and dwell time as a matter of routine. The counts tend to confirm that doorways per hundred meters predicts street life better than architectural quality does. A street can be measured in an afternoon with a tape and a counter.

WHAT THIS PASSAGE IS FOR

- Active frontage is frontage with doors and windows people use.
 - The ratio of width to height decides whether a street reads as a room.
 - The numbers conflict, which is the whole difficulty.
-

What a Judgment Needs

A critical judgment that can be examined has three parts. There is a claim, which must say something that could be false. There is evidence, drawn from the work itself and quoted or described closely enough that a reader can check it. And there is a criterion, meaning the standard against which the work is being judged, which is the part most often left out.

Each part fails in a characteristic way. A claim fails by being unfalsifiable, as when a reviewer calls a novel important without saying what would make it unimportant. Evidence fails by being about something else, which is why a review that spends its length on the author's biography is not a review of the book. The criterion fails by staying hidden, and a hidden standard produces the commonest kind of useless disagreement, in which two people argue about a work while measuring it against different things.

Making the criterion explicit changes the argument. If a critic says that a play fails because its characters are not psychologically consistent, a reader can ask whether psychological consistency is the right test for a morality play written in 1590. If another says that a film is poor because it does not satirize the system it depicts, a reader can fairly ask whether satire was ever the aim. Samuel Johnson insisted on the point in 1765. Neither question settles the matter, and both move it to the one place where it can be argued productively. Critics who name their standards can be disagreed with. Critics who do not can only be ignored or believed. Academic criticism makes the standard explicit as a matter of convention, which is part of why it reads so unlike journalism. The convention costs it readers and buys it the ability to be refuted.

WHAT THIS PASSAGE IS FOR

- A judgment needs a claim, evidence and a criterion.
 - A hidden standard produces useless disagreement.
 - Critics who name their standards can be disagreed with.
-

Social Science

Passages 91 to 100

91

SOC · S01 · mechanism

Three Ways a Survey Fails

A survey can be wrong in three separate ways, and the remedies are different in each case. The first is the sampling frame, meaning the list from which the people to be asked are drawn. A telephone poll on landlines reaches a list older and more settled than the population. No amount of courtesy on the call fixes a frame that excludes a third of the country.

The second is nonresponse, which is the failure of selected people to answer at all. Response rates to telephone polls in the United States have fallen from around seventy percent in the 1970s to under ten percent, and the people who still answer are not a small copy of those who do not. They are more interested in the subject, more trusting of institutions and older. Statisticians correct for this by weighting, which multiplies the answers of underrepresented groups, and weighting works only for characteristics that are known and measured.

The third failure is in the instrument, and it includes the wording effects from the earlier passage in this strand along with order effects and the pressure to give an acceptable answer. Respondents understate how much they drink and overstate whether they voted, by a margin large enough to be measured against official records. A bias of this kind is not removed by a larger sample. That is the central point of all three failures. Sample size governs only random error, so a poorly framed poll of fifty thousand people is worse than a well framed poll of a thousand, and more confident. Good practice therefore reports the frame, the response rate and the weighting scheme alongside the figures. Where a report gives only a margin of error, that margin describes the least of its problems.

WHAT THIS PASSAGE IS FOR

- A landline frame excludes much of the population.
 - Response rates have fallen from about seventy percent to under ten.
 - Sample size governs only random error.
-

Why the Lot Matters

Matching looks like a reasonable substitute for a lottery and is not. A researcher comparing two schools can match them on size, income, and the qualifications of the teachers, and still be defeated by something not on the list. The danger is a confounder, meaning a difference between the groups that affects the outcome as well as the treatment, and the fatal property of confounders is that the dangerous ones are the ones nobody thought of.

Randomization solves a problem that matching cannot. When a lottery is used to allocate people to a treatment and to a control group, meaning the group that does not receive it, the two groups are made alike on every characteristic at once. That includes the ones nobody has measured or imagined. That is a remarkable property and it is purely a consequence of chance. It also comes with a stated size of uncertainty, because the two groups will differ a little by luck, and the arithmetic of that luck is known.

What randomization does not do is also worth stating. It says nothing about whether the result transfers to another population, another scale or another decade. A trial in seventy-nine Tennessee schools tells us about those schools in those years. It cannot tell us whether the state would have found enough good teachers to staff small classes everywhere, which is a question about supply rather than about effect. It also does not survive people dropping out in different numbers from the two groups, which is the commonest way a good design is spoiled in practice. Medicine faced the same argument a generation earlier and settled it the same way. The streptomycin trial of 1948 is usually cited as the first properly randomized clinical trial.

WHAT THIS PASSAGE IS FOR

- The dangerous confounders are the ones nobody thought of.
 - A lottery makes groups alike on characteristics nobody measured.
 - Randomization says nothing about transfer to another population.
-

Three Reasons a Pattern Misleads

Two measurements that correlate may stand in any of four relations, and only one of them licenses action. The first possibility is the obvious one: the first thing causes the second. The second is reverse causation, meaning that the outcome is producing the supposed cause. Hospitals with more staff per bed have higher death rates, which looks alarming until one notices that the sickest patients are sent to the best staffed units.

The third possibility is a common cause, which is what the chocolate and Nobel case shows. The fourth is a selection effect, meaning a pattern produced by how the cases entered the data at all. A study of the fitness of soldiers says little about the population, because the unfit were never enlisted. A famous wartime example concerns aircraft armor. The planes that returned had most of their damage on the wings, so armoring the wings was exactly the wrong conclusion. The planes hit in the engines had not come back to be inspected.

Distinguishing the four requires something beyond the correlation. Time order helps a little, since a cause must precede its effect, though anticipation can reverse the apparent order. The strongest ordinary tool is a source of variation unrelated to everything else of interest, which is what a lottery provides and what a natural experiment imitates. Until something of that kind is in hand, a correlation is a question rather than a finding. Abraham Wald worked out the aircraft armor problem for the United States Navy in 1943. His solution was to reason about the missing planes rather than the ones in front of him. Dose response helps more, because a cause that acts in proportion is harder to fake. The strongest ordinary tool remains an outside source of variation that nobody chose.

WHAT THIS PASSAGE IS FOR

- Hospitals with more staff per bed have higher death rates.
 - A selection effect is a pattern produced by how cases entered the data.
 - A correlation on its own is a question rather than a finding.
-

Four Numbers and What They Hide

Any set of numbers can be summarized by four quantities, and each one hides something different. The mean is the total divided by the count. The median is the middle case. The standard deviation, meaning a measure of how far the typical case sits from the average, describes the spread. And the skew describes whether the distribution has a long tail on one side, which income always does and height never does.

Two groups can share a mean and a median and differ in every way that matters. A town where everyone earns thirty thousand dollars and a town where half earn ten and half earn fifty have the same mean, and a policy built for one would be useless in the other. Two schools with the same average score may differ in whether the result comes from a uniform middle or from a split between a strong half and a failing half, and only the spread shows it.

The practical rule is to ask for the shape and not the summary. A histogram, which sorts the cases into bins and draws the counts, usually settles in one glance questions that an aggregate figure leaves open. Where a shape is not available, two numbers are much better than one, and a mean with a standard deviation is the usual pairing. The worst case is a single figure with no spread and no sample size, which is also the most common way a statistic arrives in public. Francis Anscombe assembled four data sets in 1973 with identical means, variances and correlations and completely different shapes. The point of the exercise was that anyone who had plotted them would have seen the difference immediately. A school report that gives an average with no spread is withholding the more useful of the two numbers.

WHAT THIS PASSAGE IS FOR

- Each summary number hides something different.
 - Two towns can share a mean and differ in everything that matters.
 - A single figure with no spread and no sample size is the worst case.
-

Price, Salience and Substitution

An incentive works through three properties. The size of the sum is often the least important. The first is salience, meaning how noticeable the cost or benefit is at the moment of choosing. A parking charge paid at a meter before walking away is salient; the same amount added to an annual tax bill is not, and the two produce quite different behavior for identical money.

The second is the availability of a substitute, meaning an alternative that is easy to switch to. A charge of five cents on plastic bags cut their use sharply in Ireland and in Wales. A cloth bag is a close substitute and the decision is made at the till. A charge of the same size on gasoline changes very little in the short run, because nobody can substitute for the journey to work this week. The lesson is that the effect of a price depends on what else is available, not on the price alone.

The third property is timing. A cost that arrives at the moment of the decision changes behavior more than a cost that arrives in three months, and people discount the future steeply. That is why deposit schemes on bottles work and why fines for late payment often do not. Together the three properties explain why a well-placed nuisance can outperform a large sum badly placed. They also explain why many incentives fail. The money is real but arrives in the wrong form, at the wrong moment, where nothing can be switched to. Economists call the extreme cases of this inelastic demand, and they test for it by looking at what happened when prices moved. Fuel and tobacco are the standard examples, and both become more responsive over a period of years.

WHAT THIS PASSAGE IS FOR

- Salience is how noticeable a cost is at the moment of choosing.
 - The effect of a price depends on what else is available.
 - People discount the future steeply, so deferred costs change less.
-

How a Norm Is Enforced

An unwritten rule survives because breaking it costs something, and the costs are supplied by other people. The machinery has four parts. There must be observation, so that a breach is seen at all, and there must be a sanction, meaning a penalty applied by other people rather than by law, which may be as small as a look. There must be some willingness to apply it, since a sanction nobody bothers with is no sanction. And the rule itself must be known, because people cannot follow a rule they have no way of inferring.

The strength of a norm therefore depends on visibility more than on agreement. Littering falls where people can be seen littering. Tax compliance is higher where returns are public, as they are in Norway. A norm that governs private behavior has no enforcement machinery at all and survives only as a belief. This also explains why norms collapse quickly rather than fading. Once enough people visibly break a rule, the descriptive norm, meaning what people actually do as distinct from what they say should be done, has changed. Each further breach is then cheaper than the last.

The hotel towel result follows directly. A card that reports the descriptive norm is providing the information that the enforcement machinery needs and cannot otherwise get, since a guest cannot see what strangers did in other rooms. This also predicts the failure case, which has been observed: telling people that most of their neighbors do the undesirable thing makes the undesirable thing more common. The message is only as good as the fact it reports. Campaigns that advertise how widespread a problem is run exactly that risk. Several anti-littering and tax-evasion campaigns have been measured doing harm for that reason. The practical question about any unwritten rule is therefore who can see a breach.

WHAT THIS PASSAGE IS FOR

- Norm enforcement needs observation, a sanction and willingness to use it.
 - Strength depends on visibility more than on agreement.
 - Reporting an undesirable norm makes it more common.
-

The Trade a Household Makes

Where people live is largely settled by a single trade. Land near the center of a city is scarce and therefore expensive, and travel from the edge costs time and money. A household chooses a point on that line, and the economists' description of the choice is bid rent, meaning the amount a household or firm will pay for land at a given distance from the center. Bid rent falls as distance rises, and the slope of the fall is set by the cost of travel.

Two consequences follow immediately. If travel becomes cheaper or faster, the slope flattens and the city spreads, which is what streetcars, then cars, then suburban rail each did in turn. If travel becomes dearer or slower, the city tightens. The second consequence concerns who ends up where. Households differ in how much they perceive travel time to cost them. A family with two earners and expensive hours may outbid a family with cheaper hours for land near the center, or may not, depending on how each values space.

The model is crude and useful. It predicts that firms needing many workers cluster where many workers can reach them, which is why offices concentrate. It predicts that a new road reduces congestion for a while and then fills, since cheaper travel brings more distant land into play. The quantity that decides all of this is generalized cost, meaning travel time and money added together as one figure, and transport planners calculate it for every scheme they assess. The original formal version is credited to William Alonso in 1964, with earlier agricultural versions going back to von Thunen in 1826. Both were arguing about farms before anyone applied the reasoning to commuters. Every one of these predictions can be and has been tested against city data.

WHAT THIS PASSAGE IS FOR

- Bid rent falls with distance and its slope is set by travel cost.
 - Cheaper travel flattens the slope and spreads the city.
 - A new road fills because cheaper travel brings distant land into play.
-

What Sets a Wage

Adam Smith named the elements in 1776, and three of them still do the work. They pull in different directions. The first is what the work produces for the employer, which fixes a ceiling. The second is the reservation wage, meaning the lowest pay at which a person will take the work, which fixes a floor and depends on what else is available, including savings and family support. Between the floor and the ceiling the wage is settled by bargaining power, which is the third thing.

Bargaining power depends on how many alternatives each side has. A worker in a town with one employer is in a market called monopsony, meaning one buyer, or few, for a particular kind of labor, and such a worker has a weak position however scarce the skill. The dock arrangement engineered this deliberately. By hiring at a gate twice a day the companies ensured that each man competed against a crowd for that day only, and no individual could threaten to withdraw, since the next man was already present.

The arrangement also shows why wages can stick far below the ceiling. Dock work in the 1880s was productive and the trade was profitable, but the men had no way to capture any of it. What changed in 1889 was not productivity but organization, which replaced a thousand individual bargains with one. The longer-run effects are harder to see and matter as much. A cohort that enters a labor market in a bad year can be measured earning less than neighboring years for a decade or more afterward, because the first job sets the next. Studies of graduates entering work in recessions find losses persisting ten to fifteen years. The size varies by country and by how easily people can move between regions.

WHAT THIS PASSAGE IS FOR

- Productivity fixes a ceiling and the reservation wage a floor.
 - Gate hiring made each man compete against a crowd for one day.
 - What changed in 1889 was organization, not productivity.
-

Three Different Inequalities

Three quantities are routinely confused under one word. Income is a flow, measured over a year and gone at the end of it. Wealth, meaning what a household owns minus what it owes, is a stock, accumulated over a life and inherited across generations. Mobility is neither of those. Intergenerational mobility, meaning how much a child's position depends on the parents' position, can be high in a country with wide income gaps.

The three behave differently because their mechanisms differ. Wealth inequality is always larger than income inequality, in every country measured. Savings compound and so do debts, and a person on a modest income may still own a house. Income inequality responds to wages, taxes and transfers within a few years of any change to them. Wealth inequality responds over decades, and inheritance is the reason. Mobility depends on schooling, neighborhood, early childhood and the labor market, and it responds slowest of all.

A country can reduce income inequality by taxing and transferring while its wealth gaps widen, which several European countries did between 1980 and 2010. A country can have high mobility between the middle and the top and almost none out of the bottom, which is what the American data show. None of the three captures the experience of being poor, where stigma, insecurity and the cost of credit do damage that no distribution figure records. Wealth is measured from surveys that the very rich do not answer, so published wealth gaps are probably understated. Any serious argument about inequality therefore has to say which of the three quantities it is about, and a good many arguments never do. The three also respond at very different speeds, so a government can honestly report progress on income while the wealth gap widens behind it. Mobility is slower still, since a change in schooling shows up in the figures a generation later.

WHAT THIS PASSAGE IS FOR

- Income is a flow, wealth a stock, mobility neither.
 - Wealth inequality exceeds income inequality in every country measured.
 - A country can cut income gaps while its wealth gaps widen.
-

Three Errors With Names

The base rate error is one of a family, and the family members have been demonstrated often enough to be predictable. Anchoring, meaning the pull of an irrelevant number on a later estimate, is the simplest. People asked to spin a wheel and then to estimate a quantity give answers pulled toward whatever the wheel showed, and the effect survives when the subjects are told that the number is random. Estate agents given different asking prices for the same house value it differently.

Availability is the second, and it means judging how common something is by how easily examples come to mind. Deaths from causes that make the news are overestimated by large factors and deaths from common illnesses are underestimated, and the ordering of estimates follows newspaper coverage more closely than it follows the figures. The third is framing. The same medical choice described in terms of survival and in terms of death produces different decisions from the same doctors.

What makes these useful rather than merely entertaining is that they are systematic. A random error would cancel over many judgments, but these all deviate in the same direction for the same reason, which means the mistake can be anticipated and designed around. Checklists in operating theaters, standard forms for sentencing, and the practice of writing down a prediction before seeing the outcome all exist for that purpose. Whether these patterns show that people are irrational, or only that they are using cheap rules that usually work, is the subject of the last passage in this strand. Daniel Kahneman and Amos Tversky set out the first three in a paper in 1974. Much of the later work has been about which effects survive replication and how large they really are.

WHAT THIS PASSAGE IS FOR

- Anchoring pulls an estimate toward an irrelevant number.
 - Availability makes newspaper coverage drive estimates of risk.
 - These errors are systematic rather than random.
-

LEVEL 3

Target

evidence

How anyone knows: the study, the document, the measurement, and its limits.

Passage length — 288 to 312 words

Sentences — averaging 19 to 24 words

Vocabulary assumed — the top 30,000 most frequent word forms

Words above that — at most 6, explained or inferable

Clue to the reader — Read for the warrant. Someone had to find this out. These passages name the study, the document, or the measurement behind the claim, and say what it cannot show. The limit is part of the finding, not an apology for it.

History and Civics

Passages 101 to 110

101

HIS · S01 · evidence

What the Rough Draft Shows

The Declaration exists in more than one state, and the differences are evidence. What survives includes Jefferson's rough draft, meaning the working manuscript with its crossings-out and insertions still visible, several fair copies he made for friends, the printed broadside of July 4, and the engrossed parchment signed in August. Historians read these against one another in the way a textual critic reads the manuscripts of a poem, and the method is identical. Where two versions differ, the question is who altered the wording, and at what stage of the proceedings.

The most studied difference is a deletion. Jefferson's draft contained a long passage blaming George III for the slave trade, in language calling it a cruel war against human nature. Congress struck the whole passage out before approving the text. Jefferson himself recorded that delegates from South Carolina and Georgia objected, and that northern members who profited from the trade were not sorry to see it go. The deleted paragraph is therefore a double piece of evidence about the proceedings. It shows what one drafter was prepared to say, and it shows what a body of fifty-six could be brought to sign.

A draft will not bear every weight, and the limits are worth being exact about. A deletion tells us that a sentence was removed and not, by itself, why. Jefferson wrote his account of the objection decades later, when his memory had reasons of its own. The passage also blamed the king for a trade in which Virginia planters were participants, so its removal cannot simply be read as the moment when the republic repudiated its own stated principle. What the manuscript establishes is narrower and still substantial: the question was raised in 1776, in writing, by the author, and was taken out. The rough draft is held at the Library of Congress and has been photographed at high resolution.

WHAT THIS PASSAGE IS FOR

- The document survives in several states and the differences are evidence.
 - The deleted passage shows what fifty-six men could be brought to sign.
 - A deletion shows that a sentence was removed, not why.
-

Reading the Ratification Debates

The Federalist is the most quoted source on the Constitution and the least representative. It consists of eighty-five newspaper essays written to persuade New York to ratify, by three men with an interest in the outcome. Reading it as a neutral explanation of the document is like reading a campaign pamphlet as a description of a policy. It is first-class evidence of what its authors wanted understood.

The rest of the surviving record is a great deal larger and considerably messier. Each state held a ratifying convention, meaning an assembly elected to vote on the Constitution, and the debates in several of them were taken down by shorthand writers of uneven skill and published later, sometimes by partisans. Essays opposing ratification appeared under names such as Brutus and the Federal Farmer, the Pennsylvania minority published a dissent, and private letters survive in considerable quantity. Madison's own notes of the convention, the fullest account of what was said in Philadelphia, were not published until 1840, after every participant was dead.

Each kind of source carries a characteristic defect that can be stated. Published debates were edited before printing, and in one well-known case the reporter admitted that he had improved the speeches as he set them down. Private letters are candid and partial in the same breath. Madison's notes were revised by him over forty years, and the revisions can be dated by the ink and the paper. A historian who wants to show that executive power was later usurped, or that it was always intended to be large, can find support in the record for either claim. That is why the argument at the end of this strand has outlasted everyone who began it. Modern editions print the sources side by side with notes on their reliability. The Documentary History of the Ratification of the Constitution now runs to more than thirty volumes.

WHAT THIS PASSAGE IS FOR

- The Federalist is persuasive writing, not neutral explanation.
 - Madison's notes appeared in 1840, after every participant was dead.
 - The record supports either claim about executive power.
-

Signatures, Poll Books and a Trial

Four kinds of record carry the history of the suffrage campaign, and they measure different things. Petitions survive in bulk, with names, addresses and sometimes occupations, and Congress preserved them by the cartload. A signature count measures organizing capacity rather than opinion, because gathering fifty thousand names tells a historian mainly that somebody could pay for canvassers and print the sheets. Membership rolls of the suffrage associations measure that same capacity rather more directly, and they were kept with varying care from one state association to the next.

The third record is the poll book, meaning the register of who actually voted in a given election, and it supplies the hardest evidence in the whole subject. In 1872 Susan Anthony and fourteen other women registered and voted in Rochester, New York, arguing that the Fourteenth Amendment already gave them a mandate to do so. The poll book records the votes. The subsequent prosecution produced the fourth record, a court transcript, in which the judge directed the jury to find her guilty and fined her a hundred dollars, which she never paid. The case had been invited on purpose, and the judge refused to let it go further.

Each record has a boundary. Petitions say nothing about those who declined to sign, and poll books record who voted rather than what any of them thought. Transcripts record what was said in a room where everyone was performing for a result. Newspaper coverage, which survives in the greatest quantity of all, mainly measures what editors thought their readers would enjoy. The combination is still enough to answer specific questions, and it is the reason historians can state with confidence that the campaign was organized state by state rather than directed from one center. Digitization has changed the field in twenty years, since petition signatures can now be matched against census records by name and street.

WHAT THIS PASSAGE IS FOR

- A signature count measures organizing capacity rather than opinion.
 - Anthony and fourteen others voted in Rochester in 1872.
 - Poll books record who voted, not what they thought.
-

Ledgers, Narratives and Manifests

The history of slavery is written from four kinds of source, each with a distortion that can be stated precisely. Ships manifests, meaning the lists of people and goods carried on particular voyages, survive in large numbers because they were commercial and legal documents. A database assembled from them now covers more than thirty-six thousand Atlantic voyages and permits estimates of numbers, routes and mortality. What it cannot contain is anyone name, because the people listed were entered as cargo by age and sex.

Plantation ledgers supply the opposite kind of detail: individuals, with names, prices, tasks and punishments, recorded by the person applying the coercion. They are precise and they are testimony by an interested party. Published narratives by formerly enslaved people, such as those of Frederick Douglass and Harriet Jacobs, supply the inside view, and they were written to persuade, often with an abolitionist editor involved. Contemporaries disputed their accuracy, and modern historians have since verified much of the detail from other records.

The fourth source is the set of interviews recorded in the 1930s with people who had been children in slavery. There are more than two thousand of them, and they are the most difficult of all. The speakers were in their eighties and nineties, most interviewers were white, and many interviews took place in states where the speaker still depended on local goodwill. The transcripts were also written in a dialect spelling imposed by the interviewer. Historians use them, carefully, and the care consists in reading across many interviews rather than quoting one. The slave trade database is now public and can be queried by anyone with a browser. It was built over thirty years from archives in seven countries and is still being corrected.

WHAT THIS PASSAGE IS FOR

- Manifests permit estimates of numbers but contain no names.
 - Ledgers are precise testimony by an interested party.
 - The 1930s interviews are used by reading across many, not by quoting one.
-

Thirteen Volumes of Testimony

In 1871 a joint committee of Congress took sworn testimony, meaning evidence given under oath and taken down by a clerk, on violence in the southern states. The published result runs to thirteen volumes. Hundreds of witnesses appear, including people who had been attacked, people accused of attacking them, sheriffs, teachers, planters and former officers. It is one of the fullest records of political violence anywhere in the nineteenth century, and it exists because a congressional majority wanted the violence on the record.

Testimony of that kind has two uses and two problems. It establishes particular events in detail, with names and dates that can be checked against local papers and court files. It also preserves the voices of people who left no other written trace, which is rare. The problems are that witnesses were selected by committee members with purposes, and that both sides understood what the hearings were for, so the accused denied and minimized while the committee pressed for the worst cases.

Registration rolls, which are the official lists of those entitled to vote, answer a different question and answer it coldly. They show how many Black men were registered in a county in 1867 and in 1904, and the fall is not a matter of interpretation. What they cannot show is why any individual stopped registering, since a roll records an absence without a reason. Put the two sources together and the picture is reasonably firm: a pattern of violence established by testimony, followed by an entrenched legal exclusion established by the rolls, with a decade in between where the evidence is thinner than historians would like. The hearings are now searchable online, which has produced a series of local studies. Several of them have matched named witnesses to census entries and property records.

WHAT THIS PASSAGE IS FOR

- The 1871 hearings run to thirteen volumes of sworn testimony.
 - Witnesses were selected by members with purposes.
 - A roll records an absence without a reason.
-

Mortality Tables and Their Gaps

Factory legislation in Britain and in New York was argued from numbers, and those numbers came from three places. There were commission reports with their collected testimony, the annual returns of the factory inspectors, and the registers of deaths kept by the civil authorities. Each was collected for a purpose other than research, which is the normal condition of historical statistics and the first thing to hold in mind.

Three defects recur. The first is the missing denominator, meaning the figure a count must be divided by before it means anything. A report that a trade killed four hundred men in a year is uninterpretable without knowing how many men worked in it, and employment figures by trade were poor before the twentieth century. The second is cause of death coding. A weaver who died of tuberculosis after twenty years in a damp shed was recorded as dying of tuberculosis, and the shed appears nowhere in the record, so industrial disease was systematically invisible while industrial accident was not.

The third defect is reporting incentive. Accidents were reported by employers to inspectors who could prosecute them. Comparisons across years are therefore contaminated by changes in how willing employers were to report, and a fall in recorded accidents after a new law may mean fewer accidents or more concealment. Historians handle this by looking for series collected by somebody with no stake, such as burial registers or union benefit payments, and by comparing trades that were covered by a law with trades that were not. Measures that appeared to ameliorate conditions sometimes only improved the paperwork, and the distinction can usually be settled with a second source. Union sick-pay records have turned out to be among the best series available. They were kept carefully because money depended on them, and nobody involved had a reason to flatter an employer.

WHAT THIS PASSAGE IS FOR

- Historical statistics were collected for purposes other than research.
 - A count without a denominator cannot be interpreted.
 - A fall in recorded accidents may mean concealment.
-

What the Files Turned Out to Hold

The civil rights period is unusually well documented, and a large part of the documentation was created by people trying to stop it. Court transcripts and pleadings survive for thousands of cases, and registration figures by county were collected by the federal government in order to decide which areas the 1965 Act would cover, which is why they are unusually careful. Organizations kept minutes, and the movement itself produced a flood of printed material.

Then there are the surveillance files. The Federal Bureau of Investigation watched the main organizations and their leaders, tapped telephones under warrants it had obtained by misrepresenting the purpose, and compiled files that ran to thousands of pages. Much of this became available after 1974, when amendments to the freedom of information statute, which is a right to demand government documents, gave researchers a usable instrument. The files are a strange source. They record meetings in detail because an agent was present, and they record them through a lens that treated the subject as a threat.

Using them requires a particular discipline. An informer report is evidence that something was said and no evidence at all that it was true, since informers were paid and had reason to produce material. An agency summary is a partisan document whose purpose was to justify continued attention. Where a file can be matched against a minute, a newspaper account or a participant recollection, the match is often close on facts and wide apart on meaning. Historians therefore treat the files as a record of what was observed rather than as a record of what was happening. Some files are still closed, and several thousand pages remain redacted. A standing committee reviews the releases, and material has continued to appear into the present decade.

WHAT THIS PASSAGE IS FOR

- Much of the documentation was created by people trying to stop the movement.
 - Registration figures were collected to decide which areas the Act would cover.
 - An informer report shows what was said, not that it was true.
-

Whose Numbers Are the Famine Reports

British India generated statistics on an extraordinary scale, and almost all of them were produced by the administration for administrative purposes. The revenue settlement, meaning the periodic assessment that fixed what each district owed, required measurement of land, crops and population, and the resulting series run for a century. A historian therefore has a great deal of data and no independent check on it, which is a specific and awkward position.

The famines illustrate the problem. Mortality figures were compiled by the same district officers whose management was in question, and the usual measure is excess mortality, meaning deaths above the number that would have been expected without the event, which requires a baseline that is itself an estimate. Published figures for the famine of 1876 to 1878 range from five million to over eight million depending on the baseline chosen. The Famine Commission of 1880 collected evidence and made recommendations, and it was also a defense of policy, since it had been appointed by the government whose policy was at issue.

Three practices help. Comparisons between districts under different revenue systems, such as areas where the tax was fixed and areas where it was reassessed, allow something like a natural experiment. Private records kept by merchants and missionaries, who had their own reasons for accuracy, provide a partial check. Later census data can be read backward for the holes left in age structure by a famine twenty years earlier. None of this makes the official series reliable, and none makes it useless. It makes it a source with a known direction of error, which is the most that is usually available when the expedient of collecting new data is not open. Famine codes written after 1880 did change administrative behavior, and the later record shows it.

WHAT THIS PASSAGE IS FOR

- Nearly all the statistics were produced by the administration for its own purposes.
 - Excess mortality requires a baseline that is itself an estimate.
 - The result is a source with a known direction of error.
-

Two Reports That Were Not Shown

Two assessments of the loyalty of Japanese Americans were in government hands before the removal was ordered. Curtis Munson, a businessman sent by the State Department, reported in November 1941 that the community presented no substantial danger and that an immense majority were loyal. Kenneth Ringle, a naval intelligence officer who spoke Japanese and had worked in the community, reported in January 1942 that the number of genuinely dangerous individuals was small and already known, and that mass removal was unnecessary.

Neither report was given to the Supreme Court when it heard the cases in 1943 and 1944. The government argued military necessity, and the Court accepted it, upholding the exclusion in *Korematsu* by six votes to three. The dissents are worth reading precisely because they had to be written without the documents. Justice Murphy wrote that the claim of necessity rested on misinformation and half-truths, which turned out to be exactly right, and he had no proof.

Peter Irons, a legal historian, found the suppressed material in government archives in 1981 while writing a book. One memorandum from a Justice Department lawyer warned his superiors that the brief contained statements he believed to be false. Lawyers then brought *coram nobis* petitions, a rare procedure for reopening a conviction on the ground that the court was misled, and the convictions of Fred Korematsu, Gordon Hirabayashi and Minoru Yasui were vacated in the mid-1980s. The precedent set in 1944 was never formally overruled until 2018, in a sentence inside a case about something else. The National Archives holds the files and they are open to any reader. The documents also explain something otherwise puzzling about the dissents, which read as though their authors suspected exactly what was missing. Both were reasoning from the shape of the argument rather than from anything they had seen.

WHAT THIS PASSAGE IS FOR

- Both reports concluded that mass removal was unnecessary.
 - Neither report was given to the Supreme Court.
 - A historian found the suppressed material in 1981.
-

Two Million Replies and a Wrong Answer

In 1936 the Literary Digest ran the largest straw poll, meaning an informal canvass with no attempt at a representative sample, that had ever been attempted. It mailed about ten million ballots and received more than two million replies, a sample two thousand times larger than anything a modern pollster uses. It predicted that Alf Landon would beat Franklin Roosevelt by fifty-seven to forty-three. Roosevelt won by sixty-one to thirty-seven, the largest margin in a century, and the magazine was out of business within two years.

The failure had two causes and neither was size. The addresses came from telephone directories, automobile registrations and the magazine own subscription lists, which in 1936 described a population that was richer than the country. That is a frame error. On top of it, only about a fifth of those contacted replied, and people who felt strongly against the administration were more willing to spend a stamp on saying so. That is nonresponse bias. The two pushed in the same direction and the result was wrong by eighteen points.

In the same election George Gallup predicted the outcome correctly from about fifty thousand interviews, and he also predicted, in advance and in print, what the Digest poll would say and why. His method was quota sampling, meaning that interviewers were instructed to fill a set number of respondents in each category to match known proportions in the population. The episode became the founding lesson of survey research, and it has had to be relearned repeatedly, most recently where an enormous but self-selected sample is collected online and reported as though its factional enthusiasm were representative. Gallup's method failed in its turn in 1948, for a different reason, and was replaced by probability sampling. Each generation of the technique has been broken by a change in how people can be reached.

WHAT THIS PASSAGE IS FOR

- The poll had more than two million replies and was wrong by eighteen points.
 - The address lists described a population richer than the country.
 - Gallup predicted the outcome from about fifty thousand interviews.
-

Biology and Earth Science

Passages 111 to 120

111

BIO · S01 · evidence

Forty Years of Beaks

Peter and Rosemary Grant began working on Daphne Major, a volcanic cone in the Galapagos about a kilometer across, in 1973, and they kept at it for forty years. Their method was exhaustive rather than clever. Every finch on the island was caught, banded, weighed and measured, which was possible because the population ran to a few hundred birds and the island has no fresh water and almost no visitors. Beak length and depth were recorded to a tenth of a millimeter, and because the birds were individually marked the team could follow particular lineages and establish the heritability of each trait, meaning the share of its variation that is passed to offspring.

Then the weather did the experiment. In 1977 the rains failed, no seeds were set for eighteen months, and about eighty-five percent of the medium ground finches died. The small soft seeds went first, so the survivors were disproportionately the birds with deeper beaks able to crack the large hard fruits that remained, and the mean beak depth of the next generation was measurably greater than that of their parents. In 1983 an exceptional wet year reversed the conditions, small seeds became abundant, and the trait moved back.

What this study provides that no laboratory can is a measured rate of change in a wild population under natural conditions, with the selective agent identified and the inheritance established separately. Its limits are equally clear. One island, a few species, and a trait that happens to be easy to measure. Whether the same rates apply to traits that are not viable to measure in the field, or to populations too large to band individually, is not something this work can settle. The Grants published the measurements in full, which is why others have been able to reanalyze them.

WHAT THIS PASSAGE IS FOR

- Every bird on the island was caught, banded and measured.
 - A drought killed most of the population and shifted mean beak depth.
 - The study gives a measured rate in a wild population.
-

From Linkage Maps to Sequence

Three generations of method have been used to find out where genes are and what they do, and each resolves something the previous one could not. The first was the linkage map, meaning a chart of gene order built from how often two inherited traits are separated in breeding. Alfred Sturtevant drew the first one in 1913 as an undergraduate, using fruit flies, and the logic is indirect but sound: traits that are rarely separated must sit close together, and the frequency of separation can be converted into a distance.

The second generation was physical mapping, which located genes on visible chromosomes using stains, deletions and later fluorescent probes. That gave an actual position rather than a relative order. The third is sequencing, meaning reading the order of the chemical letters along a stretch of DNA, which arrived in a usable form in 1977 and became industrial in the 1990s. A draft of the human sequence was announced in 2001 and completed in stages over the following two decades.

What has not changed is the hard part. A sequence tells a reader the letters and nothing about what they do, and most of the human sequence does not code for protein at all. Establishing function still requires breeding experiments, cell studies and natural variation, which is to say that the nascent methods of 1913 are still in use. The characteristic error of the sequencing era has been to treat a correlation between a sequence variant and a trait as an explanation of it, which it is not, any more than knowing a word position in a dictionary explains its meaning. Each generation also inherited the errors of the last, and linkage maps for several organisms had to be reordered once sequences arrived.

WHAT THIS PASSAGE IS FOR

- A linkage map infers order from how often traits are separated.
 - Sequencing became usable in 1977 and industrial in the 1990s.
 - A sequence gives the letters and not what they do.
-

Poisons, Labels and Broken Cells

Nobody has ever watched a sugar molecule being dismantled. The pathway was established by indirect means, and three tactics did most of the work. The first was the inhibitor, meaning a substance that blocks one particular step. Cyanide stops one reaction near the end of the chain, and in a cell treated with it the substances that accumulate are the ones immediately before the block, so the order of steps can be deduced from what piles up behind each poison.

The second tactic was the isotope label, meaning an atom of unusual weight used to trace a substance through a sequence of reactions. Feed a cell sugar in which particular carbon atoms are heavy, collect the carbon dioxide it breathes out, and the position of the heavy atoms shows which parts of the original molecule ended up where. Rudolph Schoenheimer and others established the technique in the 1930s, and it turned metabolic chemistry from speculation into bookkeeping.

The third was to take the cell apart. In the 1940s mitochondria were separated out by spinning broken cells at carefully chosen speeds, and the isolated particles were shown to carry out the oxygen-dependent steps on their own, which located the pathway in a structure rather than in the cell at large. The limits of all three are worth naming. A poison may attenuate several steps rather than one. A label may be scrambled by reactions nobody suspected. An isolated organelle has been removed from everything it was connected to, and the question of whether it still behaves as it did inside the cell can only be answered by further experiments of the same indirect kind. The last large piece of the account, how the energy gradient is actually converted, was proposed by Peter Mitchell in 1961.

WHAT THIS PASSAGE IS FOR

- Substances pile up behind a blocked step, which reveals the order.
 - Heavy atoms show which part of a molecule ended up where.
 - An isolated organelle has been removed from everything it was connected to.
-

Fences and Isotope Ratios

A claim that one species controls another is hard to test, because the obvious experiment would require removing a predator from a landscape and waiting. Two methods make the question tractable without that. The first is the enclosure, meaning a fenced plot that keeps a particular animal out while leaving everything else the same. Fence twenty plots along a creek, leave twenty unfenced beside them, and the difference in willow growth after ten years separates the effect of browsing from the effects of water, soil and weather, because those act on both sets of plots equally.

The second method reads the animals themselves. The proportion of heavy to light atoms of nitrogen and carbon in a tissue, called the stable isotope ratio, differs systematically by diet and rises at each step up a food chain. A tuft of hair or a feather therefore carries a record of what its owner had been eating, averaged over the period in which the tissue grew. Museum specimens can be sampled, which allows comparisons with animals that died a century ago.

Both methods have boundaries that matter in the Yellowstone argument. An enclosure tests browsing and cannot test fear, so it cannot distinguish elk eating fewer willows from elk standing in different places. Isotope ratios give diet, not cause, and the baseline ratios in a river valley vary with geology. Anyone who wishes to perturb a system and watch, which is the cleanest design available, is also asking for an experiment at a scale that no institution will fund and no regulator will permit. The reintroduction itself was therefore the experiment, and it was performed without a control. That single fact is the reason the dispute in the next passage has never been settled on the evidence.

WHAT THIS PASSAGE IS FOR

- An enclosure separates browsing from water, soil and weather.
 - Isotope ratios in tissue record what the animal had been eating.
 - An enclosure tests browsing and cannot test fear.
-

Ninety Years of Fur Returns

The Hudson's Bay Company bought furs across northern Canada for three centuries and kept its accounts. Those accounts were published in the 1930s as a series of annual fur returns, meaning the number of pelts bought in each season, and for lynx and snowshoe hare they run from the 1820s to the 1930s. The series shows something striking: both animals rise and crash on a cycle of roughly ten years, the hare peaks slightly before the lynx, and the pattern repeats for more than a century. No other wild population record of that length exists.

A fur return is a proxy, meaning a measurement used to stand in for a quantity that cannot be observed directly, and it is a proxy for several things at once. It depends on how many animals there were and on how many trappers were working. It also depends on how hard they worked, on fur prices in London, on where the company posts happened to be, and on whether a war or an epidemic had reduced the local human population. Each of those varies over a century, and several vary on their own cycles.

Ecologists have therefore spent decades separating the signal from the record. Price series can be used to correct for effort. Independent counts from the twentieth century overlap the end of the series and can be compared. Tree ring records of hare browsing provide a second and entirely separate proxy, and they show the same cycle, which is the strongest argument that the pattern is real rather than an artifact of commerce. The conclusion drawn is modest and firm: these populations are genuinely cyclic, and a population of that kind is permanently precarious even when its average is healthy. The cycle is also visible in the records of other companies and in the memories of trappers.

WHAT THIS PASSAGE IS FOR

- The fur returns run from the 1820s to the 1930s for lynx and hare.
 - A fur return is a proxy for several things at once.
 - Tree ring records of browsing show the same cycle independently.
-

The Rules and the Exceptions

Robert Koch laid down conditions that a claim about the cause of a disease should satisfy, and they are still the starting point. The organism must be found in every case of the disease, and it must be isolated from a patient and grown in pure culture. The cultured organism must then produce the disease when introduced into a healthy host, and must be recovered again from that host afterward. A postulate, meaning a condition laid down in advance, is useful precisely because it can fail, and these have failed in instructive ways.

The first condition breaks on the asymptomatic carrier, meaning a person who harbors an organism without becoming ill. Koch himself found cholera in people who were perfectly well, and the organism that causes diphtheria is ubiquitous in some populations with no disease at all. The second breaks on organisms that cannot be grown outside a living cell, which includes every virus and several bacteria. The third breaks wherever the disease occurs only in humans, since the experiment cannot ethically be done.

Workers have therefore treated the postulates as a standard to be approached rather than met. Where an animal model is impossible, the third condition has occasionally been satisfied by a researcher who volunteered. Barry Marshall drank a culture of *Helicobacter pylori* in 1984, developed gastritis, and recovered the organism from his own stomach, which was one step in overturning the view that ulcers were caused by stress. Modern practice replaces the missing steps with statistical association, dose response, and the test of whether removing the organism cures the disease. Molecular versions of the postulates were proposed in 1988 for organisms that cannot be cultured. They substitute the presence of particular genes for the growing of a pure culture.

WHAT THIS PASSAGE IS FOR

- A postulate is useful because it can fail.
 - Asymptomatic carriers break the first condition.
 - Marshall drank a culture in 1984 and recovered the organism from himself.
-

The Same Field Since 1843

At Rothamsted in Hertfordshire a field called Broadbalk has been growing wheat under fixed treatments since 1843. The field is divided into strips, and one of them has received nothing at all for a hundred and eighty years. Another has received farmyard manure every year without exception, and the others receive measured combinations of nitrogen, phosphate and potash. The treatments have barely changed, the plots are in the same places, and the yields have been recorded every harvest, which makes this the longest continuous agricultural experiment in the world.

A long-term experiment answers questions that a three-year trial cannot. Soil organic matter changes over decades, so only a series of this length shows the manured strip climbing steadily while the unfertilized strip settles at about a third of the yield and stays there. Slow processes become visible: the accumulation of phosphate, the acidification of soil under ammonium fertilizer, the way weed populations proliferate and then shift in composition as management changes. Rare events are captured because the experiment happens to be running when they occur.

The second, less obvious value is the archive. Soil and grain samples have been stored since the beginning, in sealed jars, and they can be analyzed by methods that did not exist when they were collected. Those jars have since been used to measure the arrival of industrial pollution, the rise of radioactive fallout in the 1950s and its decline, and changes in soil bacteria, none of which anyone in 1843 could have intended. The limits are those of any single site. One soil, one climate, one rotation, and a set of treatments chosen by Victorians. Several other long-term sites now exist, in Illinois, Sweden and India, and comparisons between them are informative. Where two sites disagree, the disagreement is usually about soil type rather than about the chemistry.

WHAT THIS PASSAGE IS FOR

- Broadbalk has grown wheat under fixed treatments since 1843.
 - Only a long series shows soil organic matter changing.
 - Stored samples can be analyzed by methods invented later.
-

Three Proxies, One Picture

The Mauna Loa record begins in 1958. Everything earlier has to be reconstructed, and three independent proxies carry most of the weight. An ice core, meaning a cylinder drilled from a glacier, is the strongest of them, because the air trapped in its bubbles is a physical sample of the atmosphere of the year the snow fell. Cores from Antarctica now reach back about eight hundred thousand years, and the carbon dioxide in them is measured directly rather than inferred.

Tree rings provide annual resolution over the last two thousand years in suitable species, and their width and density respond to temperature and moisture. Corals grow in annual bands whose chemistry records sea surface temperature. Each proxy has a weakness that the others do not share. Ice cores smooth the record, because air diffuses in the snow before the bubbles close, and they come from two polar regions. Tree rings respond to several variables at once and to the age of the tree. Corals are tropical and marine.

Combining them depends on the calibration period, meaning the overlap in which each proxy can be checked against direct measurement. Where instruments and proxies run together for a century, the relation can be established and then extended backward, which is also where the method is most criticized, since a relation fitted in one period may not hold in another. Reconstruction is therefore reported with error bars that widen as they go back. The agreement between three methods with unrelated weaknesses is the real argument, and it is stronger than any one of them, including the one that lay dormant in a freezer for twenty years before anyone thought to measure its bubbles. Cores are stored at low temperature in national facilities and reanalyzed as techniques improve.

WHAT THIS PASSAGE IS FOR

- Air in ice core bubbles is a physical sample of ancient atmosphere.
 - Each proxy has a weakness the others do not share.
 - Agreement between methods with unrelated weaknesses is the real argument.
-

How the Age Was Settled

Before radioactivity there was no way to date a rock, and estimates for the age of the Earth ranged over two orders of magnitude. Radiometric dating, which is the dating of a rock by the proportion of a radioactive element that has decayed, changed that. A radioactive element decays at a rate that nothing in geology alters, so the proportion remaining is a clock. Arthur Holmes spent forty years applying the method to rocks and had the order of magnitude right by the 1920s, when almost nobody believed him.

The remaining problem was that the oldest rock on Earth is younger than the Earth, since the surface has been melted, buried and recycled continuously. Claire Patterson solved it in 1953 by dating meteorites instead, on the argument that they formed from the same cloud of material at the same time and have been undisturbed since. He measured lead isotopes in a fragment of the Canyon Diablo iron and obtained four and a half billion years, a figure that has moved by less than two percent in seventy years of refinement.

The method depends on an assumption that has to be checked rather than assumed: that the sample is a closed system, meaning that nothing relevant has entered or escaped since it formed. Geologists test it in two ways. They date the same rock with two different decay chains, which must agree, and they prefer minerals physically resilient enough to survive heating. Zircon crystals are the favorite, because they hold their uranium and reject lead so strongly that grains four billion years old are still usable. Where two methods disagree, the rock has been disturbed, and the disagreement itself is the evidence. Patterson had to build a clean laboratory to do the work, because ordinary dust carried enough lead to swamp the measurement.

WHAT THIS PASSAGE IS FOR

- A radioactive element decays at a rate nothing in geology alters.
 - Meteorites were dated because the oldest Earth rock is younger than the Earth.
 - Where two methods disagree, the rock has been disturbed.
-

Weighing an Ice Sheet

Two quite different instruments measure what Antarctica is doing, and they were designed by people who had no interest in each other methods. The first is gravimetry, meaning the measurement of mass by its gravitational pull. A pair of satellites launched in 2002 flew in the same orbit about two hundred kilometers apart, and the distance between them was measured continuously to within a few microns. When the leading satellite passed over a heavier region it was pulled forward slightly, and the change in separation gives the mass below. Repeating the orbit for years gives the change in mass, which for Antarctica came out as a loss of some hundreds of gigatons a year.

The second instrument is older and simpler. A tide gauge, meaning a fixed instrument recording sea level against a local benchmark, has been operating in some harbors since the nineteenth century. Gauges measure the sea where they stand, and the land they are bolted to may itself be rising or sinking, so each record must be corrected using independent measurements of ground movement.

The two methods fail in opposite directions, which is why they are used together. Gravimetry measures mass but cannot see where the water went, and it cannot separate ice loss from water moving in the rock beneath. Tide gauges measure what matters to a harbor but are clustered in the northern hemisphere and in rich countries. Satellite altimetry, which bounces radar off the sea surface, now provides a third series covering the whole ocean since 1992. Planning that aims to mitigate a rise depends on all three, since local sea level is what floods a street and global mass is what drives it. The first satellite pair was replaced in 2018 after the original instruments failed.

WHAT THIS PASSAGE IS FOR

- Satellite separation was measured to within a few microns.
 - The land a tide gauge is bolted to may itself be moving.
 - The two methods fail in opposite directions.
-

Physical Science

Passages 121 to 130

121

PHY · S01 · evidence

The Chain Back to a Standard

A reading on a laboratory balance means nothing by itself, and what makes it mean something is traceability, meaning an unbroken chain of comparisons linking the instrument to a national standard. The laboratory balance is checked against a reference mass, meaning a weight certified by comparison with a higher standard. That weight was certified against a better one held by a calibration service. The service compared its own masses against the national standard, and the national standard was compared, until recently, against a cylinder of platinum and iridium kept in a vault near Paris.

Every link in that chain adds a little uncertainty, and the whole point of the arrangement is that the additions are known and recorded. A certificate for a reference mass states its value and its uncertainty, and anyone who wants to calibrate an instrument against it inherits both. The chain also has to be maintained, because masses change: they gain a film of contamination, lose material to handling, and must be cleaned by a prescribed method that is itself part of the standard.

The platinum cylinder turned out to illustrate the weakness of defining a unit by an object. Comparisons across a century showed that the official kilogram and its official copies were drifting apart by tens of micrograms, and nobody could say which had changed. In 2019 the definition was replaced. The kilogram is now fixed by assigning an exact value to a constant of nature and building an apparatus that realizes the unit from it. A laboratory with the right equipment can therefore reproduce the standard, instead of shipping a weight to Paris. Four of the seven base units were redefined on the same day for the same reason. The arrangement is less convenient and very much harder to lose.

WHAT THIS PASSAGE IS FOR

- Traceability is an unbroken chain of comparisons to a national standard.
 - Every link adds uncertainty, and the additions are recorded.
 - The kilogram was redefined in 2019 because the metal cylinder drifted.
-

Weighing the World With a Wire

Gravity is the weakest force by an enormous margin, and measuring its strength between two ordinary objects is correspondingly difficult. Henry Cavendish did it in 1798 using a torsion balance, meaning a beam hung on a thin fiber which twists by a measurable amount under a tiny force. Two small lead spheres hung from the ends of a six-foot wooden beam. Two large lead spheres, each about three hundred and fifty pounds, were brought up beside them, and the beam turned through a fraction of a degree.

The whole experiment is a study in the control of error. The apparatus was enclosed in a case inside a shuttered room, and Cavendish observed the beam through a telescope from outside so that his body heat would not set up air currents. The twisting of the fiber was calibrated by timing the free oscillation of the beam, which converts a force measurement into a measurement of time. The large spheres were moved from one side to the other and the deflection measured in both directions, so that any constant bias canceled.

The result gave the density of the Earth to within about one percent of the modern figure, and from it the gravitational constant follows. Cavendish did not report the constant, which was not a quantity anyone used at the time, and his paper is titled as an experiment to determine the density of the Earth. The measurement remained the most accurate for about a century. The constant is still the least precisely known of the fundamental constants, because the empirical difficulty Cavendish faced has never gone away: the force is tiny, and everything in the room pulls. Modern versions use the same principle with better fibers and laser readout. Published values still disagree with one another by more than their stated uncertainties, which means somebody has an error nobody has found.

WHAT THIS PASSAGE IS FOR

- A torsion balance twists measurably under a tiny force.
 - Cavendish observed from outside the room to avoid air currents.
 - Moving the spheres both ways cancels any constant bias.
-

Where to Draw the Boundary

An efficiency figure cannot be stated until somebody chooses a system boundary, meaning the line drawn around whatever is being measured, with everything crossing it counted. Choose the boundary at the terminals of a motor made by Siemens or by any other firm and the efficiency is about ninety-five percent. Put it at the power station fence and the figure for the same motor doing the same work falls to around forty. Put it at the coal mine and it falls again.

None of those numbers is wrong. They answer different questions, and the usual dispute about any energy technology is a dispute about where the line should sit. An electric vehicle has no tailpipe, so inside a boundary drawn at the car it emits nothing at all. Drawn around the generating system it emits whatever that system emits, which varies by country and by hour. Drawn wider still, to include the mining of lithium and the manufacture of the battery, it acquires a debt before it has moved, and that debt is repaid over some number of miles that depends on everything inside the boundary.

The discipline that takes this seriously is life cycle analysis, meaning an accounting that follows energy and materials from extraction through manufacture and use to disposal. Its weakness is that the result depends on assumptions that are not themselves measurable: how long the product lasts, what it displaces, what happens at the end. Two careful analyzes of the same object can differ by a factor of two for that reason, and a difference between them that looks negligible in the summary can turn out to be the whole argument once the assumptions are printed. Standards bodies now publish rules for drawing the boundary in particular industries. The rules make studies comparable with each other without making any of them true.

WHAT THIS PASSAGE IS FOR

- An efficiency cannot be stated until a boundary is chosen.
 - The same motor is ninety-five percent or forty percent efficient.
 - Life cycle results depend on assumptions that are not measurable.
-

A Paddle Wheel and a Thermometer

In the 1840s James Joule set out to show that work and heat are two forms of the same thing, and the experiment he is remembered for is almost absurdly simple in conception. A falling weight turned a paddle wheel inside a sealed copper vessel of water. The water was stirred, it warmed, and Joule measured how much. Knowing the weight and the distance it fell gave the work done, and knowing the mass of water and the rise in temperature gave the heat produced. The ratio between them is the mechanical equivalent of heat.

The difficulty was entirely in the precision. The temperature rise in a single run was a fraction of a degree, so Joule had thermometers made that could be read to a two-hundredth. He corrected for the heat the vessel itself absorbed, for the heat leaking through the walls, and for the energy the falling weight still carried when it landed. He repeated the experiment with mercury instead of water, and by forcing water through tubes, and by compressing air, and obtained figures close to one another by three unrelated routes.

The wider significance is that it removed a category. Before Joule, heat was widely treated as a substance that could be transferred but not created, and any amount produced by rubbing had to be attributed to something being squeezed out. Afterward there was one quantity, energy, measured in one unit, and a bookkeeping rule that applies to a steam engine and a stirred bucket alike. Joule reported the figure to four significant digits in 1850, and the modern value differs from it by less than one percent. He is said to have attempted the measurement on his honeymoon, using a waterfall. The story is probably exaggerated and the thermometers were certainly real.

WHAT THIS PASSAGE IS FOR

- A falling weight turned a paddle wheel and warmed the water.
 - The temperature rise in one run was a fraction of a degree.
 - The result replaced two categories with one quantity.
-

Fifty Hertz as a Signal

An electricity grid, whether in Texas or in France, cannot store anything worth mentioning, so generation must match consumption continuously. The measurement that reveals any mismatch is the grid frequency, meaning the rate at which the alternating current reverses. Every large generator on a network turns in step. When demand exceeds supply the generators are being asked for more than they are receiving in steam or water, they slow very slightly, and the frequency falls below its nominal fifty or sixty cycles a second.

That single number is therefore a continuous readout of the balance of an entire country, and control rooms act on deviations of a few hundredths. The rate at which the frequency moves is governed by inertia, meaning the stored rotational energy of the large spinning machines, which acts like a flywheel and slows any change. A grid with many heavy turbines is forgiving. A grid supplied largely through electronics, as solar and wind are, has less of this stored motion, and the frequency can move faster than human operators can respond, which is why fast automatic reserves have become a salient part of system design.

The record is also used after the event. Because frequency wanders by tiny amounts in a pattern that is the same everywhere on a synchronous network, a recording of mains hum in the background of an audio file can be matched against the archived frequency record and dated to within a short window. Laboratories have used the method in criminal cases. A measurement maintained for the purpose of keeping lights on turns out to double as a clock that nobody designed. The frequency archive for several national grids is published and goes back decades. Its forensic use has been challenged on the ground that the archives themselves are not always complete.

WHAT THIS PASSAGE IS FOR

- Frequency falls when demand exceeds supply.
 - Inertia from heavy turbines slows any change in frequency.
 - Mains hum in a recording can be dated against the frequency archive.
-

From a Toothed Wheel to a Definition

That light takes time to travel was established before anyone could measure the time. Ole Romer noticed in 1676 that the eclipses of a moon of Jupiter ran late when the Earth was further away, and concluded that light has a finite speed, which was not the general opinion. The first terrestrial measurement came in 1849, when Hippolyte Fizeau sent a beam through the gaps of a rapidly spinning toothed wheel to a mirror five miles away and back. At the right speed the returning light struck the next tooth and vanished, which gave a time of flight, meaning a speed obtained by timing a signal over a known distance, from nothing but a measurement of rotation.

The method was refined for a century. Albert Michelson spent decades on it, using rotating mirrors over longer baselines, and by the 1920s the figure was known to a few parts in a hundred thousand. The limit was no longer the timing but the distance, because the length of a long baseline could not be measured as precisely as the time could.

That is why the problem was eventually dissolved rather than solved. In 1983 the meter was redefined as the distance light travels in a stated fraction of a second, which makes the speed of light a defined constant, meaning a quantity assigned an exact value. It is now exactly 299,792,458 meters per second by definition, and the question of measuring it does not arise. What laboratories measure instead is length, by timing light, which is the same experiment read backward. A constant of nature became a unit conversion, and the uncertainty moved from the speed to the ruler. The same move has since been made for the kilogram and for electrical units. Each redefinition takes a quantity that had been measured and makes it the thing against which measurements are made.

WHAT THIS PASSAGE IS FOR

- Romer inferred a finite speed from late eclipses in 1676.
 - Fizeau timed light with a spinning toothed wheel in 1849.
 - The speed of light is now exact by definition.
-

Gaps That Were Filled

A classification is not a discovery, and the periodic table, which is an arrangement of the elements placing chemically similar ones in the same column, could have been no more than a filing system. What made it something else was that Dmitri Mendeleev, publishing in 1869, left holes in it. His arrangement put the known elements in order of atomic weight and placed chemically similar ones in the same column, and to keep the columns consistent he had to leave three gaps and assert that elements existed to fill them.

He then did something a filing clerk could not. He predicted the properties of the missing elements from their positions, giving estimates for atomic weight, density, the formula of the oxide and the appearance of the metal. Gallium was isolated in 1875 and scandium in 1879, and germanium in 1886 matched his prediction for the gap below silicon closely enough that the chemist who found it said so in print. A table that can be wrong, and is then found to be right, is doing different work from one that merely files what is known.

The table also had defects that mattered. Ordering by atomic weight put a few elements in the wrong place, and the discrepancies were only resolved when atomic number, meaning the count of protons, replaced weight as the ordering principle in the 1910s. The underlying reason that the columns exist at all, which is that electrons occupy discrete shells with room for fixed numbers, was not available to Mendeleev in any form. He had a pattern and a nerve, and the explanation arrived half a century later. He also predicted elements that do not exist, and insisted for years on an error about the position of the rare earths. A table that generates testable claims generates wrong ones as well.

WHAT THIS PASSAGE IS FOR

- Mendeleev left gaps and asserted that elements existed to fill them.
 - He predicted the properties of the missing elements from their positions.
 - Atomic number replaced weight as the ordering principle.
-

How a Safety Factor Is Chosen

A material does not have one strength. Test a hundred nominally identical concrete cubes and they break over a range, because strength depends on flaws that are distributed by chance through the material. Engineers therefore do not design to the average. They use a characteristic strength, meaning a value below which only a small stated fraction of specimens fall, commonly five percent, which is obtained by crushing many samples and reading the result off the distribution.

The design figure is then reduced further by a safety factor, meaning the ratio between what the structure is built to withstand and what it is expected to carry. That factor is not a single allowance for ignorance. It covers several distinct things: the variability of the material, the difference between a laboratory specimen and a cast wall, uncertainty in the loads, the possibility of workmanship below specification, and the consequences of failure. Modern codes split it into separate factors applied to the load and to the material, so that each can be argued about on its own evidence.

The inherent limit of the method is that it handles only the hazards that have been listed. A factor of two on strength is no protection against a load nobody considered, which is what happened at the Tay Bridge, and no protection against a mechanism nobody knew about, which is what happened to the welded ships that fractured in cold water. Those failures were not corrected by raising the factor. They were corrected by adding the missing load case and the missing material property to the codes, after which the old factor was adequate. Codes are therefore revised after failures rather than before them, which is an uncomfortable fact about the discipline. The revisions are public, and the history of a code can be read as a list of accidents.

WHAT THIS PASSAGE IS FOR

- A material does not have one strength but a distribution.
 - The safety factor covers several distinct uncertainties.
 - A factor is no protection against a load nobody considered.
-

What Carbon Dating Can Date

Radiocarbon, meaning the heavy form of carbon produced continuously in the upper atmosphere, is taken up by every living thing through food or photosynthesis. While an organism is alive its proportion of radiocarbon matches the atmosphere. When it dies the intake stops and the radiocarbon decays with a half-life of about five thousand seven hundred years, so the proportion remaining measures the time since death.

Four conditions have to hold before a date from the Oxford or any other laboratory means anything. The sample must be organic and must have stopped exchanging carbon at the moment of interest, which is why the method dates the death of a tree rather than the building of the house. It must not be contaminated, and a trace of modern carbon makes an old sample look much younger. It must be young enough to measure, which in practice means under about fifty thousand years. And the atmospheric proportion at the time must be known, which it is not, because it has varied with solar activity and with nuclear testing.

The last condition is handled by a calibration curve, meaning a chart that converts a measurement into calendar years using samples whose age is known independently. Tree rings supply that independent age for the last twelve thousand years, one ring at a time. The curve is not a straight line, and in some periods it doubles back, so a single measurement can correspond to two or three possible calendar ranges. A date reported without its calibration and its range is therefore not a date. An anomalous result is more often a contaminated sample or a misunderstood context than a surprise about history. The curve is revised every few years as more tree ring and coral data arrive. Published dates from the 1950s and 1960s often have to be recalculated before they can be compared with modern ones.

WHAT THIS PASSAGE IS FOR

- The method dates the death of the organism, not the use of the object.
 - A trace of modern carbon makes an old sample look much younger.
 - The calibration curve doubles back, so one measurement can give several ranges.
-

A Rule Found in a Pile of Plates

Henrietta Leavitt was employed at Harvard College Observatory to measure the brightness of stars on photographic plates, meaning glass plates coated with emulsion on which images of the sky had been recorded. The work was paid by the hour and the women who did it were called computers. Leavitt was assigned the Magellanic Clouds, and she worked through thousands of plates identifying stars that varied in brightness and measuring how long each took to complete a cycle.

By 1908 she had found hundreds of them, and she noticed that the brighter ones took longer. In 1912 she stated the relation precisely for twenty-five stars: the logarithm of the period varies with the brightness along a straight line. Because all the stars in question lay in the same cloud, they were all at roughly the same distance, so differences in their apparent brightness had to be differences in their real output. That is what made the relation visible there and nowhere else.

The consequence is the single most useful tool in astronomical distance measurement. Find such a star anywhere, time its cycle, and its true brightness follows from the relation; compare that with how bright it looks and the distance follows. Edwin Hubble used it in 1924 to show that the Andromeda nebula lies far outside our own galaxy, which settled a famous argument and enlarged the known universe by a factor nobody had expected. Different calibrations of Leavitt's law converge now within a few percent. Leavitt was not given a share of that work, and she died in 1921. Her name is on the relation in modern textbooks, which was not true for most of the century. The plates she used are still held at Harvard and have been digitized. Her fundamental contribution rests on nothing more exotic than systematic measurement, repeated across thousands of photographic exposures.

WHAT THIS PASSAGE IS FOR

- Leavitt measured brightness on glass plates as paid piece work.
 - Brighter pulsating stars take longer to cycle.
 - Stars in one cloud are at the same distance, which made the relation visible.
-

Humanities

Passages 131 to 140

131

HUM · S01 · evidence

Proving a Narrator Unreliable

Calling a narrator unreliable, meaning a teller whose account the reader has reason to doubt, is a claim that has to be argued from the page rather than asserted. Three kinds of evidence are normally offered, and they differ in strength. The weakest is tone: a reader finds the voice distasteful or self-serving. That is a response and not a proof, since a writer may intend an unpleasant narrator to be entirely accurate.

The second kind is internal contradiction, meaning a place where the text disagrees with itself. Ford Madox Ford gives the narrator of *The Good Soldier* a date that cannot be right and a chronology that will not assemble, and the errors are too systematic to be accidents of composition. The third and strongest kind is a discrepancy between what the narrator reports and how other characters respond. When a narrator describes a mild remark and everybody in the room recoils, the text has supplied a second witness.

The method shows its limits in the long argument over *The Turn of the Screw*, published in 1898. Either the governess sees ghosts or she is describing her own mind, and a century of critics has produced detailed cases for both. The difficulty is that James withheld the second witness: nobody else in the story reports seeing or not seeing anything in terms that settle it, and the one piece of apparent confirmation comes through the governess herself. A text can therefore be enigmatic by construction rather than by accident, and the honest conclusion in such a case is that the question is undecidable on the evidence available. Edmund Wilson argued the psychological reading in 1934 and changed his mind twice in later editions. That record of revision is itself a useful document about how such arguments are made.

WHAT THIS PASSAGE IS FOR

- Tone is a response rather than a proof.
 - A text that disagrees with itself supplies internal evidence.
 - James withheld the second witness, so the question cannot be settled.
-

The Chapter Austen Canceled

Almost no manuscript survives of any completed Jane Austen novel, which makes one exception valuable. Two chapters of *Persuasion* exist in her own hand, and they are the chapters she threw away. The published version carries the famous scene in which Captain Wentworth writes a letter while overhearing a conversation he is not part of, and then puts the folded paper into Anne Elliot's hand before leaving the room. The canceled chapter, meaning the draft section she replaced, handles the same reconciliation through an awkward interview arranged by a third party.

Having both lets a reader see a decision being made. In the draft the feeling is reported and then discussed by the parties concerned, whereas in the revision it is enacted in a public room, with two conversations running at once and a character writing down what he cannot say aloud. The emotional content is the same and the method is entirely different, and the second version turns Wentworth from a man who explains himself into a man who is overheard. That shift is the kind of thing critics assert about writers in general. Here it can be shown.

A manuscript will not answer every question. The canceled pages are undated and the order of composition has to be inferred from the paper and the stitching. Austen died a few months after finishing the novel, so there is no later statement of intention, and the family edited what survived. A draft also shows a writer being ambivalent without recording why. What the evidence establishes is that the alternative existed and was rejected, which is a narrower claim than most of what is written about authorial intention, and very much firmer. The pages are now in the British Library and have been published in facsimile. They are the only substantial draft of a finished Austen novel known to exist.

WHAT THIS PASSAGE IS FOR

- Two canceled chapters of *Persuasion* survive in her own hand.
 - The revision turns an explanation into an overheard scene.
 - The draft shows that an alternative existed and was rejected.
-

Counting the Images

A claim that a play is full of images of disease used to be a matter of impression. The tool that changed it is the concordance, meaning an index listing every occurrence of every word in a text together with its context. Concordances to the Bible were compiled by hand in the Middle Ages, and one to Shakespeare took a Victorian clergyman and his family some sixteen years. Anyone with such an index can check whether an impression survives contact with the text.

Caroline Spurgeon did the work for Shakespeare in 1935, classifying several thousand images by subject and comparing the counts across plays. Some results confirmed what readers had felt: Hamlet does carry an unusual density of images of sickness and rot. Others corrected it. The computer made the method cheap, and a modern researcher can search a corpus, meaning a body of texts assembled to be searched as a whole, of everything printed in English before 1700 in an afternoon.

What counting establishes is presence, distribution and comparison, and those are not small things. What it cannot establish is aesthetic effect. A count shows that a word appears forty times; it cannot show that the forty occurrences work on a reader, or that an image placed at a turning point carries more weight than four placed in passing. The two methods therefore have to be used together, which is the current practice: counting to find where to look, and reading to say what is there. A claim that survives both is about as secure as literary claims get. Spurgeon's work has been criticized for categories that reflect her own period more than the plays. Her counts, however, have been checked and are substantially accurate. The combination is now standard practice in any serious study of a writer's vocabulary, and the counting is usually done first because it is cheap.

WHAT THIS PASSAGE IS FOR

- A concordance indexes every occurrence of every word with its context.
 - Spurgeon classified several thousand images and compared plays.
 - Counting establishes presence and distribution, not effect.
-

What Scansion Can Prove

Scansion, meaning the marking of stressed and unstressed syllables in a line, looks like the least consequential of literary skills, and it has produced one of the field's most solid results. Shakespeare's plays are not dated by their title pages, most of which do not exist, and the order in which he wrote them was established largely by counting metrical features.

Three measurable features change steadily across the career of William Shakespeare. The proportion of lines that run on without a pause at the end rises. The proportion of feminine endings, meaning lines that finish on an unstressed extra syllable, rises from under ten percent in the early plays to over thirty in the late ones. The proportion of lines with a break away from the middle rises as well. Counting these in plays whose dates are known from other evidence, such as records of performance, gives a curve against which an undated play can be placed. The method puts the plays in an order that agrees with external evidence wherever external evidence exists, which is the test of it.

Its limits are strict. The method dates a style, not a manuscript, so a play revised late looks late. It cannot distinguish collaboration from development, which is why the authorship of several scenes remains contested. And it says nothing about quality: an understated late style is not better or worse than an emphatic early one, only different, and the counting is silent on the question. What scansion buys is a chronology, which every other kind of argument about development then depends on. Modern work adds function word frequencies and rare collocations to the metrical measures. The results agree with the older counts in most places and have reassigned two or three plays.

WHAT THIS PASSAGE IS FOR

- Feminine endings rise from under ten percent to over thirty.
 - The order agrees with external evidence wherever it exists.
 - The method dates a style rather than a manuscript.
-

What the Promptbook Records

A printed play is a text, whereas a performance is an event, and the documentary record of events is both thinner and considerably stranger. The most useful document is the promptbook, meaning a working copy marked up by the stage manager of one production, and major collections hold thousands of them. A promptbook carries the cuts, the entrances and exits as actually performed, the cues for music and effects, the positions of the furniture, and sometimes the timings of each act written in pencil in the margin.

What it reveals most often is the extent of cutting. Nineteenth-century productions of Shakespeare routinely removed a third of the lines, rearranged scenes and sometimes changed the ending, and the promptbooks show precisely what was removed and when it came back. Hamlet was performed for a century in a version shorter than the printed play by an hour. A reader who knows only the canonical text has an incomplete picture of what audiences actually saw.

The record also has characteristic gaps. A promptbook describes what was planned and revised in rehearsal rather than what happened on a particular night. It does not record how a line was delivered, since no notation for that exists. Reviews, letters and memoirs fill some of the space, and all three are unreliable in the ordinary and familiar ways. Photographs from the later nineteenth century show poses rather than performances, because the exposures were too long for movement. The result is a documentary record good enough to establish what was done and almost never good enough to establish how it felt. Recordings change this completely from the middle of the twentieth century onward, and the record after that date is of a different kind altogether. They also create a new problem, since a filmed performance is itself an edited object.

WHAT THIS PASSAGE IS FOR

- A promptbook records cuts, cues, positions and timings.
 - Nineteenth-century productions often cut a third of the lines.
 - No notation exists for how a line was delivered.
-

Tracing Where a Form Began

The claim that a genre began with a particular book is a bibliographic claim before it is a critical one, and bibliography, meaning a systematic list of everything published of a kind in a period, is what tests it. Catalogues of early English printing list tens of thousands of items with dates, printers and formats. Trade records survive. Circulating library catalogues from the eighteenth century show what was actually borrowed, which is a different question from what was printed.

Those sources support the usual account of the gothic novel reasonably well. The Castle of Otranto appeared in 1764, presented as a translation from an Italian manuscript, and the second edition admitted the fiction. A recognizable set of imitations follows within twenty years, and library records show them being borrowed in quantity. The pattern of a first instance, a gap, and then a rush of derivative work is what one would expect if a form were being established, and it can be seen in the dates.

The hazard throughout is survival bias, meaning the distortion caused by studying only what happened to be preserved. Cheap print survives very badly. Chapbooks, broadside ballads and the serial fiction of the eighteenth century were read to pieces, and much of what remains survives because a collector thought it respectable. Any claim that a device first appeared in a particular novel therefore means that it first appears in the novels we still have. Historians of popular forms treat the record as a sample with a known bias rather than as the thing itself. Newspaper digitization has moved several supposed first appearances earlier by decades. Each revision has come from material that existed all along in uncatalogued runs of periodicals. Any history of a popular form is therefore a history of its surviving instances, and the honest versions say so in the first chapter.

WHAT THIS PASSAGE IS FOR

- Library catalogues show what was borrowed, not only what was printed.
 - A first instance, a gap, then a rush of imitation is visible in the dates.
 - Cheap print survives very badly, which biases the whole record.
-

Under the Varnish

A painting is a physical object as well as an image, and physical objects can be examined, so three laboratory techniques now do most of the work. X-radiography shows the denser pigments and the structure of the support, so a figure painted out fifty years later appears as a ghost. Infrared reflectography, meaning a method that sees through upper paint layers to the drawing underneath, reveals the artist first thoughts, including changes of position that were never meant to be seen. Pigment analysis takes a sample smaller than a full stop and identifies the materials from which the paint was made.

Each supports a different kind of claim. Pigment analysis dates: Prussian blue was not manufactured before about 1706, so a picture containing it was not painted before then, and a great many attributions have been settled by that one fact. Dendrochronology, meaning the dating of a wooden panel by matching its ring pattern against a regional record, gives the earliest year the panel could have been painted on. Reflectography attributes, because underdrawing habits are personal and hard to imitate.

The methods cannot answer the questions people most want answered. They can show that a panel is of the right date and that the materials are consistent, and they cannot show that a particular hand made a particular brushstroke. A skilled forger who uses old panels and correct pigments defeats all three, and several have. The practice is therefore to combine physical evidence with documentary history and with connoisseurship, and to treat a reverent attribution unsupported by any of the three as a hypothesis. Several paintings have changed status twice in a century as the balance of evidence moved. Museums now publish technical reports alongside catalogue entries. The reports often contain the evidence that will be used to overturn their own conclusions later.

WHAT THIS PASSAGE IS FOR

- X-radiography shows figures painted out later as ghosts.
 - Prussian blue was not manufactured before about 1706.
 - The methods cannot show that a particular hand made a brushstroke.
-

Eight Thousand Pages of Sketches

Beethoven kept his working papers, and about eight thousand pages of them survive. The sketchbooks, meaning his working notebooks of fragments and rejected versions, show melodies arriving in forms that are almost embarrassing beside the finished article, then being worked for months. The opening of the Fifth Symphony exists in several trials, none of which a listener would recognize as the figure everybody now knows. A theme from the Ninth Symphony appears in one of the notebooks fully twelve years before the work itself was written. The evidence contradicts the idea of inspiration arriving whole, and it does so in the handwriting of the person most often cited as proof of it.

Manuscripts also force decisions on everyone who performs the music. The autograph score, meaning the full score in the composer's hand, often differs from the first printed edition, and the composer sometimes corrected the proofs and sometimes did not. Where they disagree, an editor has to choose, and the choice is published as though it were the composer's wish. Beethoven's metronome marks survive for some works and have been thought impossibly fast for a century, and the argument over whether to obey them is unresolved.

The limits of the evidence are specific. A sketch shows what was tried and not why it was rejected. Dating depends on the paper, the watermarks and the ink, which is skilled work and not always decisive. And none of it records sound. A score is a set of instructions whose conventions of performance were partly unwritten and are now partly lost, which is why an elegiac tempo in one century becomes a brisk one in the next without a note being changed. Facsimiles of the main sketchbooks are now published and some are online. Several long-standing editorial readings have been revised since performers could consult them directly.

WHAT THIS PASSAGE IS FOR

- About eight thousand pages of working papers survive.
 - A theme from the Ninth appears twelve years early in a notebook.
 - A sketch shows what was tried, not why it was rejected.
-

Booth Walked Every Street

Between 1886 and 1903 Charles Booth and a team of investigators walked every street in London and recorded what they found. The result was a set of poverty maps, meaning street plans colored house by house, with seven classes from the wealthy to what Booth labeled the lowest class of occasional laborers and criminals. The maps were based on school board visitors' knowledge of individual families, on police walks, and on interviews, and they run to seventeen volumes of notes alongside the sheets.

They are the first systematic measurement of a city and they carry their period inside them. Booth's classifications mixed income with moral judgment, his categories were applied by middle-class observers walking through streets they did not live in, and the notebooks contain remarks that no modern survey would print. The maps are nonetheless accurate enough about rents, trades and overcrowding to be used by historians, and they can be compared street by street with modern deprivation figures.

Measurement today is cheaper, narrower and continuous. Footfall counts, meaning tallies of how many people pass a point, come from sensors and mobile telephone data. Transport cards record journeys. Satellite images give building footprints and tree cover. Each series is precise about one thing and silent about the rest, and none of them contains anything like Booth's notebooks, where an investigator records that a street is quiet, or respectable, or whimsical in its decoration. The modern city is measured better and described worse, and the two kinds of evidence answer different questions. Booth's maps have been digitized and can be laid over a modern map in a browser. A majority of the street classifications still predict present-day deprivation measures. Historians of the modern city therefore have better numbers and thinner description than Booth had, and have to read both kinds of record to say anything much about how a street actually worked.

WHAT THIS PASSAGE IS FOR

- Booth team walked every street in London and recorded what they found.
 - His classifications mixed income with moral judgment.
 - Each modern series is precise about one thing and silent about the rest.
-

Reputations That Moved

Critical judgments can be tested against the record of what happened to them, and the discipline that does this is reception history, meaning the record of how a work was read and valued over time. Its materials are reviews, sales figures, library borrowings, school syllabuses, editions in print, and the frequency with which a writer is quoted.

Two cases are standard. Herman Melville died in 1891 with his major novel out of print and his obituary in the New York Times misspelling his name. A deliberate campaign of rediscovery in the 1920s, driven by a few critics and a new edition, installed him within twenty years as a central American writer, and he has stayed there. John Donne was held in low regard for most of the eighteenth and nineteenth centuries and was raised by a polemic from T. S. Eliot and others after 1920. In both cases the texts had not changed. The criteria had.

This is evidence about criticism rather than about the works, and it has to be handled with care. That a judgment changed does not show that either version was wrong, and the fact that a reputation rose does not mean the critics who raised it were right. What the record does establish is that standards are historical objects with datable origins, which makes a certain kind of critical claim impossible to maintain, namely that a particular standard is simply self-evident. Anyone who wants to make that claim has to explain the record of its absence. Databases of library borrowings and school curricula now make such histories quantitative. They have confirmed several reputational stories and demolished others, including some about neglect that turned out to be overstated. The record of a standard arrival is also a record of what it displaced, which is the part most often left out of the story.

WHAT THIS PASSAGE IS FOR

- Reception history uses reviews, sales, borrowings and syllabuses.
 - Melville was out of print at his death and central twenty years later.
 - A changed judgment does not show which version was wrong.
-

Social Science

Passages 141 to 150

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How a Modern Poll Is Built

A serious poll is built in four stages, and a reader who knows the stages can judge the result. The first is the frame. Address-based sampling, used by Pew and by most serious firms, begins from a list of residential addresses, which covers almost every household, and letters are sent inviting people to answer online or by telephone. Telephone sampling generates numbers at random, including unlisted ones, which is why pollsters still ring mobile numbers at considerable expense.

The second stage is selection, and the aim is a probability sample, meaning one in which every member of the population has a known chance of being chosen. That known chance is what licenses the arithmetic afterward. The third stage is weighting, meaning the adjustment of answers so that the sample matches known population proportions for age, sex, region, education and sometimes past vote. Weighting is where judgment enters, because the pollster must decide which characteristics matter, and education turned out to matter a great deal more after 2016 than most firms had assumed.

The fourth stage is the reported margin, and it describes only the first two stages. A margin of error of three points on a thousand interviews is a statement about sampling variation under the assumption that the frame is right and the weights are right. It makes no allowance for a frame that misses people, for the ninety percent who did not answer, or for the wording of the questions. A methodical firm publishes all four stages and its weighting targets, which is the only way a reader can tell a careful poll from a confident one. Several national polling associations now require that disclosure as a condition of membership. Firms that decline to join are not thereby wrong, but they are harder to check.

WHAT THIS PASSAGE IS FOR

- A probability sample gives every member a known chance of selection.
 - Weighting is the stage at which the pollster judgment enters.
 - The reported margin describes only sampling variation.
-

Two Lotteries and a Preschool

In 2008 Oregon had money to extend health coverage to about ten thousand adults and roughly ninety thousand people applied. The state allocated the places by lottery, which created, without intending to, the cleanest comparison available anywhere: two large groups alike in every respect except coverage. Researchers followed both. Those who gained coverage used more health care, were much less likely to face catastrophic medical bills, and reported better mental health. Measured blood pressure and cholesterol after two years showed no statistically significant difference, which was not the result most people had expected and is the most argued-over part of the study.

The Perry Preschool study is older and smaller. In the 1960s a hundred and twenty-three poor children in Ypsilanti, Michigan, were assigned by lot to a high-quality preschool program or to nothing, and have been followed ever since. Differences in measured intelligence faded by the early school years, which was read at the time as failure. The follow-up period, meaning how long participants are tracked, turned out to be the crucial design choice. By age forty the treated group had higher earnings, more home ownership and fewer arrests, and the benefit-cost ratios computed from those outcomes have shaped policy for twenty years.

Both studies support causal claims because of the lottery. Both are limited by the same thing: each tells us about one program, in one place, at one time, for one population. The Perry sample is a hundred and twenty-three children in a single town in the 1960s, which is a slender base for a national argument, and the honest practice is to say so while using the result. Several later preschool trials have found smaller effects, and the differences are still being argued about. What is not in dispute is that the effects were measured after the children had grown up.

WHAT THIS PASSAGE IS FOR

- Oregon allocated coverage by lottery and created a clean comparison.
 - Perry effects on measured intelligence faded but adult outcomes differed.
 - Each study tells us about one program in one place at one time.
-

When Chance Does the Assigning

Where a lottery cannot be run, researchers look for one that has already happened by accident. A natural experiment, meaning a situation in which something outside the researchers control has allocated people almost at random, is now the main instrument of modern social science. Three kinds of situation account for most of the published work, and each carries a characteristic weakness that a careful reader can look for before accepting the result.

The first is a genuine lottery used for some other purpose. The military draft lotteries of the Vietnam era assigned young American men a number by date of birth. Because that number was random, comparing the later earnings of men with low and high numbers isolates the effect of service from everything that would otherwise confound it. The second is the discontinuity. Where a program has an arbitrary cutoff, such as a score of sixty for a scholarship, applicants who scored fifty-nine and sixty-one are indistinguishable in every respect except the award. Regression discontinuity, meaning the comparison of cases just above and just below such a line, exploits exactly that. The third kind is the border, since the two sides of a state line differ in law while remaining similar in almost every other respect.

Each rests on an assumption that has to be defended rather than assumed. The draft numbers were random, but service was not, since men could enlist or evade, so what the comparison measures is the effect of being more likely to serve. Discontinuities assume that nobody manipulated the score, which is sometimes false and can be tested by looking for a suspicious bunching of scores just above the line. Borders assume that people do not move across them in response to the policy, which they sometimes do. The spread of the method has produced a literature of cautionary cases in which the assumption failed quietly.

WHAT THIS PASSAGE IS FOR

- A natural experiment uses an allocation made by something outside the researcher.
 - Scores either side of an arbitrary cutoff are otherwise indistinguishable.
 - Each kind rests on an assumption that must be defended.
-

What a Margin of Error Says

A confidence interval is a statement about a procedure rather than about a number. When a Gallup poll reports forty-eight percent with a margin of three points, the claim is narrower than it looks. The method used, repeated many times on fresh samples, would produce an interval containing the true value in about ninety-five cases out of a hundred. It does not say that there is a ninety-five percent chance that the truth lies between forty-five and fifty-one, although almost every newspaper sentence on the subject says exactly that.

Three misuses follow from the confusion. The first is treating two overlapping intervals as showing no difference, which does not follow, and treating two separated intervals as showing one, which requires its own test. The second is reading a result that is marginal as though significance were a property of the world. Statistical significance means the judgment that a result is unlikely to have arisen from sampling variation alone, and it depends on the sample size as much as on the effect. A trivial difference becomes significant in a large enough study, and an important one stays invisible in a small one.

The third misuse is the comparison nobody declared. If twenty subgroups are examined and one shows a significant difference, that is what twenty tests produce by chance about once in every set. Trials registered in advance, stating which comparison is the main one, exist to close that door. A reader who wants to judge a reported figure therefore needs the interval, the sample size, and the number of comparisons that were made before this one was chosen for the headline. Several journals now require that the pre-registered comparison be identified in the abstract. The requirement has made a visible difference to the proportion of positive findings reported.

WHAT THIS PASSAGE IS FOR

- A confidence interval describes a procedure, not a single number.
 - Significance depends on sample size as much as on the effect.
 - Twenty tests produce one chance result per set.
-

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SOC · S05 · evidence

Measuring What a Price Did

A city such as San Francisco that changes a price and then measures what happened has not measured the effect of the price. Something else always changed in the same period: the weather, the economy, a new office building, the price of fuel. The standard remedy is difference in differences, meaning the comparison of the change in a treated place with the change in an untreated one over the same period. If parking occupancy fell eight percent in the districts with new rates and two percent in otherwise similar districts without them, the estimate of the effect is six points, not eight.

The quantity usually reported is an elasticity, meaning the percentage change in quantity produced by a one percent change in price. Elasticities transfer between settings better than absolute figures do, and they are dangerous because they are not constants. The same good is more elastic over a year than over a week. It is more elastic for people with substitutes available, and more elastic at high prices.

The method depends on a condition that can be stated and tested: that the two areas would have moved in parallel had nothing been done. Analysts test it by checking whether the two series ran parallel in the years before the change, which is weak evidence and better than none. The design fails where the untreated area is affected by the treatment, as when drivers simply move to the next district, and that spillover will make a real effect look larger than it is. A latent change already under way in one area and not the other is the other standard way such a study goes wrong. Where several places adopt a policy at different times, the comparisons can be stacked, which strengthens the design.

WHAT THIS PASSAGE IS FOR

- Measuring before and after does not isolate the effect of a price.
 - An elasticity transfers between settings better than an absolute figure.
 - The design fails when drivers move to the untreated district.
-

Three Lines on a Card

Solomon Asch asked young men to compare the length of lines, and the task he set them was as simple as it sounds. Each was shown a standard line and three others, one of which plainly matched, and was asked to say aloud which one it was. The task was easy to the point of being insulting. Seven other people in the room gave their answers first, and all seven were confederates, meaning people who appeared to be participants and were working for the researcher. On twelve of eighteen trials they gave the same wrong answer.

The figures are often reported loosely and are worth having exactly. About a third of the responses by real participants went along with the wrong majority. About a quarter of participants never conformed at all, and around three quarters conformed at least once. Asch also ran variations, and these are the most informative part of the work. Conformity collapsed when one confederate gave the correct answer, and fell sharply when participants could write their answers privately instead of speaking them.

Replication, meaning independent repetition, has qualified the finding in two directions. Studies in other countries have produced rates from near zero to above the original, and a British replication in 1980 found almost no conformity among science students, which its authors attributed to a change in how such a setting was understood. Later analysis found the effect larger in collectivist societies and smaller in later decades. What survives is not a constant but a mechanism: people will give a public answer they privately disbelieve when the normative cost of disagreeing is visible, and the size of the effect depends on who is in the room. The original work used only male college students, which was normal at the time and limits what it establishes. Several of the variations have never been repeated at all.

WHAT THIS PASSAGE IS FOR

- The confederates gave the same wrong answer on twelve of eighteen trials.
 - Conformity collapsed when one confederate answered correctly.
 - What survives is a mechanism rather than a constant.
-

Reading a Census Carefully

A census of the kind run by the United States is the best social data that exists, and it is not simple to use. Three features complicate every comparison. The first is the undercount, meaning the people a census fails to record. The net figure for a large modern census is often about one percent, which sounds negligible until it is broken down, because the undercount is not spread evenly. Young men, renters, recent migrants and children in complex households are missed at several times the average rate, so the undercount changes the apparent composition of a place and not merely its size.

The second feature is the boundary change, meaning an alteration of the areas for which figures are reported. Tracts are split as they grow and merged as they shrink. A table that compares a neighborhood in two censuses may therefore be comparing two different pieces of ground. Statistical agencies publish conversion tables for this purpose, and the conversions they allow are approximate rather than exact.

The third complication is that the questions themselves change from one census to the next. Categories for race, ethnicity, occupation and household relationship have been rewritten repeatedly, usually for good reasons, with the consequence that a series apparently running for a century is in places a series of different questions. A heterogeneous category in one year may appear as three separate categories in the next, and the two tables cannot then be placed side by side. None of this makes the data unusable, and the agencies document all of it in detail. A comparison across decades is therefore a piece of work rather than a lookup. The commonest error is to treat a table as though the words at the top of its columns had always meant the same thing, which they have not.

WHAT THIS PASSAGE IS FOR

- The undercount is about one percent and is not spread evenly.
 - A neighborhood compared across censuses may be different ground.
 - A long series is in places a series of different questions.
-

Two Sides of One State Line

New Jersey raised its minimum wage in April 1992. David Card and Alan Krueger saw a natural experiment in it, because eastern Pennsylvania sits across a state line with similar towns, similar labor markets and no change in its own minimum. They surveyed over four hundred fast food restaurants on both sides before and after the increase, asking the same questions each time, and found no fall in employment in New Jersey relative to Pennsylvania. The standard prediction from the textbook model is that employment should have fallen.

The response was immediate and technical. David Neumark and William Wascher obtained payroll records, meaning employment figures taken from the administrative accounts of firms, for a sample of the same restaurants, and found that employment had fallen. The two teams were therefore disagreeing about measurement rather than about theory. Survey data may suffer from respondents who misremember or who answer for a chain rather than a store. Payroll data covers only the firms that supply it, and the sample was assembled with help from a restaurant trade association, which the other side pointed out at length.

Neither paper settled the matter and both improved the field. The dispute drove a generation of work using administrative data, borders, and hundreds of separate minimum wage changes, and the current position is that modest increases have small employment effects whose sign depends on the local labor market. The reciprocal checking of methods that the exchange forced is now standard practice, and the two principals shared a Nobel prize in 2021 with a third economist for work of exactly that kind. Both data sets were eventually released, which is why the exchange could be adjudicated at all. Several later studies have replicated the comparison with much larger administrative samples.

WHAT THIS PASSAGE IS FOR

- Pennsylvania supplied a comparison across a state line.
 - The two teams disagreed about measurement rather than theory.
 - Each data source has its own characteristic weakness.
-

One Number for a Distribution

The standard measure of inequality begins with a graph. The Lorenz curve, meaning a plot of the cumulative share of income held by the poorest given share of the population, runs from the bottom left to the top right of a square. Perfect equality would be the diagonal of that square, where the poorest tenth hold exactly a tenth of the income, and every real distribution sags below it. The Gini coefficient, meaning a single figure summarizing how far a distribution departs from equality, is the area between the curve and the diagonal, scaled to run from zero to one.

The convenience of one number is also its defect. The same Gini value can be produced by very different distributions. A country where the top one percent hold a great deal and the rest are similar to one another can share a coefficient with a country where the gap runs through the middle, and policies that would reduce one disparity will not touch the other. The measure is also least sensitive at the two ends, which is where the political argument usually is.

Alternatives exist and are chosen for what they expose. A percentile ratio compares the income at the ninetieth percentile with the tenth, which is easy to interpret. The Palma ratio compares the top tenth with the bottom four tenths, on the argument that the middle half changes little. The Theil index can be decomposed into inequality within regions and inequality between them, which no single-number measure otherwise allows. Serious work reports more than one, and a report that gives a Gini alone is giving the least informative summary available. Comparisons across countries are harder still, because the underlying surveys differ in what they count as income. Whether housing, benefits in kind and capital gains are included moves a national figure by several points.

WHAT THIS PASSAGE IS FOR

- The Gini is the area between the Lorenz curve and the diagonal.
 - The same Gini value can come from very different distributions.
 - A Theil index can separate inequality within and between regions.
-

Which Results Held Up

The judgment and decision literature was built from small studies, often on students, analyzed after the fact. Beginning around 2011 psychologists at Virginia and elsewhere organized systematic attempts to repeat the best-known findings, and the results separated the field into parts. Large coordinated replications confirmed the core effects with reduced magnitudes. Anchoring survived robustly, framing survived, and loss aversion survived with its size disputed. Several widely taught results did not survive at all, including some that had appeared in textbooks for twenty years.

Two mechanisms were identified as responsible. The first is publication bias, meaning the tendency for positive results to be published while null results stay in drawers, and it inflates the apparent size of every effect in a literature. The second is flexibility in analysis. A researcher who may choose when to stop collecting data, which outliers to exclude, and which of several outcomes to report can produce a significant result from noise without any dishonesty. The number of available choices was shown to be large enough to make the problem endemic.

The remedies that followed were procedural rather than theoretical. Preregistration, meaning the publication of a design and analysis plan before the data are collected, removes most of that flexibility at a stroke. Registered reports go further, since a journal accepts the study on its design before any results exist, which removes the incentive to produce a positive finding at all. Published effect sizes in preregistered work are smaller and more stable than in the older literature. The episode is often described as a crisis and is better understood as a measurement of the field, carried out by the field, whose main result is that the strongest effects are real and smaller than advertised. Economics, medicine and political science have since run versions of the same exercise with much the same result.

WHAT THIS PASSAGE IS FOR

- Large coordinated replications confirmed core effects with reduced magnitudes.
 - Analytic flexibility can produce significance from noise without dishonesty.
 - Preregistered effect sizes are smaller and more stable.
-

LEVEL 4

Stretch

dispute

What is contested: two readings of one body of evidence, and the writer's stance.

Passage length — 288 to 312 words

Sentences — averaging 21 to 28 words

Vocabulary assumed — the top 40,000 most frequent word forms

Words above that — at most 6, explained or inferable

Clue to the reader — Read for the disagreement. One body of evidence, two readings of it. The writer takes a side, and the passage is built so that the side taken can be told apart from the evidence it rests on.

History and Civics

Passages 151 to 160

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HIS · S01 · dispute

One Sentence, Two Readings

The sentence about equality has been read in two incompatible ways by people who agreed about the words, and the dispute matters because the whole subsequent argument about American citizenship ran through it. On the first reading the Declaration states a universal principle to which the nation is committed, whatever its practice, and the gap between the words and the conduct is a national failure to be remedied rather than evidence about what the words mean. That was Douglass's position in 1852 and Lincoln's position at Peoria two years later.

On the second reading the clause was written by men who held slaves, for an audience that held slaves, in a document whose purpose was separation from Britain, and it therefore cannot have meant what the first reading requires. Chief Justice Taney put that argument into law in 1857, holding that the framers had not included Black people in the word men and that no contemporary would have thought otherwise. That is the position called originalism, which is the view that a text means what its authors and first readers understood it to mean, and it is not a careless reading.

Each side has to qualify something. The first must explain why no signer acted on the principle at the time, and the honest version concedes that the words outran their authors. The second must explain the deletion of the slave trade passage, the immediate use of the sentence by abolitionists within a decade, and the fact that several signers freed people in their wills. The dispute cannot be settled from the text alone, which is why it was settled by an amendment that made the first reading law in 1868. Taney's reasoning was repudiated in the Fourteenth Amendment and has never been formally overruled by a court.

WHAT THIS PASSAGE IS FOR

- The two readings agree about the words and not about the meaning.
 - Taney argued that no contemporary would have included Black people.
 - The dispute was settled by amendment rather than by interpretation.
-

Protection or Paralysis

Whether the American system of divided powers protects liberty or prevents government is the oldest live argument in American political science, and both sides are arguing from the same design. The defense is the one Madison made. Power divided among institutions with different electorates, different terms and different incentives cannot be captured at once, so a majority with a bad idea is checked at several points and a persistent majority with a good one eventually prevails. On this reading the delay is the protection, and the evidence offered is the number of occasions on which something unwise was stopped.

The criticism begins from the same veto points, meaning the places where one actor can block a decision, and counts them differently. A system with two chambers, a presidential veto, judicial review, a Senate requiring sixty votes for most business, and fifty state governments has a great many such points. Critics argue that the result is gridlock, meaning a condition in which the system cannot act although majorities exist, and that the consequence is not careful government but government by crisis, executive order and court ruling, since those are the routes that remain open.

The evidence is genuinely ambiguous, because the two sides count different things. Defenders point to measures blocked; critics point to problems unaddressed, and neither list can be compiled without a judgment about what should have happened. Comparative work is the strongest available evidence, and it shows that countries with fewer veto points legislate faster in both directions, passing both the reforms their critics feared and the ones their defenders wanted. That finding is uncomfortable for everyone and is about as far as the data reach. Political scientists have counted veto points across democracies since the 1990s and the measures broadly agree with one another.

WHAT THIS PASSAGE IS FOR

- Both sides argue from the same design.
 - On the defense, the delay is the protection.
 - Countries with fewer veto points legislate faster in both directions.
-

The Movement Argued With Itself

The suffrage campaign divided over means rather than over ends, and the division ran deep enough to produce two rival national organizations that worked separately for twenty years. One wing held that the vote should be won state by state, meaning one legislature at a time, on the argument that a federal amendment required three quarters of the states and could be blocked by a minority of them. Wyoming, Colorado, Utah and Idaho did in fact enfranchise women decades before the federal amendment passed, which remains the strongest evidence for that view.

The other wing held that state campaigns were a trap, because each one had to be fought in full and could be reversed, and that only a constitutional amendment would settle the matter nationally. Within that wing a further split opened over tactics. Carrie Chapman Catt built a disciplined lobbying operation that worked through existing parties and declined to embarrass a wartime administration. Alice Paul and the younger militants picketed the White House during the war, were arrested, went on hunger strike, and were force-fed, and the publicity that followed was considerable.

Both strategies can claim the outcome, and the historians divide accordingly. The amendment passed in 1920 after Catt's lobbying had produced the congressional votes, which supports her account. It passed after the picketing had made the issue impossible to table any longer, which supports the other. The honest reading is that the two worked on different audiences at once, and that neither wing would have accepted that description at the time. This is the normal condition of a successful movement: the question of which part won is settled afterward, by whoever writes it down. Both organizations published their own histories, and each gives itself the leading part. Paul and Catt barely spoke for the rest of their long lives.

WHAT THIS PASSAGE IS FOR

- Western states enfranchised women decades before the amendment.
 - Militants picketed, were arrested and were force-fed.
 - Which part won is settled afterward by whoever writes it down.
-

Suasion or Politics

Abolitionists agreed about the end and fought bitterly about the means. William Lloyd Garrison held that slavery was a sin, that the Constitution gave it a sanction no reformer should accept, and that the duty of an abolitionist was moral suasion, meaning the persuading of individuals to renounce the practice rather than the seeking of laws against it. He burned a copy of the Constitution in public, refused to vote, and argued that a political party which compromised in order to win was simply slavery in another dress.

The opposing wing held that moral argument had been tried for thirty years in a country where the owners were not listening, and that the practical route ran through votes, parties and federal power over the territories. That wing built the Liberty Party, then the Free Soil Party, and much of it ended inside the Republican Party, which won the presidency in 1860 on a platform that proposed only to stop the extension of slavery rather than to end it.

The outcome does not decide the question, although it is often said to. Slavery ended through a war fought by a government that had promised not to abolish it, which is an argument neither wing made. The political wing can say that only state power could have done it and that they had built the state power. Garrison can say that the war came because thirty years of moral argument had made the North unwilling to compromise further, and that without the sin there would have been no crisis. The evidence is compatible with both, and each requires the other to be in the story. Garrison dissolved his own society in 1865 on the ground that its work was done, over the objections of colleagues.

WHAT THIS PASSAGE IS FOR

- Garrison burned a copy of the Constitution and refused to vote.
 - The Republican platform of 1860 proposed only to stop extension.
 - Slavery ended through a war fought by a government that promised not to abolish it.
-

Failed or Defeated

For fifty years the standard account held that Reconstruction failed because it was unworkable. That account is the Dunning school, meaning an interpretation that treats Reconstruction as a corrupt failure, and it is named after a Columbia professor whose students wrote the individual state studies. It held that the period was one of misrule imposed by outsiders on a prostrate South, and that its collapse was the natural end of an experiment which should never have been attempted at all. That account was in the textbooks and the films, and it was written from official sources, white newspapers and the papers of the men who had overthrown the governments.

W. E. B. Du Bois attacked it in 1935 in a book of eight hundred pages with a closing chapter titled as a propaganda of history. His argument was partly about evidence: that the sources had been read selectively, that the tax and school records showed real achievement, and that the testimony of Black participants had been ignored because Black witnesses were not treated as witnesses. It was also about the exact question being asked. Reconstruction did not fail, in his account. It was defeated, by organized violence and federal withdrawal, which is a different claim with different evidence behind it.

The modern consensus lies much nearer to Du Bois, and the reason for the shift is instructive rather than sentimental, since it is not that historians simply became more sympathetic to the people they were writing about. It is that the sources widened. Once the congressional testimony, the registration rolls, the Freedmen's Bureau records and the Black press were read alongside the planters' papers, the earlier account could not be sustained on the evidence. The dispute is therefore a case study in how a historiography changes, which is rarely by argument alone. The book is now the standard starting point for the period.

WHAT THIS PASSAGE IS FOR

- The Dunning school treated the period as misrule imposed by outsiders.
 - Du Bois argued the sources had been read selectively.
 - The consensus changed because the sources widened.
-

Wages or Jobs

The effects of factory regulation in New York and in Britain are argued from the same series by people who reach opposite conclusions. The argument turns almost entirely on the counterfactual, meaning what would have happened without the measure. One reading holds that regulation raised wages and reduced deaths at a modest compliance cost, meaning the expense a firm incurs in meeting a rule. It points to the fall in industrial mortality after the laws, to the survival of the industries affected, and to the fact that employers who predicted ruin in 1911 were still trading in 1930.

The other reading accepts the fall in deaths and attributes much of it to rising capital intensity, better machinery and electric light, all of which were arriving anyway and all of which make a workplace safer without a statute. On this account the laws certified a change that was already under way and raised costs at the margin, with the burden falling on the smallest firms, which is where employment losses would appear and where records are worst. Both readings are consistent with the published figures.

What tempers the dispute is the comparative evidence. Regulation arrived at different dates in different states and countries, which allows the same event to be examined many times over. The weight of that work supports a modest version of the first reading, since deaths fell faster where the rules arrived earlier and employment effects are small and hard to find. Neither side is obliged to abandon its position on that evidence, because a modest effect leaves room for argument about whether it was worth its cost, which is a question about values and not about the series. The comparative studies also find that enforcement mattered more than the text of the statute.

WHAT THIS PASSAGE IS FOR

- The argument turns on the counterfactual.
 - The second reading credits machinery and electric light.
 - Deaths fell faster where rules arrived earlier.
-

Law or Pressure

The gains of the civil rights period are claimed by two accounts that both have strong evidence. The legal account points to a twenty-year litigation strategy, meaning the planned use of selected court cases to establish a principle, run by the organization legal fund and culminating in Brown. It points out that the statutes of 1964 and 1965 were drafted by lawyers, enacted by Congress and enforced by federal officials, and that the measurable changes in registration followed the Act rather than the marches.

The pressure account answers that the law moved when and only when the cost of not moving became intolerable, and that the cost was created by direct action, meaning protest designed to force negotiation. Birmingham in 1963 and the bridge at Selma in 1965 were followed within months by the two statutes, and the sequence is hard to dismiss. Martin Luther King made the argument explicitly from a jail cell, writing that the moderate who preferred order to justice was a greater obstacle than the intransigent opponent.

King also made the other case, which is why he is a useful witness. He worked with the lawyers, pressed for federal legislation, and spent much of his time in meetings at the Department of Justice. The two strategies were run by people who understood that neither was sufficient: a court ruling with no constituency behind it had been ignored for a decade after Brown, and a movement with no legal instrument had nothing to enforce. The dispute survives because each side can point to a period in which the other was visibly failing. The organizations involved disagreed sharply at the time about where money and attention should go. Those disagreements are documented in their own minutes, which makes the argument unusually well evidenced.

WHAT THIS PASSAGE IS FOR

- The legal account points to a twenty-year litigation strategy.
 - Birmingham and Selma were followed within months by the statutes.
 - A ruling with no constituency behind it had been ignored for a decade.
-

Granted or Taken

British official language called it a transfer of power, which is a phrase containing an argument. On that account independence was the intended conclusion of a long development: representative institutions were extended in stages from 1909, Indians entered the civil service and the officer corps, provincial self-government arrived in 1935, and the final step in 1947 was the completion of a program rather than a surrender. Clement Attlee presented it in those terms in the Commons, and the Indian Independence Act reads like administration rather than defeat.

The nationalist account treats that as a story told afterward by the side that lost. On this reading each constitutional advance was conceded under pressure and after refusal, the 1935 reforms were offered only once non-cooperation had made the provinces ungovernable, and the British did not capitulate gracefully but ran out of money, soldiers and policemen. Jawaharlal Nehru made exactly that argument, and the record of mass arrests supports it: a government confident in its program does not imprison sixty thousand people over a salt tax.

The timing is the best available test, and it favors the second account while complicating it. Every major concession followed a crisis rather than preceding one. But the British did leave by negotiation when they could have fought, as the French did in Indochina and Algeria at enormous cost, and that choice needs explaining too. The most defensible reading is that the movement made continued rule impossible and the government then chose the manner of its departure, which gives each side the half of the story it claims and neither side the whole. Both accounts also pass over the partition, which killed hundreds of thousands and displaced millions. Neither the program nor the triumph is easy to narrate alongside that.

WHAT THIS PASSAGE IS FOR

- The official phrase contains an argument.
 - A confident government does not imprison sixty thousand people over a salt tax.
 - Every major concession followed a crisis rather than preceding one.
-

Three Dissents and a Vacated Case

The majority opinion in *Korematsu* is the strongest available statement of the case for suspension, and it is worth reading on its own terms. Justice Black held that the exclusion was not based on race but on military necessity in time of war, that courts are not equipped to second-guess a commander in the field, and that hardships are part of war. The reasoning rests on judicial deference, meaning the practice of accepting an executive judgment within its own field, and nobody has produced a rule for emergencies that dispenses with deference altogether.

The three dissents attack it from three directions. Justice Roberts said the case was not about curfew but about imprisonment for refusing to leave home. Justice Murphy wrote that the claim of necessity rested on misinformation and on assumptions about racial characteristics, and used the word racism, which was unusual in a judicial opinion in 1944. Justice Jackson made the most uncomfortable argument, which is that a court cannot in practice review a military order during a war, and that it should therefore refuse to validate one rather than writing the validation into the law where it will outlive the emergency.

Jackson's point is the one the subsequent history vindicated. The convictions were vacated in the 1980s when the suppressed reports appeared, Congress apologized and paid compensation, and the Court itself stated in 2018 that *Korematsu* was gravely wrong the day it was decided. What it did not do was replace deference with a workable alternative. Strict scrutiny, meaning the most demanding standard applied to racial classifications, was announced in the same opinion that upheld the exclusion, which is a conciliatory formula rather than a constraint. Jackson had prosecuted at Nuremberg by the time the compensation question arose, which several commentators have found pointed.

WHAT THIS PASSAGE IS FOR

- The majority rested on judicial deference to a military judgment.
 - Murphy used the word racism in a judicial opinion in 1944.
 - Jackson argued a court should refuse to validate rather than review.
-

Who Follows Whom

Whether the press forms opinion or reflects it has been argued since Benjamin Day sold his first penny paper, and the two accounts are easy to state and hard to separate. On the influence account, editors decide what is covered and in what terms, and the public takes its sense of what matters from that; the strongest version claims that coverage changes votes. On the market account, a newspaper that prints what its readers dislike loses them, so coverage tracks opinion rather than leading it, and an editor who tries to lead will shortly have to retrench.

Both are supported by correlations and neither is established by them, since coverage and opinion move together on either account. The research that distinguishes them looks for cases where coverage changed for reasons unrelated to opinion. Where a newspaper is introduced to some towns and not others, where a channel reaches some cable systems first by accident of contract, or where an owner changes an editorial line abruptly, the effect on readers can be isolated. Studies of that kind find real effects on turnout and on vote share, usually modest, and larger effects on agenda setting, meaning which issues people regard as important.

They also find the market mechanism operating. Outlets move toward their audiences over time, selective exposure means the persuadable are the least exposed, and the largest measured effects appear among people with weak prior views and no alternative sources. The defensible conclusion is that influence runs in both directions on different timescales: coverage shapes attention in weeks, audiences shape coverage in years. Neither of the two simple accounts survives the evidence, and both survive in argument. The research design that settled most of this was not available before cable and newspaper entry data were digitized.

WHAT THIS PASSAGE IS FOR

- Coverage and opinion move together on either account.
 - Research looks for coverage changes unrelated to opinion.
 - Influence runs both ways on different timescales.
-

Biology and Earth Science

Passages 161 to 170

161

BIO · S01 · dispute

Selection or Drift

For a century after Darwin the working assumption was that any widespread feature of an organism had been favored by selection, and the research program consisted of finding out what for. In 1968 Motoo Kimura proposed something that sounded like heresy. Most change at the molecular level, he argued, is neutral, meaning that it neither helps nor harms, and its spread through a population is governed by genetic drift, which is change in the frequency of a variant by chance alone. His evidence was arithmetical: the rate at which protein sequences were found to differ between species was far too high to be explained by selection operating on each change.

The argument that followed was long and is worth understanding as a dispute about defaults rather than about mechanisms. Nobody denied that drift happens or that selection happens. The question was which should be assumed in the absence of evidence. Selectionists held that an apparently useless difference usually has a use that has not been found. Neutralists held that the burden of proof runs the other way, and that a novel feature should be presumed accidental until a selective advantage is demonstrated.

Sequencing data eventually settled much of it, and the settlement went mostly to the neutralists at the molecular level and mostly to the selectionists at the level of visible traits. Changes that do not alter a protein accumulate at a steady rate, as drift predicts, and that steadiness is now used as a clock. Changes that alter a protein show the signature of selection. The practical consequence is a test rather than a doctrine: a researcher comparing the two classes of change in a gene can say whether selection has acted on it. Kimura had to argue the case in English in a field dominated by English-speaking laboratories, which he said cost him years.

WHAT THIS PASSAGE IS FOR

- Kimura argued that most molecular change is neutral.
 - The dispute was about defaults rather than mechanisms.
 - Sequence data settled it differently at the two levels.
-

What Heritability Does Not Mean

Heritability is the most misunderstood number in biology, and the misunderstanding is specific. The term means the proportion of the variation in a trait, within one particular population at one time, that is associated with genetic variation in that population. It is not a statement about an individual, and it is not a statement about how much a trait can be changed. Height is highly heritable and also rose by several inches in many countries within two generations, which is not a contradiction but a demonstration that the number says nothing about the effect of changing the environment.

Twin studies of the kind run in Minnesota produce these estimates by comparing identical and non-identical pairs, and they have been attacked most seriously on the ground that identical twins are treated more alike. The genome era then produced a surprise. When the variants associated with a trait were identified directly and added up, they explained far less of the variation than the twin studies had implied, and the gap was named missing heritability, meaning the difference between the heritability estimated from twins and the variation that identified variants can explain. That gap has narrowed with larger samples and better methods and has not closed.

Two readings persist. One holds that the twin estimates were inflated by shared environment and that the direct measurements are nearer the truth. The other holds that the variation is spread across many thousands of variants of very small effect, most still undetected, and that the estimates will converge as samples grow. Both predict testable things and both have been partly borne out. What neither supports is the inference that a heritable trait is fixed, since a highly plastic trait can be heritable and a trait with low heritability can be impossible to shift. The policy misuse of these numbers has a long and ugly history.

WHAT THIS PASSAGE IS FOR

- Heritability describes variation in a population, not an individual.
 - Height is highly heritable and rose several inches in two generations.
 - A heritable trait is not thereby a fixed one.
-

Machine or Colony

The textbook picture of a cell is a factory, with compartments that have jobs, assembly lines running between them, a control room in the nucleus and an energy plant in the mitochondria. It is a useful picture and it is historically false in a way that turns out to matter. Lynn Margulis argued from the 1960s that mitochondria and chloroplasts are not compartments at all but descendants of free-living bacteria that were engulfed and never digested, a proposal called endosymbiosis. Her first paper on it was rejected by fifteen journals.

The evidence that persuaded the field is of a kind that admits little argument. Both organelles have their own circular DNA, of the bacterial type rather than the type in the nucleus. Both are enclosed in a double membrane, as an engulfed cell would be. Both make their own ribosomes, which resemble bacterial ribosomes closely enough to be shut down by antibiotics that act on bacteria and leave the rest of the cell alone. Both divide on a schedule of their own rather than with the cell that contains them. Sequence comparison places mitochondria firmly within one group of bacteria.

What remains disputed is how far the colonial picture should be pushed. Margulis argued late in her life for further cases, including the origin of cilia, which the evidence does not flag as strongly, and most of those claims have not been accepted. The question of how much of cell behavior should be described as cooperation among once-independent parts rather than as mechanism is still open, and it matters practically, because mitochondrial diseases are inherited only from the mother and behave unlike anything else in medical genetics. Margulis was also a vocal opponent of several well-supported positions in other fields, which complicates the usual story about vindicated outsiders.

WHAT THIS PASSAGE IS FOR

- The factory picture is useful and historically false.
 - Both organelles carry circular DNA of the bacterial type.
 - How far the colonial picture should be pushed remains disputed.
-

Did the Wolves Do It

The Yellowstone story is told as a clean case of a trophic cascade, meaning a chain of effects running down a food web from a predator to the plants, and a good deal of the telling has outrun the evidence. Wolves returned in 1995, elk numbers fell, willows and aspen recovered along the creeks, beaver returned, and streambanks stabilized. Each link in that chain has been measured. The difficulty is that other things changed over the same twenty years, which makes the design confounded in the strict sense.

The objections are specific and come from ecologists who work in the park. Elk numbers fell for several reasons at once, including a severe drought, heavy human hunting outside the park boundary, and a growing bear and cougar population. Willow recovery was patchy rather than general, and the strongest recovery occurred where the water table was high enough to support it, which suggests that hydrology may stem the effect as much as browsing does. Exclosure studies showed willows growing where elk were fenced out and water was available, and not where water was absent.

The defensible position is narrower than the popular one and still substantial. Wolves are one cause among several of a real change, their effect is strongest where water allows it, and the claim that they rewrote the rivers is not supported. The episode is now used in teaching as a lesson about causal inference rather than about wolves, and the lesson is that a spectacular natural experiment without a control area will produce a spectacular story that cannot be checked. Several of the most widely shared accounts of the reintroduction came from a short film that compressed the chain into a minute. Its narrator has since said that the script overstated the case.

WHAT THIS PASSAGE IS FOR

- Other things changed over the same twenty years.
 - Willow recovery was strongest where the water table allowed it.
 - A natural experiment without a control cannot be checked.
-

Food, Predators, or Both

The ten-year hare cycle generated two rival explanations that held the field for fifty years. One said food: hares eat the shrubs down, the shrubs respond with chemical defenses and take years to recover, and the hare population crashes for want of edible plants. The other said predators: lynx, coyote and owl numbers build up behind the hares, overtake them, and drive the crash. Both accounts fit the fur return series, which is the usual situation when a correlation is all anyone has.

Charles Krebs and colleagues settled it by experiment in the Yukon between 1986 and 1996, using a factorial design, meaning one in which treatments are applied separately and together. On blocks of a square kilometer each they added food, excluded predators with electric fencing, did both, and did neither. The result was that food alone roughly doubled hare density, predator exclusion alone roughly doubled it, and the two together increased it by about eleven times, which is far more than the two effects added.

That interaction is the finding, and neither camp had predicted it. A hare with enough food can afford to be vigilant and a hare in cover can afford to forage carelessly, so the two pressures are not independent, and a fecund season under both conditions produces an explosion. The result also explains why the fur records could never settle the question: a correlation cannot reveal an interaction, and the two single-cause theories were both correct and both insufficient. The experiment cost a decade and about a hundred kilometers of fence. The enclosures had to be electrified against bears as well as against the predators of interest. Several blocks were lost to fire and weather over the ten years and the design had to be rebuilt.

WHAT THIS PASSAGE IS FOR

- Both accounts fit the fur return series.
 - Food and predator exclusion together gave about eleven times the density.
 - A correlation cannot reveal an interaction.
-

The Germ or the Ground

Max von Pettenkofer agreed with Robert Koch that the cholera organism existed, and he denied that its presence was sufficient to produce an epidemic. His formula was that an epidemic required the germ and the ground, meaning the state of the soil, the water table and the local conditions, and he argued that the organism alone would not produce disease in a healthy population with good drainage. In 1892 he drank a culture of cholera in front of witnesses to prove it, developed only mild symptoms, and considered himself vindicated.

The germ theorists won the century, and they were right about the question they were answering. Identify the organism, interrupt the route, and the epidemic stops, which is what the pump handle and the water filter did. But the twentieth century also produced the figures Pettenkofer would have wanted. Most people who harbor the tuberculosis bacterium never develop the disease at all, and cholera mortality within a single outbreak varies by a large factor with nutrition and with access to clean fluids. The organism is necessary, and the condition of the person largely decides who dies of it.

The modern form of the dispute is about where to spend money, and it is sharp. One side argues for the specific intervention: the vaccine, the drug, the filter, each of which can be tested and shown to work. The other argues that housing, income and sanitation produce larger effects across more diseases at once, and points out that mortality from tuberculosis in Europe had already fallen by most of its total before any effective treatment existed. Both claims are supported. The practical question of allocation is not settled by the biology, which is why it is still an argument. His experiment proved less than he claimed, and the point he was making was not thereby wrong.

WHAT THIS PASSAGE IS FOR

- Pettenkofer required the germ and the ground together.
 - Most people carrying the tuberculosis bacterium never become ill.
 - Tuberculosis mortality fell before any effective treatment existed.
-

The Bargain of Fixed Nitrogen

Something close to half the nitrogen in the body of a person alive today was fixed industrially, which means that roughly half the world population is fed by a chemical process invented in 1909. No serious account of twentieth-century population growth can leave that process out of the explanation. Cereal yields in countries that adopted the package of fertilizer, irrigation and selected varieties rose by factors of two to four in a generation, and famine in Asia, which had been a regular event, became rare.

The costs are also measurable, which is what makes this a real argument rather than a slogan. Perhaps half the nitrogen applied to a field never reaches the crop. It leaves in water and air, and in the water it produces eutrophication, meaning enrichment that strips the water of oxygen, which has created dead zones at the mouths of major rivers, the largest of them in the Gulf of Mexico. Fixing nitrogen industrially consumes one or two percent of all the energy used in the world, and nitrous oxide released from fertilized soil is a greenhouse gas of considerable strength.

The two positions that follow are not symmetrical and both are serious. One holds that the bargain was obviously worth making, that the alternative was mass starvation, and that the losses are an engineering problem to be reduced by better timing and placement. The other holds that a system which must husband its soil is more stable than one which buys fertility by the ton, and points to the long-term plots where manure outperforms mineral nitrogen on soil structure. Neither position disputes the figures. They disagree about what to do with a technology that cannot be given up and cannot be continued unchanged. Precision application has cut losses substantially in a few wealthy countries. Global fertilizer use has gone on rising throughout the same period.

WHAT THIS PASSAGE IS FOR

- Roughly half the world population is fed by a process invented in 1909.
 - Perhaps half the nitrogen applied never reaches the crop.
 - Neither position disputes the figures; they disagree about what to do.
-

Separating a Signal From Noise

Nobody in the argument disputes that the climate of the Earth has changed before, and the question that actually matters is attribution, meaning the establishing of how much of an observed change is caused by a particular factor. The objection that the climate has always varied is therefore not a trivial one, and the answer to it is not a louder assertion. It is a method, and the method is fingerprinting: looking for a pattern of change that one cause produces and the alternatives do not.

Several such patterns exist. Greenhouse warming predicts that the lower atmosphere warms while the upper stratosphere cools, which is what is measured, and no increase in solar output can produce that combination. Greenhouse warming predicts that nights warm faster than days, which is observed. Solar warming predicts the opposite in both cases. The isotopic composition of the carbon now in the atmosphere has shifted toward the characteristic signature of fossil carbon, which is not a quiescent coincidence but a direct trace of its origin. Each of these is an independent test of a kind that could have failed and did not.

The serious objections are about magnitude rather than attribution, and they are inside the science rather than outside it. How strongly clouds respond to warming remains the largest uncertainty in every model, and it moves the projected warming for a doubling of carbon dioxide across a range of several degrees. How much heat the deep ocean takes up, and for how long it holds it, has been measured only since the profiling floats were deployed, and those two uncertainties are the live scientific arguments. A reader who encounters a dispute about whether the mechanism exists has been handed the wrong argument, and one about how large the response is has been handed the real one.

WHAT THIS PASSAGE IS FOR

- Attribution is a method, not an assertion.
 - The troposphere warms while the stratosphere cools.
 - The live argument is about magnitude, not mechanism.
-

The Floods That Were Not Allowed

Geology was founded on a methodological rule. Uniformitarianism, the principle that the processes acting now are the ones that shaped the past, was the instrument that defeated an earlier tradition of explaining landscapes by single catastrophes, and it worked. Slow erosion, slow deposition and slow uplift account for most of the surface of the Earth, and the rule gave geologists a disciplined way to read the history written in a cliff face.

J Harlen Bretz then found a landscape it could not explain. The scablands of eastern Washington are channels cut hundreds of feet into solid basalt, boulders the size of houses carried fifty miles, gravel bars the height of small hills, and ripple marks fifty feet from crest to crest. Bretz argued from 1923 that this had been made by a flood of almost unimaginable size. His colleagues rejected it, not because they had a better account, but because a single enormous flood was exactly the kind of explanation the discipline had spent a century getting rid of.

He turned out to be right, and the evidence that settled it came from somewhere else entirely. A lake dammed by ice in Montana was identified, with shorelines showing a volume comparable to Lake Ontario, and a mechanism for the dam failing repeatedly was worked out. Bretz lived to receive the discipline highest medal in 1979 at the age of ninety-six. The episode is not an argument against uniformitarianism, which remains the correct default. It is an argument about what a default is for: the rule told geologists where to look and then told them not to believe what they found. Comparable flood channels have since been identified on Mars and are interpreted using the scablands. The rule that blocked the conclusion has become one of the tools for extending it.

WHAT THIS PASSAGE IS FOR

- Uniformitarianism defeated an earlier tradition and worked.
 - His colleagues rejected it without a better account.
 - A default told them where to look and then not to believe it.
-

Whether It Can Be Stopped

Two claims about Thwaites are often run together and should be separated. The first is marine ice sheet instability, meaning the self-sustaining retreat of a glacier grounded below sea level on a slope that deepens inland. The physics is straightforward and widely accepted: as the grounding line moves back into deeper water, thicker ice is exposed, more ice flows out, and the retreat continues without any further warming. Whether that process has already begun at Thwaites is debated, and the measurements of grounding line retreat are the evidence being argued over.

The second claim is ice cliff failure, meaning the structural collapse of an exposed face too tall to hold itself up. Ice cannot support a cliff above roughly ninety meters, and if the floating shelves go, Thwaites would present a face far taller than that. A pair of modelers added this mechanism to an ice sheet model in 2016 and obtained sea level rise of a meter or more this century, which was several times the previous projections and was reported very widely.

The objections to the second claim are serious and not political. The mechanism has never been observed at the relevant scale. The model required parameters chosen to reproduce ancient sea levels that are themselves uncertain. Later work found that the same ancient levels could be reproduced without it. The honest summary is that the first claim is physics that is probably under way, and the second is a possibility that cannot be excluded and has not been demonstrated. A planner facing a distribution with a long inimical tail has a decision problem rather than a scientific one. The two claims are often reported as one, which makes the science look less settled than it is on the first and more settled than it is on the second.

WHAT THIS PASSAGE IS FOR

- Retreat on a reverse slope continues without further warming.
 - Ice cannot support a cliff above roughly ninety meters.
 - The cliff mechanism has never been observed at the relevant scale.
-

Physical Science

Passages 171 to 180

171

PHY · S01 · dispute

No Measurement Without a Theory

A thermometer reading looks like a fact about the world and is not quite one. To treat the number as a temperature requires accepting that the liquid expands in proportion to temperature, that the glass does not expand in a way that matters, and that the instrument has reached equilibrium with what it is measuring. Each of those is an auxiliary assumption, meaning a background claim that must hold for the measurement to mean what it is taken to mean, and each comes from a theory.

Philosophers from Pierre Duhem onward have called this theory-ladenness, which is the claim that what counts as a measurement depends on the framework being used, and the strong version goes further: two scientists with different theories are not measuring the same quantity when they appear to disagree. The weak version is uncontroversial and useful. The strong version has a problem, which is that measurements regularly overturn the theories that licensed them. The cosmic microwave background was found by two engineers looking for noise in an antenna, and it destroyed the model most cosmologists then held.

The working resolution is to appreciate the point without accepting the conclusion. Scientists manage theory-ladenness by making the assumptions explicit and testing them separately, which is exactly what a calibration chain does. Where two instruments resting on different theories agree, the agreement is evidence that neither theory is distorting the result, and the history of physics contains many such convergences. The objection remains live wherever only one method exists, which is why single-method results are treated with caution in every field that has learned the lesson the hard way. The dispute has practical consequences in fields where one instrument defines the quantity it measures. Several psychological constructs are in that position and the comparison is often drawn.

WHAT THIS PASSAGE IS FOR

- Each auxiliary assumption comes from a theory.
 - Measurements regularly overturn the theories that licensed them.
 - Agreement between methods resting on different theories is evidence.
-

Wrong or Incomplete

Newton's law of gravity was the most successful theory in the history of science and it is wrong. It is also still used to fly spacecraft. Both statements are true, and holding them together requires a distinction that students are rarely given: between a theory that has been refuted and a theory that has been placed inside a larger one as a limiting case, meaning a situation in which the newer theory reduces to the older.

The evidence that forced the change was small. The perihelion of Mercury, meaning the point in its orbit closest to the Sun, advances by a tiny amount each century, and after every known gravitational influence had been accounted for there remained a discrepancy of about forty-three seconds of arc per century. Astronomers tried to resolve it for sixty years by proposing an unseen planet, a flattened Sun, and a modification of the exponent in the inverse square law. General relativity predicted the figure exactly, from a theory built for quite different reasons, and that is why it was accepted rather than merely admired.

The philosophical question is whether Newton was mistaken. On one account yes: the theory says that gravity is a force acting instantaneously at a distance, and that is simply not the case. On the other account the equations remain correct within a stated domain, which is the ordinary condition of every physical law, and relativity itself is expected to fail where quantum effects matter. The second account is the one working physicists use, and it carries a discipline: a theory is not refuted by being shown to have a boundary, and is not safe for being useful inside it. Spacecraft navigation uses Newtonian calculations with relativistic corrections applied where they matter.

WHAT THIS PASSAGE IS FOR

- Newtonian gravity is wrong and is still used to fly spacecraft.
 - The discrepancy was forty-three seconds of arc per century.
 - A theory is not refuted by being shown to have a boundary.
-

Whether Efficiency Saves Anything

Lighting a room in Britain is about seven hundred times cheaper in real terms than it was two centuries ago, and the share of income that households spend on lighting has barely changed. The reason is that people bought more light. That observation is the basis of the rebound effect, meaning the increase in use of a service that follows from making it cheaper to provide, and it is the strongest argument against treating efficiency as a straightforward route to lower energy use.

The argument has a weak form that nobody disputes and a strong form that is contested. In the weak form, some of the saving is taken back: a household with a more efficient boiler keeps the house warmer, and a firm with cheaper freight ships more often. Measured direct rebound for household energy services in rich countries is usually put between ten and thirty percent, which leaves most of the saving intact. In the strong form, advanced by a few economists, rebound across a whole economy can exceed the saving entirely, because cheaper energy services raise output and output requires energy.

The evidence does not support the strong form as a general rule and does not exclude it in particular cases. Historical examples of complete rebound exist, lighting among them, and they share a feature: the service was far from saturated, so demand had room to grow. Where a service is already saturated, as domestic refrigeration is in a rich country, rebound is small. The policy conclusion is therefore commensurate with the detail rather than with the slogan. Efficiency standards save energy where demand is near its limit and save much less where it is not. The lighting figure comes from a study that reconstructed the real price of light since the candle.

WHAT THIS PASSAGE IS FOR

- Light is seven hundred times cheaper and the spending share barely changed.
 - Measured direct rebound is usually ten to thirty percent.
 - Complete rebound cases share an unsaturated service.
-

A Law of Nature or of Counting

Heat flows from hot to cold, gases mix and do not unmix, and a broken cup does not reassemble. The second law of thermodynamics states this as the claim that entropy, meaning a measure of the number of microscopic arrangements consistent with a given state, does not decrease in an isolated system. The regularity is as reliable as anything in physics. What kind of law it is has been argued since Boltzmann, and the argument is not idle.

On one account it is a law of nature in the ordinary sense, a constraint on what can happen. On the other it is a statistical law, meaning a regularity that holds because the exceptions are overwhelmingly improbable rather than impossible. The second account follows from the mechanics: the molecular equations work equally well run backward, so nothing in them forbids a gas collecting in one corner. The reason it never happens is that the number of arrangements in which the gas is spread out exceeds the number in which it is collected by a factor with more zeros than there are atoms in the galaxy.

The statistical reading is now standard and it has a consequence that is still argued over. If the law is statistical, it says nothing about an individual molecule and nothing in principle about a sufficiently small system, and experiments on single molecules and tiny beads have measured brief, genuine decreases in entropy of exactly the kind the strict reading forbids. The inexorable character of the law is therefore a property of large numbers. That matters for anyone designing a machine at the scale of a cell, where the fluctuations are the same size as the effects being engineered. The relations governing those fluctuations were derived in the 1990s and have been confirmed repeatedly.

WHAT THIS PASSAGE IS FOR

- The molecular equations work equally well run backward.
 - Spread-out arrangements outnumber collected ones enormously.
 - The inexorable character is a property of large numbers.
-

Big Grid or Small

In August 2003 a line in Ohio sagged into a tree, protective equipment disconnected it, the power it had carried transferred to neighboring lines, and those disconnected in turn. Within about eight minutes fifty-five million people in the United States and Canada had no electricity. Each piece of equipment did what it was designed to do, which is the defining feature of a cascading failure, meaning a sequence in which each component protects itself and thereby overloads the next.

Two lessons were drawn from the same event. The first is that the grid was too weakly connected and too weakly coordinated: operators could not see conditions beyond their own territory, the alarm software in the control room had failed and nobody noticed, and stronger interconnection with better shared monitoring would have allowed the disturbance to be absorbed. That argument produced the modern reliability standards and a marked increase in transmission investment.

The second lesson is the opposite. A tightly connected network transmits trouble as efficiently as it transmits power, and a continent-wide machine in which a tree in Ohio can darken New York is fragile by construction. The remedy proposed is islanding, meaning the deliberate separation of parts of a network so that each can run alone, together with local generation and storage. Both arguments are supported by engineering and by subsequent events, and they are not fully reconcilable, because the same connection that shares reserves also shares faults. Grid planning is therefore a choice about which failure mode to prefer, and it is usually presented as a technical question when it is a judgment about risk. Standards written after 2003 are mandatory and enforced with penalties, which was not previously the case. Large blackouts have become less frequent in North America and have not stopped.

WHAT THIS PASSAGE IS FOR

- Each piece of equipment did what it was designed to do.
 - A tightly connected network transmits trouble as efficiently as power.
 - The same connection that shares reserves also shares faults.
-

What the Two Slits Settled

Light passing through two narrow slits produces a pattern of bright and dark bands on a screen behind them. Thomas Young demonstrated it in 1801, and for a century it was taken as proof that light is a wave, because particles arriving at a screen could not cancel one another out. Then the photoelectric effect showed light delivering its energy in discrete lumps, which waves do not do, and the two bodies of evidence compounded the difficulty rather than resolving it.

The experiment that forced the modern account sends the light through one particle at a time. Each arrival is a single point on the detector, which is particle behavior. After enough arrivals the points have built up the interference pattern, which is wave behavior, and the pattern is there even though no two particles were ever present together to interfere with each other. Put a detector at the slits to find out which one each particle went through and the pattern disappears. The same result has been obtained with electrons, neutrons, whole atoms and molecules of several thousand atoms.

What this settled is narrower than it is usually said to be. It did not show that light is both a wave and a particle, which is a sentence rather than a claim. It showed that neither picture, both of which come from the behavior of ordinary objects, applies to the thing being described, and that the mathematics predicts every observed result without needing either. The remaining argument concerns what the mathematics is about, and it has run for a century without a decisive experiment, which is a different situation from the one in which the physics is unsettled. Several interpretations make identical predictions for every experiment performed so far.

WHAT THIS PASSAGE IS FOR

- One particle at a time still builds the interference pattern.
 - Detecting which slit destroys the pattern.
 - Neither ordinary picture applies to the thing being described.
-

Whether Chemistry Is Physics

The strong claim about chemistry is reduction, meaning the proposition that one science is in principle derivable from another, and in this case that chemistry follows from the quantum mechanics worked out by Erwin Schrodinger and others. The claim is not idle, because the derivation has been carried out for small systems. The energy levels of a hydrogen molecule can be computed from first principles to an accuracy better than measurement. Degenerate orbitals, meaning states that share the same energy, fall out of the mathematics without being put in.

The difficulty is scale, and it is not a difficulty of principle in the usual sense. The computational cost of an exact solution grows so fast with the number of electrons that a molecule of forty atoms is already beyond any possible computer, and every practical method is an approximation whose errors are controlled empirically, by comparison with measurement. A chemist therefore uses rules, such as the ones about acids, bonds and reaction rates, that are not derived from the underlying theory and could not be.

Two readings of that situation are current. The reductionist says the limit is practical, that the equations are correct, and that better algorithms have been steadily moving the boundary for fifty years. The opposing position is emergence, meaning the appearance at one scale of regularities that are not stated at the scale below. Its advocates hold that a science whose working concepts are not derivable in practice is not reducible in any sense that matters, and that regularities at the molecular scale are genuine features of the world rather than conveniences. The disagreement is partly about the word reducible, and it has a practical edge: funding goes to the people who claim their level of description is the fundamental one. The boundary has moved from three atoms to several dozen in half a century.

WHAT THIS PASSAGE IS FOR

- Hydrogen molecule energy levels are computed better than measurement.
 - A molecule of forty atoms is already beyond any possible computer.
 - The disagreement is partly about the word reducible.
-

Two Inquiries, One Bridge

Inquiries into engineering failures reliably produce two kinds of explanation, and the choice between them is not a matter of evidence alone. The technical account names a component and a mechanism: cast iron columns with cast bolt holes, no wind load in the calculation, a seal that stiffens below a certain temperature. The organizational account names a decision process: reports that went no further than a foreman, a schedule that made a delay expensive, an engineer whose objection was overruled by people who had heard objections before.

Both accounts are usually correct and they answer different questions. The technical account identifies what broke. The organizational account explains why the thing that broke was allowed to be in that condition, and it is generally the more useful of the two for preventing the next failure, because the specific component will not recur and the decision process will. Diane Vaughan's analysis of the first shuttle loss gave this the name normalization of deviance, meaning the process by which a repeated warning sign comes to be treated as acceptable because nothing had yet gone wrong.

The dispute matters because the two accounts imply different remedies and different liabilities. A technical finding produces a design change, which is cheap and verifiable. An organizational finding implies that the management structure was unsafe, which is expensive, contestable in court, and refractory to engineering solutions. Inquiries with a narrow remit tend to produce technical findings, and the second shuttle inquiry was unusual in saying so explicitly: it found that the organizational causes identified seventeen years earlier had not been addressed. Both shuttle inquiry reports are public and are read together in engineering ethics courses. The second quotes the first at length on exactly this point.

WHAT THIS PASSAGE IS FOR

- The technical account names what broke.
 - The organizational cause recurs where the component will not.
 - Narrow inquiries tend to produce technical findings.
-

Whether Small Doses Matter

At high doses the effects of radiation on tissue are not in dispute. The survivor studies from Hiroshima and Nagasaki are the basis of every regulation in force. The argument is entirely about the shape of the curve at the bottom, where the effects are too small to detect in any feasible study, and the practical stakes are large, because almost all regulation concerns that region.

The standard assumption is linear no-threshold, meaning that risk falls in proportion to dose all the way down with no safe level. Its support comes from extrapolating the survivor data, from the mechanism, since a single track through a cell can damage DNA, and from the convenience of a rule that allows doses to be added across a population. The alternative position holds that repair mechanisms handle low doses completely, so that a threshold exists below which there is no effect, and some argue for hormesis, which is the proposal that very small doses are protective by stimulating repair.

The honest position is that the question is not currently decidable. To distinguish the curves at the doses in question would require a study of millions of people with exposure measured individually, and the confounding factors are larger than the effect. Both sides therefore argue from mechanism and from the shape of data collected elsewhere. The policy consequence is asymmetric, which is what keeps the dispute alive: assuming no threshold makes cleanup and shielding extremely expensive, while assuming a threshold risks harm that would never be sensible to detect afterward. Regulators have mostly kept the linear assumption on the ground that it errs in the safer direction, and the cost of that choice is a real argument rather than a settled one. The largest worker study is consistent with the linear assumption and does not exclude a small threshold.

WHAT THIS PASSAGE IS FOR

- The dispute concerns the bottom of the curve only.
 - Distinguishing the curves would require millions of individually measured people.
 - The policy consequence is asymmetric, which keeps the dispute alive.
-

Two Good Numbers That Disagree

Two methods measure the Hubble constant, which is the present rate at which the universe is expanding, and they have been disagreeing with each other for more than a decade. The first works outward along the distance ladder, using pulsating stars and exploding stars to measure distances to galaxies whose recession speeds are known. The second works from the pattern in the cosmic microwave background, fitting a model of the early universe and calculating what the present rate must be. Both have been refined by large teams with every known error accounted for. Both have been checked by rival teams using the same data.

The figures are about seventy-three kilometers per second per megaparsec from the ladder and about sixty-seven from the background. The gross agreement between two completely independent methods is itself remarkable and conceals a disagreement of about five sigma, meaning five times the combined uncertainty, which in most of physics is the threshold for taking a discrepancy seriously rather than waiting for better data.

Three resolutions are possible and each has advocates. There is an unidentified systematic error in the ladder, most likely in the calibration of the pulsating stars, and new instruments have been deployed to look for it. There is an error in the model fitted to the background, which would require a component of the early universe that nobody has established. Or the standard cosmological model is incomplete, in which case the disagreement is the first measurement of something new. Most working cosmologists have been betting on the first for a decade, and the disagreement has survived every improvement in both methods, which is roughly what a real effect does. Independent methods not using either route have begun to report, with values between the two. Their uncertainties are still too large to decide the question either way.

WHAT THIS PASSAGE IS FOR

- Two independent methods disagree by about five sigma.
 - The figures are about seventy-three and about sixty-seven.
 - The disagreement has survived every improvement in both methods.
-

Humanities

Passages 181 to 190

181

HUM · S01 · dispute

Whose Judgment Is It

When a narrator makes a judgment, three candidates exist for whose judgment it is. It may be the narrator alone, which is the case whenever a reader can see the narrator being wrong. It may be the implied author, meaning the sensibility a reader reconstructs from the whole text, which is neither the narrator nor the person who wrote it. Or it may be the writer, who had opinions and a biography, and whose letters may say the opposite.

The distinction matters because it decides what an arch or ironic sentence commits anybody to. A narrator who praises a slave owner in an abolitionist novel is not the novel praising him, and a reader who collapses the three levels will read the book as its own opposite. Wayne Booth made the implied author the standard tool for handling this in 1961, and its usefulness is that it can be derived from the text alone, by asking what set of values would make the whole arrangement coherent.

The objection is that this licenses almost any reading. If a sentence that appears to endorse something can be read as ironic, and irony is established by what the reader takes the implied author to think, the argument is circular, and critics have used the move to rescue admired writers from positions the writers plainly held. The discipline that controls it is textual: irony has to be marked somewhere, by exaggeration, by a detail that will not fit, or by a pattern across the book. Where nothing in the text marks it, the sentence stands, and some admired writers believed unpleasant things. The dispute is sharpest in teaching, where a class has to decide what a text is doing before it can discuss it.

WHAT THIS PASSAGE IS FOR

- Three candidates exist for whose judgment a narrator makes.
- The implied author can be derived from the text alone.
- Where nothing marks irony, the sentence stands.

Psychology or Function

Vladimir Propp analyzed a hundred Russian folk tales and found that the characters could be reduced to seven roles: the villain, the donor, the helper, the princess, the dispatcher, the hero and the false hero. The tales differ in who fills each role and in almost nothing else. That is formalism, meaning the study of a work as a structure of parts, and on this account a character is a character function, meaning the role a figure performs in a plot, with personality as decoration.

The humanist reply is that this works for folk tales and fails for novels. Readers argue about whether a character would really have done something, which is a psychological question, and novelists revise in order to make motive consistent, which is a psychological labor. A critic who says that Dorothea Brooke is the dispatcher has described a function and left out everything the book spends eight hundred pages on. Readers also form attachments that no account of function predicts.

The sharpest version of the dispute concerns what a reader is doing when making inferences about a character inner life. The formalist says the reader is constructing from textual cues and that the construction is an effect the author arranged, so talk of a character's psychology is a convenient shorthand for a pattern on the page. The humanist says the shorthand is the point, that fiction is an instrument for modeling other minds, and that an account which cannot say why anyone would be affected by a novel has explained the wrong thing. Both positions are held by serious people and the disagreement is about what literary study is for. Cognitive approaches have given the humanist side experimental support about mind-reading in readers. The formalist side points out that the same experiments work on tales with no psychology in them at all.

WHAT THIS PASSAGE IS FOR

- Propp reduced a hundred tales to seven character roles.
 - Novelists revise in order to make motive consistent.
 - The disagreement is about what literary study is for.
-

Meaning or Mood

The strong claim about imagery is that images are symbols, meaning that they stand for something beyond themselves in a way that can be stated: the river is time, the garden is innocence, the fog is moral confusion. The claim licenses a whole genre of criticism and a whole genre of examination answer, and it has an obvious problem, which is that the equations are supplied by the critic and cannot be checked against anything.

The deflationary alternative holds that most images work on mood rather than on meaning. Fog in a novel makes a reader uneasy, which is an effect in the reader and requires no translation. On this account an involved symbolic reading is a way of converting an experience into a proposition, and something is lost rather than gained in the conversion. Writers have often said so bluntly when asked what their images mean, and the usual critical answer is that an author's statement does not settle the question, a position known as the intentional fallacy.

That answer is good as far as it goes and does not resolve the dispute, since it removes one constraint without supplying another. What constrains a symbolic reading in practice is pattern and consistency: an image that recurs, that is placed at the turns of the plot, and that other elements of the text treat as significant is doing more work than one that appears once. The best current position is that the two accounts are about different images rather than rival accounts of all of them, and that the critical skill lies in telling which kind is on the page. Cleanth Brooks and his successors made the stronger case and the weaker one has worn better. Examination boards have quietly moved toward requiring evidence of pattern rather than assertion of symbolism. That shift is visible in published mark schemes over twenty years.

WHAT THIS PASSAGE IS FOR

- Symbolic equations are supplied by the critic.
 - Fog makes a reader uneasy without needing translation.
 - Pattern and consistency constrain a symbolic reading.
-

Does Free Verse Have Form

Walt Whitman published verse with no regular meter and no rhyme in 1855, and the objection was immediate: without a counted pattern there is nothing to depart from, so the line breaks are arbitrary and the result is prose with ragged edges. The objection is not frivolous. If the reader has no expectation, the devices that depend on expectation cannot work, and a line ending that suspends a word cannot suspend anything.

The defense has two parts. The first is that free verse substitutes other patterns for meter: the length of the breath, the repetition of a grammatical shape, parallel clauses that accumulate, and cadence, meaning the rhythmic shape of a phrase independent of a counted meter. Whitman catalogues are built on grammatical repetition as strictly as a sonnet is built on a rhyme scheme. The second part is empirical: readers do hear the line breaks, and rearranging the lines of a good free verse poem damages it in ways that can be demonstrated to a class.

The strongest version of the criticism concedes both points and narrows the claim. Free verse has form, it says, but the form is established inside each poem rather than inherited, which means that a reader must learn the pattern from the poem itself and cannot bring one to it. That makes the poet's task harder and the reader's task less reliable, and it explains why the worst free verse is worse than the worst metrical verse. Both sides now accept something like that, and the argument has moved to whether a studied irregularity in metrical verse achieves the same thing more economically. Eliot wrote that no verse is free for the poet who wants to do a good job, which both sides quote.

WHAT THIS PASSAGE IS FOR

- Without expectation the devices that depend on it cannot work.
 - Free verse substitutes grammatical repetition and cadence for meter.
 - The form is established inside each poem rather than inherited.
-

The Play or the Production

A play exists in two states and the question of which one is the work has produced a century of argument. On the text-centered account the play is what the author wrote, productions are interpretations of it, and a production that contradicts the text has failed in the way a translation that reverses a sentence has failed. On the stage-centered account a play is a score for performance, incomplete until performed, and what a reader holds is not the work but the instructions.

Directors' theater, meaning a tradition in which the director's interpretation is the primary creative act, pushed the second account hard from the middle of the twentieth century. Productions set the plays in asylums and boardrooms, cut and reordered freely, cast against the text, and in several celebrated cases produced something that could arrest an audience that had seen the play twenty times. Defenders argued that this is what the plays had always received, since the Victorians cut a third and nobody now calls that fidelity.

The text-centered reply has a narrow and strong version. It concedes that staging is creative and insists that a production which reverses the meaning of the text is making a different work and should say so, because the alternative is a situation in which a student who has seen the play has not encountered it. That seems to be roughly where practice has landed. Programs now routinely describe what has been done, adaptations are labeled as such, and the serious disagreements have moved to specific cases rather than to the principle. Copyright gives living playwrights a veto that dead ones do not have, which shapes the argument in practice. Several estates have withdrawn permission from productions they considered unfaithful. The principle is now largely agreed and the cases are fought one at a time.

WHAT THIS PASSAGE IS FOR

- A play exists in two states.
 - The Victorians cut a third and nobody calls that fidelity.
 - A production that reverses the text is making a different work.
-

Limit or Licence

The case against genre, put most sharply by Edmund Wilson, is that it is a marketing category masquerading as a kind, that it rewards repetition, and that a writer working inside it is solving a puzzle rather than making something. The literary hierarchy, meaning the ranking of kinds of writing by assumed seriousness, has placed detective fiction, romance, science fiction and the western near the bottom for a century, and the reasons given usually amount to the observation that such books resemble one another.

The case for genre begins by pointing out that the observation is true of everything. Sonnets resemble one another, tragedies resemble one another, and nobody holds that against them. A form that a reader knows is a form a writer can use: the expectation is a resource, and the most admired works in several genres are the ones that bend a known rule at a known point. Pastiche, meaning open imitation of a style without deceit or mockery, is only legible where the style is familiar enough to be recognized.

What makes this a real dispute rather than a misunderstanding is that both sides can point to the same books. A detective novel that keeps every rule is competent and forgettable, which supports the critics. A detective novel that breaks one rule precisely is remarkable, which supports the defenders and depends on the rules the critics complain about. The position that survives both is unexciting: a genre is a set of constraints, constraints are what make significant choices possible, and the quality of the writing is a separate question from the kind of book it is. Prize juries have moved substantially on this in thirty years, and several genre writers now appear on literary lists. The hierarchy has not disappeared and the arguments for it have become quieter.

WHAT THIS PASSAGE IS FOR

- Sonnets resemble one another and nobody holds it against them.
 - Pastiche is legible only where the style is familiar.
 - Constraints are what make a significant choice possible at all.
-

Reading With Period Eyes

A seventeenth-century Dutch painting of a woman at a window may be a scene, or it may be a warning about vanity, and the choice between those readings is the oldest methodological dispute in art history. Iconography, meaning the study of the conventional meanings attached to images in a period, holds that such pictures used a shared code: the mirror, the dog, the open window and the extinguished candle all carried meanings that a contemporary viewer read without effort. The evidence is emblem books, sermons and household inventories, of which thousands survive from the seventeenth century.

The objection is not that the code did not exist but that it is being applied too freely. A dog in a painting may signify fidelity and may also be a dog, and the available emblem books supply several meanings for most objects, which allows a determined scholar to find a recondite reading for anything. The formal alternative attends to how a picture is made and to what it does to the eye, and treats meaning as arising from arrangement rather than from vocabulary.

Michael Baxandall proposed the useful middle term with the period eye, meaning the habits of seeing that a viewer of the time brought to a work, including skills unrelated to art such as gauging volumes in a trade that required it. That reframes the question as empirical: what did this audience know how to do. Answering it requires evidence about schooling, commerce and devotion rather than about symbols, and it has produced the most durable work in the field. The dispute survives because a picture whose audience left no record cannot be read that way at all. Both camps use the same conservation reports and the same inventories. Where they differ is in what counts as a legitimate inference from them.

WHAT THIS PASSAGE IS FOR

- A dog may signify fidelity and may also be a dog.
 - Emblem books supply several meanings for most objects.
 - The period eye reframes the question as empirical.
-

The Score or the Sound

From the 1960s a movement associated with Nikolaus Harnoncourt and others argued for historically informed performance, meaning the playing of old music on the instruments and in the manner of its own period. That meant instruments of the period, the pitch of the period, the numbers the composer actually had, and the ornaments and tempos that contemporary treatises describe. Historically informed performance produced recordings that sounded startlingly unlike the orchestral tradition, with smaller forces, faster tempos, less vibrato and a different economy of sound altogether. It also produced urtext editions, meaning editions that present what the composer wrote without later editorial additions.

The objections came quickly and some have held up. There is no recording of anything before 1877, so the claim rests on written sources which are incomplete and often disagree. A modern audience is not a period audience, so reproducing the acoustic conditions does not reproduce the experience. Composers frequently wrote for forces they considered inadequate, and several said so in letters, which makes the appeal to original practice an appeal to a compromise rather than to an intention.

The movement's answer has largely been to drop the strongest claim. Few practitioners now argue that period performance is authentic in the sense of recovering a lost original. What they argue instead is that the instruments and the treatises are evidence about choices the music assumes, and that a performer who knows them has more options rather than fewer. That is a defensible position and it has changed mainstream practice, which now uses smaller forces and faster tempos than it did, without anybody conceding the original argument. The two traditions have converged in the concert hall more than in the published polemics. Several conductors move between them without comment, which would have been unthinkable in 1975.

WHAT THIS PASSAGE IS FOR

- There is no recording of anything before 1877.
 - Composers often wrote for forces they considered inadequate.
 - The strongest claim was dropped and the practice changed anyway.
-

The Block or the Tower

Le Corbusier proposed replacing the dense city with towers in parkland, and the argument was not foolish. Nineteenth-century slums were dark, overcrowded and lethal, and a tower on a small footprint can house the same people with light, air, trees and no traffic at the door. The superblock, meaning a large site planned as a single unit with towers set in open ground, was built in that spirit across Europe, Asia and America for forty years.

Jane Jacobs attacked it in 1961 from observation rather than from doctrine. Her claim was that the things that make a street safe and useful, which are short blocks, many doorways, mixed uses and people on the pavement at different hours, are destroyed by the design, and that the open ground is not a park but a space nobody is responsible for. Her book is laconic about architecture and specific about behavior, and most of its observations have since been measured and confirmed.

The honest verdict has to be split, because both doctrines produced disasters and successes. Superblocks in Vienna and Singapore work, with the same form and different management, maintenance and tenure. Dense traditional streets produce gentrification that prices out exactly the populations Jacobs defended. The variable that best predicts success in the studies is not the form but whether the residents have security, maintenance and some control, which is a conclusion that pleases neither party and has been available in the evidence for thirty years. Several demolished American projects had maintenance budgets cut to a fraction of what their designs required. Comparable buildings elsewhere with funded maintenance are still standing and sought after. That conclusion also explains the pattern of demolitions, which fell heavily on projects whose management had been cut and lightly on identical buildings that were maintained.

WHAT THIS PASSAGE IS FOR

- A tower on a small footprint can house people with light and air.
 - Jacobs attacked from observation rather than doctrine.
 - Form predicts success less well than security and maintenance.
-

Can a Judgment Be True

The question of whether a judgment of art can be true is not an idle one, since the whole practice of criticism assumes an answer. Relativism, meaning the position that such a judgment can be true only relative to a standard, has the virtue of explaining the historical record: standards demonstrably change, as the earlier passages in this strand showed, and a claim that one period simply saw correctly looks hard to defend.

The opposing tradition tries to anchor judgment in agreement rather than in the work. David Hume observed that the verdicts of people with wide experience of an art form converge more than the verdicts of beginners, and proposed the ideal critic, meaning a hypothetical judge with full knowledge and no prejudice, as the standard against which a verdict is correct. The argument is not that experts cannot be wrong but that convergence under those conditions is the only thing that could make a judgment true, and that it does.

Each position has a hard case. The relativist has to explain why the convergence happens at all, and why disagreements about technique and coherence are settled so routinely in practice. The Humean has to explain away the cases where informed opinion converged and later reversed, which are numerous enough to be embarrassing and are the subject of a sardonic literature of its own. The most defensible position concedes that judgments about craft can be argued to a conclusion, that judgments about value cannot be settled in the same way, and that criticism is worth doing anyway. Both positions have been defended in detail by philosophers who agree about almost every particular verdict. The practical consequence of the choice is smaller than the heat of the argument suggests.

WHAT THIS PASSAGE IS FOR

- Standards demonstrably change over periods, which relativism explains.
 - Hume observed that informed verdicts converge.
 - The Humean must explain reversals of informed opinion.
-

Social Science

Passages 191 to 200

191

SOC · S01 · dispute

Whether Polling Can Be Repaired

National polls in the United States were close to the popular vote in 2016 and badly wrong in several states, and in 2020 the error was larger and ran in the same direction again. Two diagnoses were offered. The first diagnosis is technical and fixable. It holds that polls had underweighted voters without college degrees, who were numerous and disproportionately supporting one candidate, and the remedy was to weight on education, which most firms promptly did. That correction was not enough in 2020, and the error in several states ran the same way again.

The second diagnosis is differential nonresponse, meaning that one group answers surveys at a different rate from another, in a way that correlates with how they vote. If people who distrust institutions are both less likely to answer a pollster and more likely to support a particular candidate, then no weighting on demographics can repair the sample, because the people missing are not missing at random within their categories. That is a structural problem rather than a technical one, and it cannot be fixed by obtaining more interviews.

Several remedies are under test. Weighting on recalled past vote helps and introduces its own error, because recall is unreliable and drifts toward the winner. Registration-based sampling allows respondents to be verified against the voter files, at the cost of excluding anyone not registered. The most honest position available is that polls remain the best instrument for measuring opinion between elections and have a known and currently unrepaired weakness in forecasting close ones, and that the appropriate response is to widen the stated uncertainty rather than to pretend the problem has been solved. Several national associations published post-mortems after each election with largely consistent findings, and the 2020 report concluded that the cause could not be identified from the data available to it.

WHAT THIS PASSAGE IS FOR

- Weighting on education was adopted and was not enough.
 - The people missing are not missing at random within their categories.
 - The appropriate response is to widen the stated uncertainty.
-

Gold Standard or Narrow Tool

A randomized trial establishes that a treatment worked in the place it was run, with the people who took part, under the conditions that prevailed. The claim that it is the gold standard asserts something more: that this design sits above others in a hierarchy of evidence regardless of the question being asked. Angus Deaton and Nancy Cartwright have pressed the objection hardest, and it concerns external validity, meaning whether a result obtained in one setting holds in another.

Their argument is that transferring a trial result requires knowledge the trial does not supply. A deworming program that raised school attendance in one Kenyan district may do nothing in a district with different worm species, different schools or different distances to a clinic, and knowing which features matter requires a mechanism rather than another trial. Without that, the basis for applying the finding anywhere else is tenuous, and a hierarchy that ranks trials above the mechanistic evidence needed to use them has the order wrong.

The defense concedes the point about transfer and denies the conclusion. Trials settle the one thing other designs cannot, which is whether the effect exists at all in a known population, and the history of medicine is a long record of treatments that mechanism and observation endorsed and trials destroyed. The position most practitioners now hold is that a trial is the best available instrument for a narrow question and that the hierarchy was always a slogan. Replication across settings, with the mechanism investigated in each, is what makes a finding usable, and that program is slower and less quotable than a single result. Multi-site trials with pre-specified comparisons across sites are one response to the objection. They are expensive and the number of them remains small.

WHAT THIS PASSAGE IS FOR

- A trial establishes that a treatment worked where it was run.
 - Transferring a result requires knowledge the trial does not supply.
 - Trials settle whether the effect exists at all in a known population.
-

Fisher, Hill and the Cigarette

Ronald Fisher, the greatest statistician of his generation, argued until his death in 1962 that the link between smoking and lung cancer had not been shown to be causal. His objection was not foolish and is still taught in courses on causal inference. He proposed the constitutional hypothesis, meaning that a third factor, in this case an inherited disposition, might cause both the taste for tobacco and the susceptibility to the disease, which would produce the observed association with no causal arrow between the two. He also pointed out, entirely correctly, that nobody could run the experiment that would settle the question directly.

Austin Bradford Hill answered not with a single study but with a list. The Bradford Hill's considerations, meaning the features that strengthen a causal reading, include the size of the association, its consistency across populations and designs, a dose response, the right time order, a plausible mechanism, and what happens when the exposure is removed. Smoking satisfied all of them, and the agreement was unusually complete. Heavy smokers had about twenty times the risk, the finding appeared in every country examined, risk rose with consumption and fell after quitting, and lung tissue from smokers showed the damage directly.

Fisher's objection was answered rather than dismissed, and the answer took fifteen years. Twin studies addressed the constitutional hypothesis directly, animal experiments supplied the mechanism, and the removal of exposure in people who quit supplied something close to an experiment. What the episode establishes is that observational evidence can carry a causal claim, that doing so requires many independent lines rather than one strong correlation, and that a sound objection from a serious opponent is the most useful thing a field can be given. Fisher had been a paid consultant to the tobacco industry, which is often mentioned and does not dispose of his argument.

WHAT THIS PASSAGE IS FOR

- Fisher proposed an inherited disposition causing both.
 - Hill answered with a list of considerations rather than one study.
 - The answer required many independent lines of evidence.
-

The Average Patient

In 1982 Stephen Jay Gould was told he had a cancer with a median survival, meaning the time by which half of patients have died, of eight months. He was a statistician by training and read the sentence correctly: a median is a location, the distribution had a long right tail, he was young and well treated and otherwise healthy, and he had reason to think he belonged in that tail. He lived twenty years. His essay on the subject is the clearest statement of the case against taking an average as a description of a person.

The general problem is the heterogeneous treatment effect, meaning a situation in which a treatment helps some people and harms others while the average looks small. A drug that saves one patient in ten and kills one in twenty reports a modest net benefit, and the figure is correct and useless for the decision in front of a doctor. Trials are powered to detect averages, and the subgroups that would reveal the structure are exactly the comparisons that produce false positives if examined without discipline.

The defense of the average is practical and strong. It is the only quantity that can be estimated reliably from a feasible sample, subgroup claims have a poor record, and personalized predictions from small data perform badly in tests. The position that has emerged is not a choice but a sequence: estimate the average first, pre-specify a small number of subgroups on mechanistic grounds, and treat anything else as a hypothesis. The want of good individual-level evidence is a real limitation and not an argument for guessing. Modern trial protocols register the subgroup analyzes in advance for exactly this reason. Where they are registered, the proportion of positive subgroup findings falls sharply.

WHAT THIS PASSAGE IS FOR

- A median is a location, not a prediction about a person.
 - A treatment can help some and harm others while the average is small.
 - The position is a sequence rather than a choice.
-

When Paying Makes Things Worse

Richard Titmuss argued in 1970 that paying for blood produces less blood and worse blood than asking for it. His reasoning was that payment changes the meaning of the act, driving out donors who gave from duty while attracting those with most reason to conceal an infection. Kenneth Arrow replied that the argument confused two claims and that a market and a gift system could coexist. The data at the time were not good enough to settle it, and the dispute became the standard case for crowding out, meaning the displacement of an internal motive by an external reward.

The experiment usually cited against incentives is smaller and cleaner. Ten Israeli day care centers introduced a fine for parents who collected their children late, and lateness increased, substantially and permanently, and did not fall back when the fine was removed. The explanation offered is that the fine converted a social obligation into a price, and once a parent could buy an extra twenty minutes the obligation was gone and did not return.

The honest reading of the literature is narrower than either slogan. Crowding out is real, is documented in a modest number of settings, and depends on what the payment signals: a payment that reads as recognition does not displace the motive, while one that reads as a transaction does. Most incentives in most places work in the ordinary direction. The pernicious cases share a feature worth remembering, which is that the activity being paid for already carried a meaning that the payment replaced, and that meanings are harder to restore than prices. Blood systems in most rich countries are unpaid, which is consistent with Titmuss and does not demonstrate him. Plasma collection, which is paid in several countries, has a different safety record and a different donor population.

WHAT THIS PASSAGE IS FOR

- Titmuss argued that payment changes the meaning of the act.
 - Lateness increased when a fine was introduced and did not fall back.
 - Crowding out depends on what the payment signals.
-

Chosen or Inherited

Two accounts of norms are current, one from game theory and one from anthropology, and they make different predictions. The equilibrium account holds that a norm persists because it is in each person's interest to follow it once others do: driving on one side of the road is the clear case, and the content of the rule is arbitrary while the coordination is not. On this view norms are chosen, in the weak sense that anyone would choose them given what everyone else is doing, and they change rapidly when the incentives change.

The transmission account, meaning the view that norms are learned in childhood and reproduced largely unexamined, holds that they are carried by people who could not say why they hold them. On this view a norm is inherited rather than chosen, and the evidence is that norms survive the conditions that produced them. Rules about food, cleanliness and family obligation persist for generations after migration, and differences between neighboring and insular communities with identical economic conditions can be large and stable.

The distinguishing prediction concerns speed. Equilibrium norms should change fast when the payoffs do, and several have: smoking indoors became unacceptable in many countries within about fifteen years, and so did seat belt use. Transmitted norms should change at the speed of generations, and several have done that instead, including attitudes to marriage and to authority, which track birth cohort more closely than they track current conditions. The useful conclusion is that both mechanisms exist and that the observed speed of change is the evidence that identifies which one is operating, which also means that a policy working on one of them will not work on the other. Studies of migrant communities in Toronto and elsewhere are the standard testing ground for the second account. They are confounded by selection, since migrants are not a random sample.

WHAT THIS PASSAGE IS FOR

- The equilibrium account makes the content arbitrary and coordination essential.
 - Norms survive the conditions that produced them.
 - The observed speed of change tells you which mechanism is at work.
-

Does Building Lower Rents

The textbook position is that rents are high where housing is scarce, so building more housing reduces rents. The skeptical reply, common among tenant organizations in expensive cities, is that new buildings are luxury buildings, that they arrive with cafes and rising rents around them, and that supply in practice produces induced demand, meaning that new units attract additional people and the price does not fall.

Both claims have evidence and the evidence operates at different scales. Studies of individual new buildings in San Francisco, Helsinki and New York find that rents in the immediately surrounding blocks rise, which supports the skeptics on the thing residents actually observe. Studies of metropolitan areas find that places which built more have lower rent growth, which supports the textbook. The reconciling mechanism is filtering, meaning the process by which new expensive housing frees older cheaper housing for others, and tracking individual moves shows chains of six or seven vacancies following one new building, reaching down the income distribution within a few years.

The dispute that remains is about who bears the costs and over what period. A tenant displaced this year is not compensated by lower regional rents in five years, and the chains work through a market in which the first beneficiaries are better off than the last. The defensible position is that supply does reduce rents at the scale of a region, that it does not protect the people nearest the construction, and that these two facts are not in conflict, which is why the policy argument is about compensation and timing rather than about economics. Both literatures have grown quickly since administrative rent data became available. The local and regional findings have each been replicated in several countries.

WHAT THIS PASSAGE IS FOR

- Rents rise in the blocks immediately around a new building.
 - Areas that built more have lower rent growth.
 - The two facts are not in conflict.
-

Thirty Years of Estimates

The minimum wage literature now contains hundreds of estimates and the central finding is that modest increases have small employment effects. That sentence conceals thirty years of argument, and the argument is instructive about how a contested empirical question gets settled or fails to. Few empirical questions in economics have been fought over for as long or with as much data.

Three things changed the field. The first was data: administrative employment records replaced surveys, and the number of separate minimum wage changes available for study rose into the hundreds as states and cities set their own rates. The second was method: comparisons began to use only nearby areas, or only the same industry, or the distribution of wages rather than total employment, which can discriminate between a job lost and a job that simply pays more. The third was meta-analysis, meaning the statistical combination of many studies, which revealed that the literature had a funnel plot consistent with small null results going unpublished.

What remains disputed is the boundary. Everyone agrees that a very high minimum would cost jobs and that a very low one would not bind at all, and the argument is about where the turn comes, which depends on the local wage distribution and cannot be answered in general. Monopsony provides the theory for why small increases may not cost jobs, since an employer with wage-setting power is already paying below the competitive level. The honest summary is that the simple textbook prediction was wrong about the range of increases actually legislated, that nobody has established the upper boundary, and that recent increases in some American cities are the largest natural experiment yet run. Several cities have raised minimum wages above half the local median wage, which is beyond the range previously studied. Results from those cases are being published now and are not yet consistent.

WHAT THIS PASSAGE IS FOR

- Administrative records replaced surveys and the number of cases rose.
 - Meta-analysis found evidence that null results went unpublished.
 - The disputed question is where the turn comes.
-

Inequality or Poverty

Two targets are available to anyone who wants to act on the distribution of income, and they are not the same target. One is absolute poverty, meaning a standard fixed at a level of goods and services rather than relative to others. On that measure the world has improved enormously in forty years, with the share of people below the international line falling from something near four in ten to under one in ten. The other is inequality, measured as the gap between the top and the rest, and on that measure many rich countries have moved sharply the other way.

The priority dispute follows directly. One side holds that poverty is what matters, that a gap is only bad if it makes someone worse off, and that the ostensible concern about inequality is often envy with a theory attached. The other side holds that inequality does make people worse off, through mechanisms that can be specified. Political influence buys rules, and high inequality raises the price of positional goods such as housing near good schools. Relative standing has also been shown to affect health and trust in ways that can be measured across populations.

The empirical question is whether those mechanisms are large, and the evidence is mixed in a way that does not favor either side decisively. Correlations between inequality and poor health across countries are real and shrink substantially when income levels are controlled for. The political mechanism is the best supported of the three and is hard to quantify. What can be said is that the two targets sometimes conflict in policy, since growth has reduced poverty in several countries while increasing inequality, and anyone who declines to choose between them is not facing the arithmetic. Both measures depend on choices about price indices and household size, as the World Bank has documented, that move the headline figures.

WHAT THIS PASSAGE IS FOR

- Absolute poverty has fallen enormously in forty years.
 - The two targets sometimes conflict in policy.
 - Health correlations shrink substantially when income is controlled for.
-

Irrational or Economical

The heuristics literature can be read in two ways, and the reading chosen decides what follows from it. The first reading treats each documented departure from a formal model as a bias, meaning an error, and it concludes that human judgment is systematically defective and in need of correction by institutions and by experts. The second holds that the rules people use are adapted to the environments in which they normally operate, a position called ecological rationality, and that performance in an artificial problem is not evidence about performance in life.

The second reading has experimental support that is easy to state. Several famous errors shrink or vanish when the same problem is posed in frequencies rather than in single-event probabilities, which is how information arrives in experience. Doctors who fail the base rate problem in percentages mostly succeed when it is posed as counts out of a thousand. Simple rules that ignore most of the available information outperform complex models in some forecasting tasks, which is not what a defect account predicts.

Neither reading explains everything. Anchoring is robust across formats and is hard to call adaptive. Framing effects persist in frequency presentations. Against that, the practical lesson from the second reading has proved more useful than the practical lesson from the first: redesigning the presentation of information works better and more reliably than training people out of errors. The most telling evidence is therefore about remedies rather than about theory, and the two camps have converged on practice while continuing to disagree about what the practice shows. Gigerenzer and Kahneman debated the question in print for two decades, and both endorsed the frequency presentation of medical risk. That convergence on practice is probably the most useful thing the whole argument has produced, and it arrived without either side conceding the theoretical point at issue.

WHAT THIS PASSAGE IS FOR

- One reading treats each departure as an error in need of correction.
 - Doctors succeed with counts out of a thousand where they fail with percentages.
 - The camps converged on practice while disagreeing about theory.
-

Appendix A

Strand index

The four numbers are the passages for that strand, in level order.

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Appendix B

The field terms explained

Every term this book defines, with the passage in which it is defined.

- abolitionist** campaigning for the ending of slavery (3)
- absolute poverty** a standard fixed at a level of goods and services rather than relative to others (199)
- absolute temperature** temperature counted from the coldest possible point rather than from the freezing point of water (74)
- activation energy** the push a reaction needs before it will start, even when it releases energy overall (77)
- active frontage** frontage with doors and windows that people use (89)
- agenda setting** influencing which issues people regard as important rather than what they think about them (160)
- all-white primaries** party contests from which Black voters were excluded (55)
- allele** one of the alternative versions of a gene (62)
- anchoring** the pull of an irrelevant number on a later estimate (100)
- appropriation** a vote that releases funds (2)
- aside** a line spoken to the audience and not to the other characters (85)
- asymptomatic carrier** a person who harbors an organism without becoming ill (116)
- atomic weight** the mass of an atom of an element relative to a standard (127)
- ATP** the molecule a cell uses as its working currency of energy (63)
- attribution** establishing how much of an observed change is caused by a particular factor (168)
- autograph score** the full score written out in the composer's own hand (138)
- auxiliary assumption** a background claim that has to hold for a measurement to mean what it is taken to mean (171)
- availability** judging how common a thing is by how easily examples come to mind (100)
- balance** an instrument that compares an unknown weight against a known one (21)
- basalt** rock that was once liquid (19)
- base rate** the rate at which a condition occurs before any test is run (50)
- beadles** uniformed officers employed to keep order (39)
- beat** an assigned area such as the courts, the city hall or the police (60)
- bias** a departure from the answer a formal model prescribes (200)
- bibliography** a systematic list of everything published of a particular kind in a particular period (136)
- bid rent** the amount a household or firm will pay for land at a given distance from the center (97)
- biomass** the total weight of living material (64)
- boundary change** an alteration of the areas for which figures are reported (147)

boycott an organized refusal to ride (7)

Bradford Hill's considerations a set of features that strengthen a causal reading of an association (193)

bred true that tall plants from those lines always gave tall offspring (12)

cadence a closing formula that signals an ending (88)

caesura a strong break inside a line (84)

calibration curve a chart converting measured radiocarbon into calendar years, built from samples of known age (129)

calibration period the overlap in which a proxy can be checked against direct measurement (118)

canceled chapter a draft section that the author replaced before publication (132)

carpools shared cars run from church basements (7)

carrying capacity the number the habitat can support indefinitely (65)

cascading failure a sequence in which each component protects itself and thereby overloads the next (175)

cast iron a material that is strong when squeezed and weak when pulled (28)

casual hiring taking men on by the half day with no contract (48)

catalyst a substance that lowers the barrier without being consumed (77)

cause of death coding the system by which a death is assigned to a single cause (106)

cavity the space under the floating ice (20)

cesspit a pit for household waste (16)

character function the role a figure performs in a plot, independent of personality (182)

characteristic strength a value below which only a small stated fraction of test specimens fail (128)

cholera a disease that killed quickly, often within a day (16)

circuit breaker a device that opens the circuit when too much current flows through it (25)

classification the withholding of the government's own evidence from examination (59)

closed system a sample from which nothing relevant has entered or escaped since it formed (119)

clover a low plant of the pea family grown for hay (17)

commission a body set up to investigate and report (6)

compliance cost the expense a firm incurs in meeting a regulation (156)

composition the arrangement of shapes and spaces within the frame (87)

concordance an index listing every occurrence of every word in a text with its context (133)

confederate a person in an experiment who appears to be a participant but is working for the researcher (146)

confidence interval a range calculated so that the procedure captures the true value in a stated share of repetitions (144)

- confounded design** a study in which two possible causes changed at the same time (164)
- confounder** a difference between groups that affects the outcome as well (92)
- conservation of energy** the rule that energy is never created or destroyed, only moved or converted (73)
- constitutional hypothesis** the proposal that a third factor causes both the exposure and the disease (193)
- control group** the group that does not receive the treatment (92)
- convention** a rule a kind of story is expected to follow (86)
- coram nobis** a rare procedure for reopening a conviction on the ground that the court was misled (109)
- corpus** a body of texts assembled so that it can be searched as a whole (133)
- counterfactual** what would have happened in the absence of the measure being assessed (156)
- covenants** clauses in deeds restricting who could buy a house (49)
- criterion** the standard against which a work is being judged (90)
- crowding out** the displacement of an internal motive by an external reward or penalty (195)
- crust** the outer rock layer of the Earth (19)
- defined constant** a quantity assigned an exact value, so that it is no longer measured (126)
- delegation** a grant of broad authority from a legislature to an executive (59)
- dendrochronology** dating a wooden panel by matching its ring pattern to a regional record (137)
- denominator** the figure a count must be divided by before it means anything (106)
- descriptive norm** what people actually do, as distinct from what they say should be done (96)
- difference in differences** comparing the change in a treated place with the change in an untreated one over the same period (145)
- differential nonresponse** a situation in which one group answers surveys at a different rate from another (191)
- diffraction** the spreading of a wave after it passes an edge or a narrow gap (76)
- direct action** protest intended to create a crisis that forces negotiation (157)
- direct rebound** the extra consumption of the same service by the same user (173)
- directors' theater** a tradition in which the director's interpretation is the primary creative act (185)
- discovery space** a curtained opening at the back of the stage (85)
- dominant** the version that shows whenever it is present (62)
- dose** the energy absorbed per unit of body mass (79)
- Dunning school** an early twentieth-century interpretation treating Reconstruction as a corrupt failure (155)
- earthenware** baked clay, unglazed or roughly glazed (37)

- ecological rationality** the view that a rule is rational if it performs well in the environment where it is used (200)
- efficiency** the fraction of the energy put in that comes out in the useful form (73)
- elasticity** the percentage change in quantity produced by a one percent change in price (145)
- elk** the large deer that wolves hunt there (14)
- endosymbiosis** the proposal that some organelles descend from bacteria taken up by another cell (163)
- enjambment** running a sentence past the end of a line without a pause (84)
- entropy** a measure of the number of microscopic arrangements consistent with a given state (174)
- equilibrium account** the view that a norm persists because it is in each persons interest once others follow it (196)
- eutrophication** the enrichment of water by nutrients, which strips it of oxygen (167)
- excess mortality** deaths above the number that would have been expected without the event (108)
- exclosure** a fenced plot that keeps a particular animal out while leaving everything else the same (114)
- executive order** a written instruction from a president that carries the force of law (9)
- exponential growth** growth in which the increase is proportional to the number already present (65)
- exposition** direct statement of background by the narrator (82)
- external validity** whether a result obtained in one setting holds in another (192)
- faction** a group united by a common interest that is adverse to the rights of others (52)
- factorial experiment** a design in which two or more treatments are applied separately and together (165)
- fair play** the rule that every clue is shown to the reader before the solution (86)
- false positive** a positive result in a person who does not have the condition (50)
- federal examiners** officials sent to register voters where local registrars refused (57)
- feminine ending** a line that finishes on an unstressed extra syllable (134)
- fermentation** the short cut the cell takes when no oxygen is available (13)
- field experiment** a trial run in a real setting rather than a laboratory (46)
- filtering** the process by which new expensive housing frees older cheaper housing for others (197)
- fingerprint** a pattern of change that a particular cause produces and others do not (168)
- fixation** the conversion of nitrogen gas into a form that living things can use (67)
- follow-up period** the length of time over which participants are tracked after the trial (142)
- foot warmer** a small wooden box for hot coals (37)
- footfall count** a tally of how many people pass a point in a given period (139)

- formalism** the study of a work as a structure of parts rather than as a representation of people (182)
- franchise** the right to vote (3)
- free indirect style** a third-person voice that slides into a character's way of thinking (81)
- free verse** poetry without a regular meter or rhyme scheme (184)
- Freedmen's Bureau** a federal agency that set up schools and ran courts (5)
- freedom of information** a statutory right to demand government documents (107)
- friction** the force between the tires and the road surface (72)
- frontage** the length of building face that meets the pavement (89)
- funnel plot** a chart used to detect whether small studies with null results are missing from a literature (198)
- fur return** the number of pelts a trading company bought in a season (115)
- generalized cost** travel time and money added together as one figure (97)
- genetic drift** change in the frequency of a variant by chance alone, without any advantage (161)
- Gini coefficient** a single figure summarizing how far a distribution departs from complete equality (149)
- glycolysis** the first stage, which splits the sugar in half (63)
- gravimetry** measuring mass by its gravitational pull on a satellite (120)
- grid frequency** the rate at which alternating current reverses, which falls when demand exceeds supply (125)
- gridlock** a condition in which the system cannot act although majorities exist (152)
- grievances** formal complaints (1)
- grounding line** the line where the ice lifts off the sea bed (20)
- half-life** the time in which half of a quantity of a radioactive element decays (79)
- heritability** the share of the variation in a trait that is passed to offspring (111)
- heritable** passed from parent to offspring through the genes (61)
- heterogeneous treatment effect** a situation in which a treatment helps some people and harms others (194)
- historically informed performance** playing music on the instruments and in the manner of its own period (188)
- historiography** the study of how history has been written, as distinct from what happened (155)
- hormesis** the proposal that very small doses are harmless or beneficial (179)
- Hubble constant** the present rate at which the universe is expanding (180)
- iambic pentameter** ten syllables with the stress falling on every second one (34)
- ice cliff failure** the structural collapse of an exposed ice face too tall to support itself (170)
- ice core** a cylinder of ice drilled from a glacier, with annual layers and trapped air (118)

- iconography** the study of the conventional meanings attached to images in a particular period (187)
- ideal critic** a hypothetical judge with full knowledge and no prejudice, used to define a correct verdict (190)
- implied author** the sensibility a reader reconstructs from the whole text, distinct from both narrator and writer (181)
- incubation period** the gap between infection and the first symptom (66)
- induced demand** new supply attracting additional people and so sustaining the price (197)
- inertia** the stored rotational energy of large generators, which slows any change in frequency (125)
- infrared** the long wavelengths at which a warm surface radiates (68)
- infrared reflectography** a method that sees through paint layers to the drawing underneath (137)
- inhibitor** a substance that blocks one particular step in a pathway (113)
- intelligence assessment** a report evaluating a supposed threat for the government (109)
- intentional fallacy** the claim that an authors stated purpose does not settle what a work means (183)
- interaction** an effect of two treatments together that is larger than their effects added (165)
- interference pattern** the pattern of bright and dark bands produced when two waves overlap (176)
- intergenerational mobility** how much a child's position depends on the parents' position (99)
- interiority** direct access to what a character is thinking (82)
- internal contradiction** a place where a text disagrees with itself (131)
- iron oxide** which is rust (27)
- irony** saying something while giving the reader grounds to take the opposite (181)
- islanding** deliberately separating part of a network so that it can run on its own (175)
- isotope label** an atom of unusual weight used to trace a substance through a sequence of reactions (113)
- judicial deference** the practice of courts accepting an executive judgment in its own field (159)
- judicial review** the power to refuse to apply a law that conflicts with the Constitution (2)
- kinetic energy** the energy an object has because it is moving (72)
- land revenue** the tax on agricultural land that funded most of the administration (58)
- latent heat** the energy absorbed when ice turns to water without changing temperature (70)
- laureate** a person awarded the prize (43)
- leaching** the washing of dissolved nitrogen down below the roots (67)
- liability** legal responsibility for the cost of an injury (56)

- lichen** the low crusty growth that reindeer eat (15)
- lichens** the crusty growths that cover healthy bark (11)
- life cycle analysis** an accounting that follows the energy used from extraction through manufacture to disposal (123)
- life expectancy** the average number of years a person born now is expected to live (49)
- limiting case** a situation in which a newer theory reduces to an older one (172)
- linear no-threshold** the assumption that risk falls in proportion to dose all the way down with no safe level (179)
- linkage map** a chart of gene order built from how often inherited traits are separated (112)
- literary hierarchy** the ranking of kinds of writing by assumed seriousness (186)
- litigation strategy** the planned use of selected court cases to establish a legal principle (157)
- long-term experiment** a trial continued for decades so that slow effects become visible (117)
- Lorenz curve** a graph of the cumulative share of income held by the poorest given share of the population (149)
- magma** molten rock below the surface (69)
- manifest** the list of people and goods carried on a particular voyage (104)
- marine ice sheet instability** self-sustaining retreat of a glacier grounded below sea level on a reverse slope (170)
- mark and recapture** the practice of tagging individuals so that the same animal can be identified again (111)
- mechanical equivalent of heat** the amount of work that produces a given quantity of heat (124)
- mechanism** the account of how a treatment produces its effect, which may differ between settings (192)
- median** the middle value when the cases are put in order (44)
- median survival** the time by which half of a group of patients has died (194)
- meta-analysis** a statistical combination of the results of many separate studies (198)
- metonymy** naming a thing by something associated with it (83)
- missing heritability** the gap between heritability estimated from twins and the variation explained by identified variants (162)
- mixed use** buildings and streets that combine housing, work and shops in the same area (189)
- monopoly** the sole right to produce and sell (8)
- monopsony** a market with one buyer, or few, for a particular kind of labor (98)
- moral suasion** persuading individuals to renounce a practice rather than seeking to outlaw it (154)
- nationalist historiography** accounts written from the standpoint of the independence movements (158)

- natural experiment** a situation in which something outside the researchers control has allocated people almost at random (143)
- neutral theory** the proposal that most molecular change has no effect on survival (161)
- nodules** small swellings on the roots of clover (17)
- nonresponse** the failure of selected people to answer at all (91)
- normalization of deviance** the process by which a repeated warning sign comes to be treated as acceptable (178)
- omnibus** a large horse-drawn carriage running a fixed route (47)
- omniscient** a narrator with access to every mind in the book (81)
- organelle** a structure inside a cell that performs a particular function (163)
- originalism** the view that a text means what its authors and first readers understood it to mean (151)
- pamphlet** a short work sold cheaply (4)
- parallax** the shift in a star's position as the Earth moves from one side of its orbit to the other (80)
- parts per million** the unit in which the measurement is reported (18)
- payroll records** employment figures taken from the administrative accounts of firms (148)
- performance text** the version of a play that a particular production actually realizes (185)
- perihelion** the point in an orbit closest to the Sun (172)
- period** the time a pulsating star takes to complete one cycle of brightening and dimming (130)
- period eye** the habits of seeing that a viewer of the time would have brought to a work (187)
- periodic table** an arrangement of the elements in which chemically similar ones fall in the same column (127)
- photographic plate** a glass plate coated with emulsion, used to record an image before film or sensors (130)
- political abolitionism** pursuing the end of slavery through parties, votes and legislation (154)
- poll book** the register of who actually voted in a given election (103)
- poll tax** a charge that had to be paid before a person could vote (53)
- poll taxes** charges that had to be paid before a person could vote (5)
- positional good** something whose value depends on how much of it others have (199)
- postulate** a condition laid down in advance that a claim must satisfy (116)
- poverty map** a street plan colored to show the economic condition of each house (139)
- power** the rate at which energy is delivered or lost (75)
- preamble** the opening section that states a purpose (1)
- premises** statements offered without proof because the writer expects agreement (51)

preregistration publishing the design and analysis plan before the data are collected (150)

probability sample a sample in which every member of the population has a known chance of selection (141)

promptbook a working copy of a play marked up by the stage manager for one production (135)

proxy a measurement used to stand in for a quantity that cannot be observed directly (115)

publication bias the tendency for positive results to be published and null results not (150)

quota sampling choosing respondents to match known proportions in the population (110)

radiocarbon a heavy form of carbon produced in the atmosphere and taken up by living things (129)

radiometric dating dating a rock by the proportion of a radioactive element that has decayed (119)

radium an element discovered by Marie and Pierre Curie in 1898 (29)

random error scatter that falls on both sides of the true value (71)

randomized trial a study in which the groups are formed by lot (42)

ratifying convention a state assembly elected to vote on the Constitution (102)

rebound effect the increase in use of a service that follows from making it cheaper to provide (173)

reception history the record of how a work was read and valued over time (140)

reduction the claim that one science is in principle derivable from another (177)

reference mass a weight whose value has been certified by comparison with a higher standard (121)

refraction the bending of light as it passes from one material into another (26)

registration roll the official list of those entitled to vote (105)

regression discontinuity comparing cases just above and just below an arbitrary cutoff (143)

relativism the position that a judgment of value can be true only relative to a standard (190)

replication an independent repetition of a study to see whether the result holds (146)

reproduction number the average number of new cases that one case produces (66)

reservation wage the lowest pay at which a person will take the work (98)

residence time how long a molecule stays in the air before being taken up elsewhere (68)

resistance the opposition a conductor offers to current (75)

resolutions formal proposals put to a vote (3)

respondents the people who answer a survey (41)

retrograde slope a sea bed that gets deeper inland rather than shallower (70)

revenue settlement the periodic assessment that fixed what each district owed (108)

reverse causation the outcome producing the supposed cause rather than the other way round (93)

rough draft the working manuscript with its crossings-out and insertions still visible (101)

safety factor the ratio between what a structure is built to withstand and what it is expected to carry (128)

salience how noticeable a cost or benefit is at the moment of choosing (95)

sampling frame the list from which the people to be asked are drawn (91)

sanction a penalty applied by other people rather than by law (96)

scansion marking the pattern of stressed and unstressed syllables in a line (134)

scrivener a clerk employed to copy documents by hand (32)

selection effect a pattern produced by how cases entered the data (93)

selective exposure the tendency of people to attend to sources that agree with them already (160)

sensors devices that reported whether the space was occupied (45)

sequencing reading the order of chemical letters along a stretch of DNA (112)

sigma a unit expressing how far two results are apart in terms of their own uncertainties (180)

signal the information that the introduction of a payment conveys about the activity (195)

simile a comparison that announces itself with like or as (83)

sketchbook a composers working notebook of fragments, trials and rejected versions (138)

skew a distribution with a long tail on one side (94)

soot the black powder left by burning coal (11)

sprinklers ceiling pipes that release water in a fire (6)

stable isotope ratio the proportion of heavy to light atoms in a tissue, which differs by diet (114)

standard candle an object whose true brightness is known, so that its dimness gives its distance (80)

standard deviation a measure of how far the typical case sits from the average (94)

state by state strategy winning the vote one legislature at a time rather than by federal amendment (153)

statistical law a regularity that holds because exceptions are overwhelmingly improbable rather than impossible (174)

statistical significance the judgment that a result is unlikely to have arisen from sampling variation alone (144)

sterling balances debts to India recorded in British currency and held in London (58)

straw poll an informal canvass with no attempt at a representative sample (110)

stress the force carried by each unit of area (78)

strict scrutiny the most demanding standard a court applies before allowing a classification by race (159)

subduction the sliding of one plate down beneath another (69)

- subscription** payment in advance for a year of issues (10)
- substitute** an alternative that is easy to switch to (95)
- superblock** a large site planned as a single unit, with towers set in open ground (189)
- surveillance file** a record compiled by an agency about a person it was watching (107)
- survey data** figures collected by asking employers or workers directly (148)
- survival bias** the distortion caused by studying only what happened to be preserved (136)
- sworn testimony** evidence given under oath and taken down by a clerk (105)
- symbol** an image taken to stand for something beyond itself in a determinate way (183)
- system boundary** the line drawn around whatever is being measured, with everything crossing it counted (123)
- systematic error** an error that pushes every reading the same way (71)
- test case** a prosecution deliberately invited in order to get a ruling (103)
- theory-ladenness** the claim that what counts as a measurement depends on the theory being used (171)
- tide gauge** a fixed instrument recording sea level against a local benchmark (120)
- time of flight** measuring a speed by timing a signal over a known distance (126)
- tonal contrast** the difference between the lightest and darkest areas (87)
- tonic** the home note of a key, where a piece sounds settled (88)
- torsion balance** a beam hung on a thin fiber, which twists by a measurable amount under a tiny force (122)
- traceability** an unbroken chain of comparisons linking an instrument to a national standard (121)
- transfer of power** the official British term for the end of colonial rule (158)
- transmission account** the view that norms are learned in childhood and reproduced largely unexamined (196)
- trophic cascade** a chain of effects running down a food web from a predator to the plants (164)
- trophic level** a step in the chain of who eats whom (64)
- turnips** a root crop lifted for winter feed (17)
- unalienable** incapable of being transferred or surrendered (54)
- undercount** the number of people a census fails to record (147)
- uniformitarianism** the principle that the processes acting now are the ones that shaped the past (169)
- unreliable narrator** a teller whose account the reader has reason to doubt (131)
- urtext** an edition that attempts to present what the composer wrote, stripped of later editorial additions (188)
- veto** return a bill to Congress without signing (2)
- veto point** a place in a political system where one actor can block a decision (152)
- wavefront** the moving line along which a wave has the same phase (76)

wealth what a household owns minus what it owes (99)

weighting adjusting the answers so that the sample matches known population proportions (141)

workers' compensation a schedule of automatic payments replacing the lawsuit (56)

yield gap the difference between what a field produces and what it could produce (167)

yield point the load beyond which a material keeps the shape it is bent into (78)

Appendix C

Where the vocabulary words appear

The two hundred target words of Words in Context, each with the passage in this book that uses it in context.

abolished 60	dormant 118	praised 40
absorbs 26	economy 188	precarious 115
abundant 64	elegiac 138	precedent 109
accelerate 72	emerge 62	predict 49
adapted 11	emit 79	prefer 47
admired 39	empirical 122	prevailed 57
advocated 54	enacted 55	proliferate 117
aesthetic 133	endemic 150	qualify 151
affected 182	enigmatic 131	quiescent 168
aggregate 94	entrenched 105	ratified 51
allocate 92	erode 69	reciprocal 148
allusion 86	established 1	recondite 187
ambivalent 132	evokes 88	recovered 14
ameliorate 106	exact 155	reduce 29
anomalous 129	expand 3	refine 71
appreciate 171	expedient 108	reflects 37
arch 181	express 33	reform 6
arrest 185	factional 110	refractory 178
assume 43	fecund 165	rejected 8
attenuate 113	finite 126	release 13
attribute 124	flag 163	renders 85
austere 89	flows 25	repudiated 101
authority 2	fluctuate 65	resilient 119
bear 197	gross 180	resist 28
behave 46	harbor 166	resolve 172
benefit 48	heterogeneous 147	respond 50
bias 91	husband 167	restrict 9
calibrate 121	imitate 36	retrench 160
canonical 135	imposed 56	reveals 31
capitulate 158	incentive 95	reverent 137
causal 142	inexorable 174	reward 45
checked 152	influences 41	salient 125
coercion 104	inhabit 68	sanction 154
cohort 98	inherent 128	sardonic 190
collapse 78	inimical 170	satirize 90
commensurate 173	inspired 38	sensible 179
compare 42	insular 196	shaped 35
composed 34	intransigent 157	shelter 20
compound 176	involved 183	shifted 18
compress 74	laconic 189	sound 193
conceded 58	latent 145	spent 169
conciliatory 159	lyrical 84	spread 16
conduct 75	mandate 103	stable 27
conflict 10	marginal 144	stem 164
confound 143	marked 175	stigma 99
consume 17	measure 21	studied 184

contain 30
converge 130
convert 23
convey 81
correlate 93
declined 5
degenerate 177
delegated 52
densely 80
depends 12
depleted 67
derivative 136
detect 22
deviate 100
discrete 127
discriminate 198
disparity 149
disperse 66
dissent 59
dissolve 77
distort 76

methodical 141
migrate 70
mitigate 120
nascent 112
negligible 123
norm 96
normative 146
novel 161
obtain 191
opposed 4
ornate 87
ostensible 199
partisan 107
pastiche 186
perceive 97
pernicious 195
perturb 114
petitioned 53
plastic 162
polemic 140
portrays 32

subtle 82
survive 15
sustain 63
table 153
telling 200
temper 156
tenuous 192
threatened 19
thrive 61
transfer 24
trend 44
ubiquitous 116
understated 134
unite 7
usurped 102
viable 111
vivid 83
want 194
whimsical 139
yield 73

Appendix D

Reading log

Twenty rows of ten. Mark a passage when the three lines at the end of it hold no surprises.

Level 1 — Foundation

	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10
History and Civics	1	2	3	4	5	6	7	8	9	10
Biology and Earth Science	11	12	13	14	15	16	17	18	19	20
Physical Science	21	22	23	24	25	26	27	28	29	30
Humanities	31	32	33	34	35	36	37	38	39	40
Social Science	41	42	43	44	45	46	47	48	49	50

Level 2 — Developing

	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10
History and Civics	51	52	53	54	55	56	57	58	59	60
Biology and Earth Science	61	62	63	64	65	66	67	68	69	70
Physical Science	71	72	73	74	75	76	77	78	79	80
Humanities	81	82	83	84	85	86	87	88	89	90
Social Science	91	92	93	94	95	96	97	98	99	100

Level 3 — Target

	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10
History and Civics	101	102	103	104	105	106	107	108	109	110
Biology and Earth Science	111	112	113	114	115	116	117	118	119	120
Physical Science	121	122	123	124	125	126	127	128	129	130
Humanities	131	132	133	134	135	136	137	138	139	140
Social Science	141	142	143	144	145	146	147	148	149	150

Level 4 — Stretch

	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10
History and Civics	151	152	153	154	155	156	157	158	159	160
Biology and Earth Science	161	162	163	164	165	166	167	168	169	170
Physical Science	171	172	173	174	175	176	177	178	179	180
Humanities	181	182	183	184	185	186	187	188	189	190
Social Science	191	192	193	194	195	196	197	198	199	200

Appendix E

Answer key

Two thousand questions in passage order. Each row gives the key, what makes it right, and the wrong answer most likely to attract. The trap line is the one worth reading twice.

1 · Four Days in Philadelphia HIS · S01 · phenomenon

- 1.1 B** central idea · easy Most of the document is a list of particular grievances because Congress was assembling a legal case against the king, which the third paragraph states outright. *Trap: A is true of the text but reports a detail rather than its main idea.*
- 1.2 D** detail · easy The second paragraph states that Congress cut about a quarter of what Jefferson produced after the other committee members left most of the writing to him. *Trap: A reverses the record, since the draft was shortened rather than approved untouched.*
- 1.3 A** evidence · medium The quotation says directly that Congress was assembling a legal case and that such a case requires particulars, which is the claim at issue. *Trap: D quotes the grievances themselves, which show what the case contained rather than that it was a legal case.*
- 1.4 C** inference · medium Congress voted to separate before it had either an army fund or a treasury and then had to build both within the year, so the vote committed it to work it could not yet do. *Trap: A inverts the sequence, since the institutions followed the vote rather than preceding it.*
- 1.5 B** words in context · easy Congress had neither an army fund nor a treasury and then had both, so here the word means brought into being rather than proved. *Trap: A gives the other common sense of the word, as in establishing that something is true.*
- 1.6 A** structure · medium The paragraph opens by noting that the grievances take up most of the page and then supplies the reason, which is that a legal case needs particulars. *Trap: B describes work the second paragraph has already done rather than the third.*
- 1.7 D** cross text · hard Text 2 holds that the premises are offered without proof while the grievances are evidence for a factual claim, so the evidence is what needs the space. *Trap: C uses a real detail, the cut of a quarter, to explain a proportion it did not produce.*
- 1.8 C** transitions · easy Congress lacked both an army fund and a treasury and then built both within the year, so the two sentences stand in contrast. *Trap: D would fit if the second sentence restated the first, but it reports a change instead.*
- 1.9 A** boundaries · easy Both halves are independent clauses, so a semicolon is needed rather than a comma, a colon, or no mark at all. *Trap: B is the comma splice, joining two independent clauses with only a comma.*
- 1.10 C** synthesis · medium Only this choice gives the reason for the proportion, tying the space taken by the grievances to the legal case that required named particulars. *Trap: B repeats two of the notes without explaining the proportion the student set out to explain.*